



UNIVERSITEIT VAN AMSTERDAM

Stochastic Processes

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INTRODUCTION. This course is a measure-theoretic introduction to the theory of continuous-time stochastic processes. We intend to treat some classical, fundamental results and give an overview of two important classes of processes. These processes are so-called *martingales* and *Markov processes*. The main part of the course is devoted to developing fundamental results in martingale theory and Markov process theory, with an emphasis on the interplay between the two worlds. The general results will then be used to study the fascinating properties of *Brownian motion*, an important process that is both a martingale and a Markov process. We also plan to study some applications in *queueing theory*.

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Stochastic Processes

Loosely speaking, a *stochastic process* is a phenomenon that can be thought of as evolving in time in a random manner.

EXAMPLE 1.1. (i) *Poisson process*: Arriving process of customers to queue.
(ii) *Brownian motion*: The erratic movement of pollen grains suspended in water.

NOTATION 1.2.

The basic assumptions for this model (in dimension 1) are the following:

- (i) The motion is continuous, moreover, in a time-interval $[t, t + \tau]$, τ small.
- (ii) Particle movements in two non-overlapping time intervals of length τ are mutually independent.
- (iii) The relative proportion of particles experiencing a displacement of size $(\delta, \delta + d\delta)$ is approximately $\phi_\tau(\delta)$ with
 - the probability of *some* displacement is 1:

$$\int_{-\infty}^{\infty} \phi_\tau(\delta) d\delta = 1;$$

- the *average* displacement is 0:

$$\int_{-\infty}^{\infty} \delta \phi_\tau(\delta) d\delta = 0;$$

- the variation in displacement is linear in the length of the time interval:

$$\int_{-\infty}^{\infty} \delta^2 \phi_\tau(\delta) d\delta = D\tau,$$

where $D \geq 0$ is called the *diffusion coefficient*.

Denote by $f(x, t)$ the density of particles over unit volume at position x and time t . Under differentiable assumptions, we get by a Taylor expansion that

$$\begin{aligned} f(x, t + \tau) &\approx f(x, t) + \tau \frac{\partial f(x, t)}{\partial t} \\ &= \int_{-\infty}^{\infty} f(x - \delta, t) \phi_\tau(\delta) d\delta \\ &\approx \int_{-\infty}^{\infty} \left[f(x, t) - \delta \frac{\partial f}{\partial x}(x, t) + \frac{1}{2} \delta^2 \frac{\partial^2 f}{\partial x^2}(x, t) \right] \phi_\tau(\delta) d\delta \end{aligned}$$

$$\approx f(x, t) + \frac{1}{2} D \tau \frac{\partial^2 f(x, t)}{\partial x^2}.$$

Equating gives rise to the *heat equation* in one dimension:

$$\frac{\partial f}{\partial t} = \frac{1}{2} D \frac{\partial^2 f}{\partial x^2},$$

which has the solution

$$f(x, t) = \frac{\text{\#particles}}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}.$$

So $f(x, t)$ is the density of a $\mathcal{N}(0, 4Dt)$ -distributed random variable multiplied by the number of particles.

Remark. BM has connection with partial differential equations: the Dirichlet problem, for instance, if there exists a function u such that

$$\begin{cases} \Delta u = 0, & \text{on } D \\ u = f, & \text{on } \partial D \end{cases}$$

with u continuously differentiable on D and continuous on ∂D . We solve it by running BM starting at point x till hits ∂D , say at time τ .

DEFINITION 1.3 (Stochastic Process).

Let T be a $[0, \infty)$ set in $\mathbb{Z}^+$ (or a subinterval) and $(E, \mathcal{E})$ a measurable space. A *stochastic process* indexed by T , with values in $(E, \mathcal{E})$, is a collection $X = (X_t)_{t \in T}$ of measurable maps

$$X_t : (\Omega, \mathcal{F}, P) \rightarrow (E, \mathcal{E}).$$

DEFINITION 1.4.

Let a space E be given. A collection $\mathcal{A}$ of subsets of E is called an *algebra* (or a *field*) if the following three conditions hold:

- (i) $E \in \mathcal{A}$.
- (ii) $A \in \mathcal{A} \Rightarrow A^c = E \setminus A \in \mathcal{A}$.
- (iii) $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$.

A collection $\mathcal{A}$ is called a σ -algebra on E if $\mathcal{A}$ is an algebra, such that

$$A_1, A_2, \dots \in \mathcal{A} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}.$$

DEFINITION 1.5 (π -System).

A collection $\mathcal{A}$ of subsets of E is called a π -system if $\mathcal{A}$ is invariant under finite intersections, i.e., $A, B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A}$.

LEMMA 1.6. (i) Let $\mathcal{I}$ be a π -system on E generating the σ -algebra $\mathcal{A}$. Suppose that μ_1, μ_2 are measures on $(E, \sigma(\mathcal{I}))$, such that $\mu_1(E) = \mu_2(E) < \infty$, and $\mu_1(A) = \mu_2(A)$ for all $A \in \mathcal{I}$. Then $\mu_1(A) = \mu_2(A)$ for all $A \in \sigma(\mathcal{I})$.

(ii) Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ be given. Let $\mathcal{I}, \mathcal{J}$ be π -systems on E with $\mathcal{A} = \sigma(\mathcal{I})$ and $\mathcal{F} = \sigma(\mathcal{J})$. Then $\mathcal{A}$ and $\mathcal{F}$ are independent if $\mathcal{I}$ and $\mathcal{J}$ are independent, that is, if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for all $A \in \mathcal{I}$, $B \in \mathcal{J}$.

Suppose that we have measure spaces $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$. Then the function $X : \Omega \rightarrow E$ is called a *random element* if X is $\mathcal{F}/\mathcal{E}$ -measurable. In other words, if $X^{-1}(A) \in \mathcal{F}$, for all $A \in \mathcal{E}$. The following lemma is helpful in checking measurability.

LEMMA 1.7.

Let $(\Omega, \mathcal{F})$, $(E, \mathcal{E})$ be measurable spaces. Let C be a collection of subsets of E , such that $\sigma(C) = \mathcal{E}$. Let $X : \Omega \rightarrow E$ be a map. Then X is $\mathcal{F}/\mathcal{E}$ -measurable if $X^{-1}(A) \in \mathcal{F}$ for all $A \in C$.

Given $\omega \in \Omega$, the corresponding realisation $x(\omega) = (x_t(\omega))_{t \in T}$ of the stochastic process X is called a *sample path* or *trajectory* of X . One can also view $X(\omega)$ as an E -valued function of T , i.e., there exists $f : T \rightarrow E$, such that $X(\omega) = f$, and $X_t(\omega) = f(t)$, $t \in T$. Hence, the process X takes values in the function space E^T .

It is natural to view such sets as the building blocks of our σ -algebra, so we define

$$\mathcal{E}^T = \sigma\{\text{1-dimensional rectangles}\} = \{\sigma\text{-cylinder}\}.$$

For understanding the structure of this σ -algebra, let us recall the concept of product- σ -algebras $\mathcal{E}^n$, $n = 1, 2, \dots, \infty$. For n finite, this is simply the σ -algebra generated by all n -dimensional rectangles:

$$\mathcal{E}^n = \sigma\{E_1 \times \dots \times E_n \mid E_1, \dots, E_n \in \mathcal{E}\}.$$

For $n = \infty$, it is the σ -algebra generated by all finite-dimensional rectangles (defined in E^∞ in a natural way):

$$\mathcal{E}^\infty = \sigma\{E_1 \times \dots \times E_n \times E \times E \times \dots \mid E_1, \dots, E_n \in \mathcal{E}, n = 1, 2, \dots\}.$$

Now we shall call a set $A \in \mathcal{E}^T$ a *cylinder*, if there exists a finite subset $\{t_1, \dots, t_n\}$ of T and a set $B \in \mathcal{E}^n$, such that

$$A = \{x \mid (x_{t_1}, \dots, x_{t_n}) \in B\}. \quad (1.1)$$

It will be called a σ -cylinder, if there exists a (at most) non-ordered countable subset $\{t_1, \dots\}$ of T and a set $B \in \mathcal{E}^\infty$, such that (1.1) holds.

LEMMA 1.8.

$\mathcal{E}^T$ is precisely the collection of σ -cylinder sets.

COROLLARY 1.9.

Let $T = [0, \infty)$, $E = \mathbb{R}$, $\mathcal{E} = \mathcal{B}(\mathbb{R})$, and

$$C = \{f : T \rightarrow E \mid f \text{ continuous}\}.$$

Then $C \notin \mathcal{E}^T$.

Proof. By definition, the σ -algebra

$$\mathcal{E}^T := \sigma\{\text{1-dimensional rectangles}\} = \{\sigma\text{-cylinders}\}$$

generated by sets of the form

$$A_t = \{f : T \rightarrow E \mid f(t) \in B\}, \quad \forall t \in T, \quad \forall B \in \mathcal{E}$$

is the smallest σ -algebra on the space of functions from $T = [0, \infty)$ to $E = \mathbb{R}$. The sets in $\mathcal{E}^T$ are constructed by countable operations involving only finite-dimensional conditions (finite collections of points).

Assume that $C \in \mathcal{E}^T$, then there exists a sequence of cylindrical sets (based on finite-dimensional Borel sets) whose countable operations define C . For a function $f : T \rightarrow E$ in C to be continuous, we have for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|t - \tau| < \delta \Rightarrow |f(t) - f(\tau)| < \varepsilon, \quad \forall \tau \in T,$$

requiring the function to behave in a certain way over an infinite neighbourhood around every point t , and it is impossible for a finite-dimensional Borel set to capture this continuity property. Therefore, we have proved that $C \notin \mathcal{E}^T$. $\square$

DEFINITION 1.10 (Brownian Motion).

The stochastic process $W = (W_t)_{t \geq 0}$ is called a (*standard*) *Brownian Motion (BM)* or *Wiener process*, if

- (i) $W_0 = 0$, *a.s.*
- (ii) W is a stochastic process with stationary, independent increments, i.e., $\sigma(W_t - W_s)$ and $\sigma(W_u, u \leq s)$ are independent.
- (iii) $W_t - W_s \stackrel{\mathcal{D}}{=} \mathcal{N}(0, t - s)$.
- (iv) almost all sample paths are continuous.

DEFINITION 1.11 (Poisson Process).

A real-valued stochastic process $N = (N_t)_{t \geq 0}$ is called a *Poisson process* if

- (i) $N_0 = 0$, *a.s.*
- (ii) N is a stochastic process with stationary, independent increments.
- (iii) N is a counting process, in other words:
 - N_t takes only values in $(\mathbb{Z}^+, 2^{\mathbb{Z}^+})$, $t \geq 0$;
 - $t \mapsto N_t$ is increasing, i.e., $N_s \leq N_t$, $t \geq s$.
- (iv) *a.s.* right-continuous of sample paths:

$$N_t = \lim_{s \downarrow t} N_s \leq \lim_{s \uparrow t} (N_s + 1).$$

The construction of a Poisson process is simpler than the construction of BM. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ be given. We construct a sequence of *i.i.d.* $(\mathbb{R}^+, \mathcal{B}(\mathbb{R}^+))$ -measurable random variables X_n , $n = 1, \dots$, on this space, such that $X_n \stackrel{\mathcal{D}}{=} \exp(\lambda)$. Next define

$$N_t = \max \left\{ n \left| S_n = \sum_{i=1}^n X_i \leq t \right. \right\}.$$

This is a Poisson process.

DEFINITION 1.12 (Multivariate Normal Distribution).

The random vector $X = (X_1, \dots, X_n)^T$ has a *multivariate normal distribution* if there exists a $n \times k$ matrix B , a vector $a \in \mathbb{R}^n$ and independent random variables $Z_1, \dots, Z_k$, with

$$Z_i \stackrel{\mathcal{D}}{=} \mathcal{N}(0, \sigma^2(Z_i)), \quad \sigma^2(Z_i) \geq 0,$$

defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$, such that $X = a + BZ$, where we write $Z = (Z_1, Z_2, \dots, Z_k)^T$. In this case, X is said to have the *multivariate normal distribution*

$\mathcal{N}(a, \Sigma)$, where $a = (\mathbb{E}X_1, \dots, \mathbb{E}X_n)^T$ is the *mean vector* and

$$\Sigma = \text{cov}(X_i, X_j) = B \begin{pmatrix} \sigma^2(Z_1) & & & \\ & \sigma^2(Z_2) & & \\ & & \ddots & \\ & & & \sigma^2(Z_k) \end{pmatrix} B^T,$$

the *covariance matrix*.

EXAMPLE 1.13. In general it is not true that two normally distributed random variables X and Y with $\text{cov}(X, Y) = 0$, are independent. Let $X \stackrel{\mathcal{D}}{=} \mathcal{N}(0, 1)$, and

$$Y^c = \begin{cases} X, & |X| \leq c \\ -X, & |X| > c \end{cases}, \quad c \geq 0,$$

then $Y \stackrel{\mathcal{D}}{=} \mathcal{N}(0, 1)$. The covariance is given by

$$\text{cov}(X, Y) = \mathbb{E}XY^c = \mathbb{E}X^2 \mathbb{1}_{\{|X| \leq c\}} - \mathbb{E}X^2 \mathbb{1}_{\{|X| > c\}} = \begin{cases} -1, & c = 0 \\ 1, & c \rightarrow \infty \end{cases}.$$

By continuity, there exists a c such that $\mathbb{E}XY^c = 0$, but X and Y^c are not independent, since $\underbrace{\mathbb{P}(X > c, Y^c > c)}_{=0} \neq \mathbb{P}(X > c)\mathbb{P}(Y^c > c)$.

SECTION 1.1

Finite-Dimensional Distributions

DEFINITION 1.14 (Finite-Dimensional Distributions).

Let

$$X = (X_t)_{t \in T} : (\Omega, \mathcal{F}, P) \rightarrow (E^T, \mathcal{E}^T)$$

be an $(E, \mathcal{E})$ -valued stochastic process. The distributions of the finite-dimensional vectors of the form

$$(X_{t_1}, X_{t_2}, \dots, X_{t_n}), \quad t_1 < t_2 < \dots < t_n,$$

are called the *finite-dimensional distributions (fdd's)* of the process.

It is easily verified that the *fdd's* of a stochastic process form a consistent system of measures in the sense of the following definition.

DEFINITION 1.15 (Consistency).

Let $T \subset \mathbb{R}$ and let $(E, \mathcal{E})$ be a measurable space. For all $n \in \mathbb{Z}^+$ and all $t_1 < \dots < t_n$, $t_i \in T$, $i = 1, \dots, n$, let $\mu_{t_1, \dots, t_n}$ be a probability measure on $(E^n, \mathcal{E}^n)$. This collection of measures is called *consistent* if it has the property that

$$\begin{aligned} \mu_{t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_n}(A_1 \times \dots \times A_{i-1} \times E \times A_{i+1} \times \dots \times A_n) \\ = \mu_{t_1, \dots, t_n}(A_1 \times \dots \times A_{i-1} \times A_i \times A_{i+1} \times \dots \times A_n) \end{aligned}$$

for all $A_1, \dots, A_{i-1}, A_{i+1}, \dots, A_n \in \mathcal{E}$.

Some assumptions are needed on the state space $(E, \mathcal{E})$. We will assume that E is a *Polish space*. This is a topological space, on which we can define a metric that consistent with the topology, and which makes the space complete and separable. As $\mathcal{E}$ we take the Borel- σ -algebra, i.e., the σ -algebra generated by the open sets. Clearly, the Euclidean spaces $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ in this framework.

THEOREM 1.16 (Kolmogorov's Consistency Theorem).

Suppose that E is a Polish space and $\mathcal{E}$ its Borel- σ -algebra. Let $T \subset \mathbb{R}$ and for all $n \in \mathbb{Z}^+$, $t_1 < \dots < t_n \in T$, let $\mu_{t_1, \dots, t_n}$ be a probability measure on $(E^n, \mathcal{E}^n)$. If the measures $\mu_{t_1, \dots, t_n}$ form a consistent system, then there exists a probability measure $\mathbb{P}$ on E^T , such that the canonical (or coordinate) process $(X_t)_t$ on $(\Omega = E^T, \mathcal{F} = \mathcal{E}^T, \mathbb{P})$, defined by $X(\omega) = \omega$, $X_t(\omega) = \omega_t$, has fdd's $\mu_{t_1, \dots, t_n}$.

Proof.

□

COROLLARY 1.17.

There exists a probability measure P on the space

$$(\Omega = \mathbb{R}^{[0, \infty)}, \mathcal{F} = \mathcal{B}(\mathbb{R})^{[0, \infty)}),$$

such that the coordinate process $W = (W_t)_{t \geq 0}$ on $(\Omega, \mathcal{F}, P)$ has properties (i), (ii) and (iii) of Definition 1.10.

Hint. The proof could contain the following ingredients:

- (i) Show that for $0 \leq t_0 \leq t_1 \leq \dots \leq t_n$, there exist multivariate normal distributions with covariance matrices

$$\Sigma = \begin{pmatrix} t_0 & 0 & \dots & \dots & 0 \\ 0 & t_1 - t_0 & 0 & \dots & 0 \\ 0 & 0 & t_2 - t_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & \dots & t_n - t_{n-1} \end{pmatrix},$$

and

$$\Sigma_{t_0, \dots, t_n} = \begin{pmatrix} t_0 & t_0 & \dots & \dots & t_0 \\ t_0 & t_1 & t_1 & \dots & t_1 \\ t_0 & t_1 & t_2 & \dots & t_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_0 & t_1 & t_2 & \dots & t_n \end{pmatrix}.$$

- (ii) Show that a stochastic process W has properties (i), (ii) and (iii) if and only if for all $n \in \mathbb{Z}$, $0 \leq t_0 < \dots < t_n$ the vector

$$(W_{t_0}, W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}) \stackrel{\mathcal{D}}{=} \mathcal{N}(0, \Sigma).$$

- (iii) Show that for a stochastic process W the conditions below are equivalent:

- for all $n \in \mathbb{Z}$, $0 \leq t_0 < \dots < t_n$ the vector

$$(W_{t_0}, W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}) \stackrel{\mathcal{D}}{=} \mathcal{N}(0, \Sigma);$$

- for all $n \in \mathbb{Z}$, $0 \leq t_0 < \dots < t_n$ the vector

$$(W_{t_0}, W_{t_1}, \dots, W_{t_n}) \stackrel{\mathcal{D}}{=} \mathcal{N}(0, \Sigma_{t_0, \dots, t_n}).$$

Proof. For any $0 \leq t_0 \leq t_1 \leq \dots \leq t_n$ and for all $n \in \mathbb{Z}$, we define the increments of the stochastic process W as $W_{t_0}, W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$. Consider the covariance matrix Σ for the increments:

$$\Sigma = \begin{pmatrix} t_0 & 0 & \dots & \dots & 0 \\ 0 & t_1 - t_0 & 0 & \dots & 0 \\ 0 & 0 & t_2 - t_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & \dots & t_n - t_{n-1} \end{pmatrix},$$

where the off-diagonal elements are 0 and the i -th diagonal element is the variance of $W_{t_i} - W_{t_{i-1}}$ which is $t_i - t_{i-1}$, showing that $W_{t_i} - W_{t_{i-1}}$ is normally distributed with variance $t_i - t_{i-1}$, therefore $W_t - W_s \stackrel{d}{=} \mathcal{N}(0, t - s)$. To achieve such covariance matrix Σ , we require the increments to be stationary, independent, and the process starts at zero *a.s.*, i.e., $W_0 = W_{t_0} = 0$.

Then, consider the covariance matrix $\Sigma_{t_0, \dots, t_n}$ of the vector $(W_{t_0}, W_{t_1}, \dots, W_{t_n})$:

$$\Sigma_{t_0, \dots, t_n} = \begin{pmatrix} t_0 & t_0 & \dots & \dots & t_0 \\ t_0 & t_1 & t_1 & \dots & t_1 \\ t_0 & t_1 & t_2 & \dots & t_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_0 & t_1 & t_2 & \dots & t_n \end{pmatrix},$$

where the i, j -th element is $\min(t_{i-1}, t_{j-1})$, indicating that the increments of the process are independent: the shared history of the process up to the minimum of two times contributes to their covariance.

By now, we have shown that

$$(W_{t_0}, W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}) \stackrel{d}{=} \mathcal{N}(0, \Sigma)$$

implies the properties (i,ii,iii). Conversely, if W has properties (i,ii,iii), the increments $W_{t_i} - W_{t_{i-1}}$ are independent and normally distributed with mean 0 and variance $t_i - t_{i-1}$, so their joint distribution is a multivariate normal distribution with mean vector 0 and covariance matrix Σ . Therefore, a stochastic process W has properties (i), (ii) and (iii) if and only if for all $n \in \mathbb{Z}$, $0 \leq t_0 < \dots < t_n$ the vector

$$(W_{t_0}, W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}) \stackrel{d}{=} \mathcal{N}(0, \Sigma).$$

We have to show that: for a stochastic process W the (a) and (b) below are equivalent:

- (a) for all $n \in \mathbb{Z}$, $0 \leq t_0 < \dots < t_n$ the vector $(W_{t_0}, W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}) \stackrel{d}{=} \mathcal{N}(0, \Sigma)$;
- (b) for all $n \in \mathbb{Z}$, $0 \leq t_0 < \dots < t_n$ the vector $(W_{t_0}, W_{t_1}, \dots, W_{t_n}) \stackrel{d}{=} \mathcal{N}(0, \Sigma_{t_0, \dots, t_n})$.

“(a) $\Rightarrow$ (b)”: We can construct the vector $(W_{t_0}, W_{t_1}, \dots, W_{t_n})$ from the independent increments $(W_{t_0}, W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}})$. The variance of W_i is t_i , and for $i \leq j$, $W_{t_j} = W_{t_i} + (W_{t_j} - W_{t_i})$, where $W_{t_j} - W_{t_i}$ is independent of W_{t_i} . By the properties of $\mathbf{BM}$, the sum of independent normal random variables is also normally distributed, implying that the vector $(W_{t_0}, W_{t_1}, \dots, W_{t_n})$ is multivariate normal. Given that the increments are independent and stationary, the covariance between W_{t_i} and W_{t_j} is determined by the overlap in their paths, therefore $\Sigma_{t_0, \dots, t_n}$ where the i, j -th element is $\min(t_{i-1}, t_{j-1})$ corresponds to the shared path length.

“(b) $\Rightarrow$ (a)”: Since the vector $(W_{t_0}, W_{t_1}, \dots, W_{t_n})$ is multivariate normal, these differences also follow a multivariate normal distribution due to the closure properties of the normal distribution under linear transformations. The covariance between $W_{t_i} - W_{t_{i-1}}$ and $W_{t_j} - W_{t_{j-1}}$ for $i \neq j$ is 0 because of the independent increments property of $\mathbf{BM}$, which changes $\Sigma_{t_0, \dots, t_n}$ into the diagonal matrix Σ .

Finally, we have verified the conditions of *Kolmogorov's consistency theorem*, then there exists such a probability measure P on the space $(\Omega = \mathbb{R}^{[0, \infty)}, \mathcal{F} = \mathcal{B}(\mathbb{R})^{[0, \infty)})$, under which the coordinate process $W = (W_t)_{t \geq 0}$ on $(\Omega, \mathcal{F}, P)$ satisfies properties (i,ii,iii) in [Definition 1.10](#). $\square$

In order to construct $\mathbf{w}_M$, we compute the finite-dimensional distributions (fdd's):

$$\mathbb{P}((W_{t_1}, W_{t_2}, \dots, W_{t_n}) \in A), \quad t_1 < t_2 < \dots < t_n,$$

where $A \in (\mathcal{B}(\mathbb{R}))^n$, $n \geq 1$. By *Kolmogorov's consistency theorem* for $T = [0, \infty)$:

$$W : (\Omega = \mathbb{R}^T, \mathcal{F} = (\mathcal{B}(\mathbb{R}))^T, \mathbb{P}) \rightarrow (\mathbb{R}^T, (\mathcal{B}(\mathbb{R}))^T).$$

Check the properties (i), (ii) and (iii) of [Definition 1.10](#), and use the *Kolmogorov's continuity criterion* to make suitable adaptations.

SECTION 1.2

| Kolmogorov's Continuity Criterion

DEFINITION 1.18 (Versions).

Let X and Y be two stochastic processes, indexed by the same set T and with the same state space $(E, \mathcal{E})$, defined on probability spaces $(\Omega, \mathcal{F}, \mathbb{P})$ and $(\Omega', \mathcal{F}', \mathbb{P}')$, respectively. The processes are called *versions* of each other, if they have the same *fdd's*. In other words, if for all $n \in \mathbb{Z}^+$, $t_1, \dots, t_n \in T$ and $B_1, \dots, B_n \in \mathcal{E}$:

$$\mathbb{P}(X_{t_1} \in B_1, X_{t_2} \in B_2, \dots, X_{t_n} \in B_n) = \mathbb{P}'(Y_{t_1} \in B_1, Y_{t_2} \in B_2, \dots, Y_{t_n} \in B_n).$$

DEFINITION 1.19.

Let X and Y be two stochastic processes, indexed by the same set T and with the same state space $(E, \mathcal{E})$, defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- (i) The processes are called *modifications* of each other, if for every $t \in T$: $X_t = Y_t$ a.s.
- (ii) The processes are called *indistinguishable*, if there exists a set $\Omega^* \in \mathcal{F}$, with $\mathbb{P}(\Omega^*) = 1$, such that for every $\omega \in \Omega^*$ the paths $t \mapsto X_t(\omega)$ and $t \mapsto Y_t(\omega)$ are equal.

Remark. If processes are indistinguishable, then they are modifications of each other. If they are modifications of each other, then they certainly are versions of each other. The reverse is not true in general.

EXAMPLE 1.20. (i) *Versions of each other but not modifications of each other.* Consider the space

$$(\Omega = \{\omega\}, \mathcal{F} = \{\emptyset, \{\omega\}\}, \mathbb{P}(\{\omega\}) = 1).$$

Let $T = [0, \infty)$ and let $(E = \{0, 1\}, \mathcal{E} = \{\emptyset, E\})$ be a measurable space. Let $X_t(\omega) = 0$ for all t and let $Y_t(\omega) = 1$. Then

$$\underbrace{\mathbb{P}(X_{t_1} \in A_1, X_{t_2} \in A_2, \dots, X_{t_n} \in A_n)}_{=\mathbb{P}(Y_{t_1} \in A_1, \dots, Y_{t_n} \in A_n)} = \begin{cases} 1, & A_i = \mathcal{E} \text{ for all } i \\ 0, & \text{otherwise} \end{cases}.$$

(ii) *Modifications of each other but not indistinguishable.* Consider the space

$$(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], \mathbb{P} = \lambda)$$

. Let $T = [0, \infty)$ and let

$$(E = \{0, 1\}, \mathcal{E} = 2^E = \{\mathcal{E}, \emptyset, \{0\}, \{1\}\})$$

be a measurable space. Let $X_t(\omega) = 0$ for all t, ω and let

$$Y_t(\omega) = \begin{cases} 0, & t - \lfloor t \rfloor \neq \omega \\ 1, & t - \lfloor t \rfloor = \omega \end{cases}.$$

Then

$$\mathbb{P}(X_t \neq Y_t) = \mathbb{P}(\omega : X_t(\omega) \neq Y_t(\omega)) = \mathbb{P}(\omega = t - \lfloor t \rfloor) = \lambda(\{t - \lfloor t \rfloor\}) = 0.$$

But the set $\{\omega : X_t(\omega) \neq Y_t(\omega)\} = \emptyset$.

THEOREM 1.21 (Kolmogorov's Continuity Criterion).

Let $X = (X_t)_{t \in [0, T]}$ be an $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ -valued stochastic process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that there exist constants $\alpha, \beta, K > 0$ such that

$$\mathbb{E} \|X_t - X_s\|^\alpha \leq K |t - s|^{1+\beta},$$

for all $s, t \in [0, T]$. Then there exists a (everywhere) continuous modification $\hat{X}$ of X , in other words, $\hat{X}(\omega)$ is a continuous function on $[0, T]$ for each $\omega \in \Omega$. Thus the map

$$\hat{X} : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \left(C^d[0, T], C^d[0, T] \cap \mathcal{B}(\mathbb{R}^d)^{[0, T]} \right)$$

is an $\mathcal{F} / C^d[0, T] \cap \mathcal{B}(\mathbb{R}^d)^{[0, T]}$ -measurable map.

Proof. The proof consists of the following steps.

- (i) X_t is continuous in probability on $[0, T]$.

LEMMA 1.22.

$X_n \xrightarrow{\mathbb{P}} X$ if and only if every subsequence $\{X_{n_k}\}_{k=1,2,\dots}$ of $\{X_n\}_n$ contains a further subsequence $\{X_{n'_k}\}_k$ such that $X_{n'_k} \rightarrow X$, a.s.

- (ii) X_t is a.s. uniformly continuous on a countable, dense subset $D \subset [0, T]$.
- (iii) Extend X to a continuous process Y on all of $[0, T]$.
- (iv) Show that Y is a well-defined stochastic process, and a continuous modification of X .

□

SECTION 1.3

| Gaussian Processes

DEFINITION 1.23 (Gaussian Processes).

A $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ -valued stochastic process $X = \{X_t\}_{t \in T}$ is called *Gaussian* if all its *fdd's* are Gaussian, in other words, if they are multivariate normal distributions.

Let X be a Gaussian process indexed by the set T . Then

$$m(t) = \mathbb{E}X_t, \quad t \in T$$

is the *mean function* of the process. The function

$$r(s, t) = \text{cov}(X_s, X_t) = \mathbb{E}[X_s X_t] - \mathbb{E}X_s \mathbb{E}X_t, \quad (s, t) \in T \times T$$

is the *covariance function*.

LEMMA 1.24.

Two Gaussian processes with the same mean and covariance functions are versions of each other.

Proof. Use the fact that the *fdd's* of multivariate normal distributions are completely determined by the mean function and the covariance function. Notice that X has a density whenever $\det \Sigma \neq 0$, given by

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp \left\{ -\frac{1}{2} (x - a)^T \Sigma^{-1} (x - a) \right\},$$

where $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$. □

LEMMA 1.25.

A continuous or *a.s. continuous* Gaussian process $X = (X_t)_{t \geq 0}$ is a BM process if and only if it has the same mean function $m(t) = \mathbb{E}X_t = 0$ and covariance function $r(s, t) = \mathbb{E}[X_s X_t] = s \wedge t$.

THEOREM 1.26.

Let W be a BM process on $(\Omega, \mathcal{F}, \mathbb{P})$. Then the following are BM processes as well:

- (i) *Time-homogeneity:* For every $s \geq 0$ the shifted process $W(s) = (W_{t+s} - W_s)_{t \geq 0}$.
- (ii) *Symmetry:* The process $-W = (-W_t)_{t \geq 0}$.
- (iii) *Scaling:* For every $a > 0$, the process W^a defined by $W_t^a = a^{-\frac{1}{2}} W_{at}$.
- (iv) *Time inversion:* The process $X = (X_t)_{t \geq 0}$ with $X_0 = 0$ and $X_t = tW_{\frac{1}{t}}$, $t > 0$.

Proof. We have to check that: the defined processes are Gaussian, (almost all) sample paths are continuous, and they have the same mean and covariance functions as BM.

- (i) Since W is a BM, it is a Gaussian process. The difference of two Gaussian random variables is also Gaussian, and thus $W(s)_t$ is Gaussian for every $t \geq 0$. The finite-dimensional distributions of $W(s)$ are determined by the linear combinations of $W(s)_t$, which remain Gaussian due to the closure properties of Gaussian distributions under linear transformations.
Since W is a BM, its sample paths are *a.s. continuous*. Shifting and subtracting W_s from W_{t+s} does not change the continuity of the paths. Therefore, the sample paths of $W(s)$ are also *a.s. continuous*.

By the linearity of expectation, we have

$$\mathbb{E}[W(s)_t] = \mathbb{E}[W_{t+s} - W_s] = \mathbb{E}[W_{t+s}] - \mathbb{E}[W_s] = 0$$

since W is a BM implies $\mathbb{E}[W_{t+s}] = \mathbb{E}[W_s] = 0$.

Consider the covariance function, we have the mean function

$$\begin{aligned} \mathbb{E}[W(s)_t W(s)_u] &= \mathbb{E}[(W_{t+s} - W_s)(W_{u+s} - W_s)] \\ &= \mathbb{E}[W_{t+s} W_{u+s}] - \mathbb{E}[W_{t+s} W_s] - \mathbb{E}[W_{u+s} W_s] + \mathbb{E}[W_s^2] \\ &= (t+s) \wedge (u+s) - s - s + s = \min(t, u) \end{aligned}$$

by the independence of the increments. Therefore, the shifted process $W(s) = (W_{t+s} - W_s)_{t \geq 0}$ satisfies all the conditions of [Lemma 1.25](#), proving that it is indeed a BM.

- (ii) Since W is a BM, similarly, the negation of a Gaussian random variable is also Gaussian, then $-W$ is a Gaussian process. And the continuity of $-W$ follows directly from the continuity of W . Calculate the mean function

$$\mathbb{E}[-W_t] = -\mathbb{E}[W_t] = 0,$$

since the mean function of W is zero. Consider the covariance function:

$$\mathbb{E}[(-W_s)(-W_t)] = \mathbb{E}[W_s W_t] = \min(s, t),$$

where the covariance function for the original BM is

$$r(s, t) = \mathbb{E}[W_s W_t] = s \wedge t = \min(s, t).$$

Therefore, $-W = (-W_t)_{t \geq 0}$ satisfies all the conditions of [Lemma 1.25](#), proving that it is a BM.

- (iii) Since W is a Gaussian process, any linear transformation of W including scaling, results in a process that is also Gaussian. Therefore, W^a is a Gaussian process. And the continuity of W^a follows directly from the continuity of W . Calculate the mean function

$$\mathbb{E}[W_t^a] = \mathbb{E}[a^{-\frac{1}{2}} W_{at}] = a^{-\frac{1}{2}} \mathbb{E}[W_{at}] = 0,$$

since $\mathbb{E}[W_{at}] = 0$ for BM W . Consider the covariance function:

$$\mathbb{E}[W_s^a W_t^a] = \mathbb{E}[a^{-\frac{1}{2}} W_{as} a^{-\frac{1}{2}} W_{at}] = a^{-1} \mathbb{E}[W_{as} W_{at}] = a^{-1} \min(as, at) = \min(s, t).$$

Therefore, $W_t^a = a^{-\frac{1}{2}} W_{at}$ satisfies all the conditions of [Lemma 1.25](#), proving that it is a BM.

- (iv) Let us show that almost all sample paths of X are continuous. By time inversion, it is immediate that $(X_t)_{t > 0}$ (a.s.) has continuous sample paths if W has. We only need show a.s. continuity at $t = 0$, that is, we need to show that $\lim_{t \downarrow 0} X_t = 0$, a.s. Let

$$\Omega^* = \{\omega \in \Omega \mid (W_t(\omega))_{t \geq 0} \text{ continuous}, W_0(\omega) = 0\}.$$

By assumption $\Omega \setminus \Omega^*$ is contained in a $\mathbb{P}$ -null set. Further, $(X_t)_{t > 0}$ has continuous paths on Ω^* . Then $\lim_{t \downarrow 0} X_t(\omega) = 0$ if and only if for all $\epsilon > 0$ there exists $\delta_\omega > 0$ such that $|X_t(\omega)| < \epsilon$ for all $t \leq \delta_\omega$. This is true if and only if for all integers $m \geq 1$, there exists an integer n_ω , such that $|X_q(\omega)| < \frac{1}{m}$ for all $q \in \mathbb{Q}$ with $q < \frac{1}{n_\omega}$, because of continuity of $X_t(\omega)$, $t > 0$. Check that this implies

$$\left\{ \omega : \lim_{t \downarrow 0} X_t(\omega) = 0 \right\} \cap \Omega^* = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{q \in (0, \frac{1}{n}] \cap \mathbb{Q}} \left\{ \omega : |X_q(\omega)| < \frac{1}{m} \right\} \cap \Omega^*.$$

The fdd's of X and W are equal. Hence

$$\mathbb{P} \left(\bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{q \in (0, \frac{1}{n}] \cap \mathbb{Q}} \left\{ \omega : |X_q(\omega)| < \frac{1}{m} \right\} \right) = \mathbb{P} \left(\bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{q \in (0, \frac{1}{n}] \cap \mathbb{Q}} \left\{ \omega : |W_q(\omega)| < \frac{1}{m} \right\} \right) = 1.$$

As a consequence $\mathbb{P}(\omega : \lim_{t \downarrow 0} X_t(\omega) = 0) = 1$.

□

COROLLARY 1.27.

Let W be a BM with the property that all paths are continuous. Then

$$\mathbb{P}\left\{\sup_{t \geq 0} W_t = \infty, \inf_{t \geq 0} W_t = -\infty\right\} = 1.$$

Proof. It is sufficient to show that

$$\mathbb{P}\left\{\sup_{t \geq 0} W_t = \infty\right\} = 1. \quad (1.2)$$

The symmetry property implies

$$\sup_{t \geq 0} W_t \stackrel{\mathcal{D}}{=} \sup_{t \geq 0} (-W_t) = -\inf_{t \geq 0} W_t.$$

Hence (1.2) implies that $\mathbb{P}\{\inf_{t \geq 0} W_t = -\infty\} = 1$. As a consequence, the probability of the intersection equals 1.

First of all, notice that $\sup_t W_t$ is well-defined. We need to show that $\sup_t W_t$ is a measurable function. This is true if $\{\sup_t W_t \leq x\}$ is measurable for all $x \in \mathbb{R}$ ($\mathbb{Q}$ is sufficient of course). This follows from

$$\left\{\sup_t W_t \leq x\right\} = \bigcap_{q \in \mathbb{Q}} \{W_q \leq x\}.$$

Here we use that all paths are continuous. We cannot make any assertions on measurability of $\{W_q \leq x\}$ restricted to the set of discontinuous paths, unless $\mathcal{F}$ is $\mathbb{P}$ -complete.

By the scaling property we have for all $a > 0$:

$$\sup_t W_t^a \stackrel{\mathcal{D}}{=} \sup_t \frac{1}{\sqrt{a}} W_{at} = \frac{1}{\sqrt{a}} \sup_t W_t.$$

It follows for $n \in \mathbb{Z}^+$ that

$$\mathbb{P}\left\{\sup_t W_t \leq n\right\} = \mathbb{P}\left\{n^2 \sup_t W_t \leq n\right\} = \mathbb{P}\left\{\sup_t W_t \leq \frac{1}{n}\right\}.$$

By letting n tend to infinity, we see that

$$\mathbb{P}\left\{\sup_t W_t < \infty\right\} = \mathbb{P}\left\{\sup_t W_t \leq 0\right\}.$$

Thus, for (1.2) it is sufficient to show that $\mathbb{P}\{\sup_t W_t \leq 0\} = 0$. We have

$$\begin{aligned} \mathbb{P}\left\{\sup_t W_t \leq 0\right\} &\leq \mathbb{P}\left\{W_1 \leq 0, \sup_{t \geq 1} W_t \leq 0\right\} \\ &\leq \mathbb{P}\left\{W_1 \leq 0, \sup_{t \geq 1} W_t - W_1 < \infty\right\} \\ &= \mathbb{P}\{W_1 \leq 0\} \mathbb{P}\left\{\sup_{t \geq 1} W_t - W_1 < \infty\right\}, \end{aligned}$$

by the independence of BM increments. By the time-homogeneity of BM, the latter probability equals the probability that the supremum of BM is finite. We have just shown that this equals $\mathbb{P}\{\sup_t W_t \leq 0\}$. And so we find

$$\mathbb{P}\left\{\sup_t W_t \leq 0\right\} \leq \frac{1}{2} \mathbb{P}\left\{\sup_t W_t \leq 0\right\}.$$

This shows that $\mathbb{P}\{\sup_t W_t \leq 0\} = 0$ and so we have shown (1.2). □

SECTION 1.4

| Filtrations and Stopping Times

The probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is fixed again and we suppose that T is a subinterval of $\mathbb{Z}^+$ or $\mathbb{R}^+$.

DEFINITION 1.28 (Filtrations).

A collection $(\mathcal{F}_t)_{t \in T}$ of sub- σ -algebras is called a *filtration* if $\mathcal{F}_s \subset \mathcal{F}_t$ for all $s \leq t$. A stochastic process $X = (X_t)_{t \in T}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$ and indexed by T is called *adapted* to the filtration if for every $t \in T$, the random variable X_t is $\mathcal{F}_t$ -measurable. Then $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in T}, \mathbb{P})$ is a *filtered probability space*.

Then we can consider the filtration

$$(\mathcal{F}_t^X)_{t \in T} = \sigma(X_s, s \leq t).$$

We call this the *filtration generated by X* , or the *natural filtration* of X . It is the *smallest* filtration, to which X is adapted. Intuitively, the natural filtration of a process keeps track of the history of the process. A stochastic process is always adapted to its natural filtration.

If $(\mathcal{F}_t)_{t \in T}$ is a filtration, then for $t \in T$ we may define the σ -algebra

$$\mathcal{F}_{t+} = \bigcap_{n=1}^{\infty} \mathcal{F}_{t+\frac{1}{n}}.$$

This is the σ -algebra $\mathcal{F}_t$, augmented with the events that *happen immediately after time t* . The collection $(\mathcal{F}_{t+})_{t \in T}$ is again a filtration.

DEFINITION 1.29 (Right-Continuous).

We call a filtration $(\mathcal{F}_t)_{t \in T}$ *right-continuous* if $\mathcal{F}_t = \mathcal{F}_{t+}$ for all $t \in T$.

DEFINITION 1.30 (Usual Conditions).

A filtration $(\mathcal{F}_t)_{t \in T}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is said to satisfy the *usual conditions* if it is right-continuous and $\mathcal{F}_0$ contains all $\mathbb{P}$ -negligible events of $\mathcal{F}_{\infty}$.

We now introduce a very important class of *random times* that can be associated with a filtration.

DEFINITION 1.31 (Stopping Times).

An $[0, \infty]$ -valued random variable τ is called a $(\mathcal{F}_t)_t$ -*stopping time*, if for every $t \in T$ it holds that the event $\{\tau \leq t\}$ is $\mathcal{F}_t$ -measurable. We may drop the dependence on the filtration in the notation, whenever filtration at play is clear.

If $\tau < \infty$, we call τ a *finite* stopping time, and if $\mathbb{P}(\tau < \infty) = 1$, we call τ *a.s. finite*. Similarly, if there exists a constant K such that $\tau(\omega) \leq K$, then τ is said to be *bounded*, and if $\mathbb{P}(\tau \leq K) = 1$, τ is *a.s. bounded*.

With a stopping time τ we can associate the σ -algebra σ^τ generated by τ . However, this σ -algebra only contains the information about *when* τ occurred. If τ is associated with an adapted process X , then σ^τ contains no further information on the history of the process up

to the stopping time. For this reason we associate with τ the (generally) larger σ -algebra $\mathcal{F}_\tau$ defined by

$$\mathcal{F}_\tau = \{A \in \mathcal{F}_\infty : A \cap \{\tau \leq t\} \in \mathcal{F}_t, \forall t \in T\}.$$

LEMMA 1.32.

Let $\Omega \subset E^T$ have the property that for each $t \in T$ and $\omega \in \Omega$ there exists $\bar{\omega} \in \Omega$ with $\bar{\omega}_s = \omega_{s \wedge t}$ for all $s \in T$. Let $\mathcal{F} = \mathcal{E}^T \cap \Omega$. Let X be the canonical process on $(\Omega, \mathcal{F})$. Define $\mathcal{F}_\infty^X = \sigma(X_t, t \in T)$. Let $A \subset \Omega$. Then $A \in \mathcal{F}_t^X$ if and only if

- (i) $A \in \mathcal{F}_\infty^X$;
- (ii) $\omega \in A$ and $X_s(\omega) = X_s(\omega')$ for all $s \in T$ with $s \leq t$ imply that $\omega' \in A$.

DEFINITION 1.33 (Optional Times).

A $[0, \infty]$ -valued random variable τ is called an *optional time* with respect to the filtration $\mathcal{F}_t$ if for every $t \in T$ it holds that $\{\tau < t\} \in \mathcal{F}_t$. If $\tau < \infty$ almost surely, we call the optional time *finite*.

EXAMPLE 1.34.

LEMMA 1.35.

τ is an $\mathcal{F}_t$ -optional time if and only if it is a $\mathcal{F}_{t+}$ -stopping time with respect to $\mathcal{F}_t$. Every $\mathcal{F}_t$ -stopping time is a $\mathcal{F}_t$ -optional time.

Proof.

□

The so-called *hitting* and *first entrance times* form an important class of stopping times and optional times. They are related to the first time that the process visits a set B .

DEFINITION 1.36.

Let $X = (X_t)_{t \geq 0}$ be an $(E, \mathcal{E})$ -valued stochastic process defined on the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $B \in \mathcal{E}$. The *first entrance time* of B is defined by

$$\sigma_B = \inf \{t \geq 0 \mid X_t \in B\}.$$

The *first hitting time* of B is defined by

$$\tau_B = \inf \{t > 0 \mid X_t \in B\}.$$

LEMMA 1.37.

If $x : [0, \infty) \rightarrow E$ is right-continuous, then x is $\mathcal{B}(\mathbb{R}^+)/\mathcal{E}$ -measurable.

LEMMA 1.38.

Let (E, d) be a metric space and let $\mathcal{B}(E)$ be the Borel- σ -algebra of open sets compatible with the metric d . Suppose that $X = (X_t)_{t \geq 0}$ is a continuous, $(E, \mathcal{B}(E))$ -valued stochastic process and that B is closed in E . Then σ_B is an $\mathcal{F}_t^X$ -stopping time.¹

Proof. Denote the distance of a point $x \in E$ to the set B by

$$d(x, B) = \inf \{d(x, y) \mid y \in B\}.$$

¹ As is usual, we define $\inf \emptyset = \infty$.

First note that $x \rightarrow d(x, B)$ is a continuous function. Hence it is $\mathcal{B}(E)$ -measurable. It follows that $Y_t = d(X_t, B)$ is $\mathcal{F}_t^X$ -measurable as a composition of measurable maps. Since X_t is continuous, the real-valued process $(Y_t)_t$ is continuous as well. Moreover, since B is closed, it holds that $X_t \in B$ if and only if $Y_t = 0$. By continuity of Y_t , it follows that $\sigma_B > t$ if and only if $Y_s > 0$ for all $s \leq t$. This means that

$$\begin{aligned} \{\sigma_B > t\} &= \{Y_s > 0, 0 \leq s \leq t\} \\ &= \bigcup_{n=1}^{\infty} \bigcap_{q \in \mathbb{Q}_t} \left\{ Y_q > \frac{1}{n} \right\} \\ &= \bigcup_{n=1}^{\infty} \bigcap_{q \in \mathbb{Q}_t} \left\{ d(X_q, B) > \frac{1}{n} \right\} \in \mathcal{F}_t^X, \end{aligned}$$

where $\mathbb{Q}_t = \{tq \mid q \in \mathbb{Q} \cap [0, 1]\}$. □

EXAMPLE 1.39. Let W be a BM with continuous paths and, for $x > 0$, consider the random variable

$$\tau_x = \inf \{t > 0 \mid W_t = x\}.$$

Note that

$$\{\tau_x > t\} = \{W_0 \neq x\} \cap \{\sigma_x > t\}, \quad t > 0.$$

Since W is continuous, $\{\sigma_x > t\} \in \mathcal{F}_t^W$, for $t \geq 0$, by Lemma 1.38. Additionally, $\{W_0 \neq x\} \in \mathcal{F}_0^W \subset \mathcal{F}_t^W$, hence $\{\tau_x > t\} \in \mathcal{F}_t^W$, for all $t \geq 0$. We conclude that τ_x is an $\mathcal{F}_t^W$ -stopping time.

Furthermore, as $\mathbb{P}\{W_0 = 0\} = 1$, it holds that $\mathbb{P}\{\tau_x = \sigma_x\} = 1$. It is therefore sufficient to show that $\{\sigma_x < \infty\}$ is measurable and $\mathbb{P}\{\sigma_x < \infty\} = 1$.

First,

$$\{\sigma_x < \infty\} = \bigcup_{n=1}^{\infty} \{\sigma_x \leq n\}$$

is a measurable set. Next, consider

$$A = \left\{ \omega : \sup_{t \geq 0} W_t = \infty, \inf_{t \geq 0} W_t = -\infty \right\}.$$

By Corollary 1.27, this set has probability 1.

Let $T > |x|$. For each $\omega \in A$, there exist T_ω, T'_ω , such that $W_{T_\omega} \geq T, W_{T'_\omega} \leq -T$. By continuity of paths, there exists $t_\omega \in (T_\omega \wedge T'_\omega, T_\omega \vee T'_\omega)$, such that $W_{t_\omega} = x$. It follows that $A \subset \{\sigma_x < \infty\}$. Hence $\mathbb{P}\{\sigma_x < \infty\} = 1$. Thus, $\mathbb{P}\{\sigma_x < \infty\} = 1$.

LEMMA 1.40.

Let (E, d) be a metric space and let $\mathcal{B}(E)$ be the Borel- σ -algebra of open sets compatible with the metric d . Suppose that $X = (X_t)_{t \geq 0}$ is a (everywhere) cadlag, $(E, \mathcal{B}(E))$ -valued stochastic process and that B is a closed set in E . Then

$$\gamma_B = \inf \{t > 0 \mid X_t \in B, \text{ or } X_{t-} \in B\},$$

is an $\mathcal{F}_t^X$ -stopping time.

We often would like to consider the stochastic process X evaluated at a finite stopping time τ . However, it is not a priori clear that the map $\omega \rightarrow X_{\tau(\omega)}(\omega)$ is measurable. In other words, that X_τ is a random variable. We need measurability of X in both parameters t and ω . This motivates the following definition.

DEFINITION 1.41 (Progressively Measurable).

An $(E, \mathcal{E})$ -valued stochastic process is called *progressively measurable* with respect to the filtration $\mathcal{F}_t$ if for every $t \in T$ the map $(s, \omega) \rightarrow X_s(\omega)$ is measurable as a map from $([0, t] \times \Omega, \mathcal{B}([0, t]) \times \mathcal{F}_t)$ to $(E, \mathcal{E})$.

LEMMA 1.42.

Suppose that X is a progressively measurable process. Let τ be a stopping time. Then $\mathbb{1}_{\{\tau < \infty\}} X_\tau$ is an $\mathcal{F}_\tau \setminus \mathcal{E}$ -measurable random element.

Proof. We have to show that

$$\{\mathbb{1}_{\{\tau < \infty\}} X_\tau \in B\} \cap \{\tau \leq t\} = \{X_\tau \in B\} \cap \{\tau \leq t\} \in \mathcal{F}_t,$$

for every $B \in \mathcal{E}$ and every $t \geq 0$. Now note that

$$\{X_{\tau \wedge t} \in B\} = \{X_\tau \in B, \tau \leq t\} \cup \{X_t \in B, \tau > t\}.$$

Clearly $\{X_t \in B, \tau > t\} \in \mathcal{F}_t$. If we can show that $\{X_{\tau \wedge t} \in B\} \in \mathcal{F}_t$, it easily follows that $\{X_\tau \in B, \tau \leq t\} \in \mathcal{F}_t$. Hence, it suffices to show that the map $\omega \rightarrow X_{\tau(\omega) \wedge t}(\omega)$ is $\mathcal{F}_t$ -measurable. To this end consider the map

$$\phi : ([0, t] \times \Omega, \mathcal{B}([0, t]) \times \mathcal{F}_t) \rightarrow ([0, t] \times \Omega, \mathcal{B}([0, t]) \times \mathcal{F}_t)$$

given by $\phi(s, \omega) = (\tau(\omega) \wedge s, \omega)$ is measurable. Using that X is progressively measurable, it follows that the composition map

$$(s, \omega) \rightarrow X(\phi(s, \omega)) = X_{\tau(\omega) \wedge s}(\omega)$$

is measurable. By Fubini's theorem the section map $\omega \rightarrow X_{\tau(\omega) \wedge t}(\omega)$ is $\mathcal{F}_t$ -measurable. $\square$

LEMMA 1.43.

Let τ be a stopping time. Then there exist stopping times τ_n that only take either finitely or countably many values and such $\tau_n \downarrow \tau$.

Hint. Define

$$\tau_n = \sum_{k=1}^{n2^n} \frac{k}{2^n} \mathbb{1}_{\{\tau \in (\frac{k-1}{2^n}, \frac{k}{2^n}]\}} + \infty \mathbb{1}_{\{\tau > n\}}, \quad (1.3)$$

to obtain an approximating stopping time taking only finitely many values. Or

$$\tau_n = \sum_{k=1}^{\infty} \frac{k}{2^n} \mathbb{1}_{\{\tau \in (\frac{k-1}{2^n}, \frac{k}{2^n}]\}} + \infty \mathbb{1}_{\{\tau = \infty\}}, \quad (1.4)$$

for an approximating stopping time taking countable many values. Then τ_n is a stopping time and $\tau_n \downarrow \tau$.

Proof. From the definition, to show an $[0, \infty]$ -valued random variable τ_n is a $(\mathcal{F}_t)_t$ -stopping time, we have to show that for every $t \in T$ it holds that the event $\{\tau_n \leq t\}$ is $\mathcal{F}_t$ -measurable. For (1.3), if we let $k \in [1, n2^n]$, the range of the set $(\frac{k-1}{2^n}, \frac{k}{2^n}]$ will be $(0, n]$. Then for any $t \in T$, the event $\{\tau_n \leq t\}$ is equivalent to consider the event $\{\frac{k}{2^n} \leq t\}$ when $\tau \in (\frac{k-1}{2^n}, \frac{k}{2^n}]$. Otherwise τ will be greater than n and hence $\tau_n = \infty$, which would not satisfy $\tau_n \leq t$. Given that τ is a stopping time, the event $\{\tau \in (\frac{k-1}{2^n}, \frac{k}{2^n}]\}$ is $\mathcal{F}_{\frac{k}{2^n}}$ -measurable. Since $\mathcal{F}_t$ is a filtration, any event determined by τ within these intervals for each k is also $\mathcal{F}_t$ -measurable. Consequently, the union of these events $\{\tau_n \leq t\}$ for which $\frac{k}{2^n} \leq t$ is $\mathcal{F}_t$ -measurable and hence τ_n is indeed a stopping time.

To prove that $\tau_n \downarrow \tau$, we first notice that as n increases, the partition of the interval $(0, n]$ into $\left(\frac{k-1}{2^n}, \frac{k}{2^n}\right]$ becomes smaller. Given that τ_n is taking the value $\frac{k}{2^n}$, we have for all $\varepsilon > 0$, there exists N such that for all $n > N$:

$$\tau_n - \tau < \frac{k}{2^n} - \frac{k-1}{2^n} = \frac{1}{2^n} < \varepsilon,$$

proving that τ_n converges to τ . (1.4) extends the summation to ∞ , allowing τ_n to take countably many values, then the range of the set $\left(\frac{k-1}{2^n}, \frac{k}{2^n}\right]$ will be $(0, \infty)$. Again, when $t < \infty$, the event $\left\{\tau \in \left(\frac{k-1}{2^n}, \frac{k}{2^n}\right]\right\}$ is $\mathcal{F}_{\frac{k}{2^n}}$ -measurable for any k , together with the proposition of $\mathcal{F}_t$ being filtration, the union of events $\{\tau_n \leq t\}$ is $\mathcal{F}_t$ -measurable. Therefore τ_n is a stopping time, and the proof of τ_n converging to τ also follows a similar way as before. $\square$

Exercises

EXERCISE 1.1. Suppose that X and Y are modifications of each other with values in a Polish space $(E, \mathcal{E})$, and for both X and Y all sample paths are either left or right continuous. Let T be an interval in $\mathbb{R}$. Show that

$$P\{X_t = Y_t, \forall t \in T\} = 1.$$

SOLUTION. Given that X and Y are modifications of each other, this means

$$X_t = Y_t, \quad a.s. \quad \forall t \in T.$$

Since the sample paths of both X and Y are either left or right continuous, we can assume, without loss of generality, that they are left-continuous for the sake of this proof (the argument would be similar for right continuity by considering limits from the right).

Define the set

$$\begin{aligned} A &= \{\omega : X_t(\omega) = Y_t(\omega), \forall t \in T \cap \mathbb{Q}\} \\ &= \bigcap_{t \in T \cap \mathbb{Q}} \{\omega : X_t(\omega) = Y_t(\omega)\} \\ &= \left(\bigcup_{t \in T \cap \mathbb{Q}} \{\omega : X_t(\omega) \neq Y_t(\omega)\} \right)^c. \end{aligned}$$

Since a countable intersection of *a.s.* events is *a.s.*, we have $P(A) = 1$. Furthermore, we define

$$B = \{\omega : X(w) \text{ left-continuous}\}, \quad C = \{\omega : Y(w) \text{ left-continuous}\},$$

with $P(B) = P(C) = 1$. Therefore, the intersection $A' = A \cap B \cap C$ denoting the collection of those paths which are left-continuous and the two processes are modifications of each other with $P(A') = 1$. Because $T \cap \mathbb{Q}$ is dense in T , there exists a sequence $(q_n) \subset T \cap \mathbb{Q}$ converging to t from the left. Due to the left continuity of X and Y , we have

$$\lim_{n \rightarrow \infty} X_{q_n}(w) = X_t(w), \quad \lim_{n \rightarrow \infty} Y_{q_n}(w) = Y_t(w), \quad \forall \omega \in A'.$$

Since $X_{q_n}(w) = Y_{q_n}(w)$, it follows that $X_t(w) = Y_t(w)$ for all $t \in T$ and $\omega \in A'$. Hence, we conclude that

$$P\{X_t = Y_t, \quad \forall t \in T\} = 1,$$

which completes the proof. $\square$

EXERCISE 1.2. Let $H \in (0, 1)$ be given. A continuous, zero-mean Gaussian process X with covariance function

$$2\mathbb{E}X_s X_t = t^{2H} + s^{2H} - |t - s|^{2H}$$

is called a *fractional Brownian motion* (FBM) with Hurst index H . Show that the FBM with Hurst index $\frac{1}{2}$ is simply the BM. Show that if X is a FBM with Hurst index H , then for all $a > 0$ the process $a^{-H}X_{at}$ is a FBM with Hurst index H as well.

SOLUTION. – When $H = \frac{1}{2}$, the covariance function

$$2\mathbb{E}X_s X_t = t + s - |t - s|.$$

Since X is a zero-mean Gaussian process by assumption, $\mathbb{E}X_t = 0$. Then we verify

$$\mathbb{E}X_s X_t = \frac{1}{2}(t + s - |t - s|) = \begin{cases} s, & t \geq s \\ t, & t \leq s \end{cases} = \min\{s, t\}.$$

Therefore by [Lemma 1.25](#), the FBM with $H = \frac{1}{2}$ is a standard BM.

- Given that X is a FBM with Hurst index H , then for any fixed $t \geq 0$, the random variable X_{at} is zero-mean Gaussian. Multiplying a constant a^{-H} , the process $Y_t := a^{-H}X_{at}$ is also zero-mean Gaussian. Then we verify

$$\begin{aligned} 2\mathbb{E}Y_s Y_t &= 2\mathbb{E}[a^{-H}X_{as} \cdot a^{-H}X_{at}] \\ &= 2a^{-2H}\mathbb{E}[X_{as}X_{at}] \\ &= a^{-2H}\left((at)^{2H} + (as)^{2H} - |at - as|^{2H}\right) = t^{2H} + s^{2H} - |t - s|^{2H}, \end{aligned}$$

which is exactly the covariance function required for a FBM with Hurst index H .

□

EXERCISE 1.3. Consider Brownian motion $W = (W_t)_{t \geq 0}$ with continuous paths. Assume that W is defined on the underlying filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P})$. Let $Z : \Omega \rightarrow \mathcal{B}([0, \infty))$ be given by $Z(\omega) = \{t \geq 0 \mid W_t(\omega) = 0\}$.

Show (by an interchange argument) that

$$\int_{\Omega} \lambda(Z(\omega)) d\mathbb{P}(\omega) = 0,$$

for λ the Lebesgue measure on $([0, \infty), \mathcal{B}([0, \infty)))$. Argue from this that $\lambda(Z(\omega)) = 0$ for almost all $\omega \in \Omega$.

EXERCISE 1.4. We define the last exit time of x :

$$L_x = \sup \{t > 0 : W_t = x\},$$

where $\sup \{\emptyset\} = 0$.

- Show that τ_0 is measurable (τ_0 is defined in [Definition 1.36](#)).
- Show that L_x is measurable for all x . Derive first that

$$\{L_x < t\} = \bigcap_{n > t} \{|W_s - x| > 0, t \leq s \leq n\}.$$

(iii) Show that $L_x = \infty$ a.s. for all x , by considering the set

$$\left\{ \sup_{t \geq 0} W_t = \infty, \inf_{t \geq 0} W_t = -\infty \right\}$$

as in the proof of [Example 1.39](#).

(iv) Show that for almost all $\omega \in \Omega$ there exists a strictly decreasing sequence $\{t_n(\omega)\}_n$, $\lim_n t_n(\omega) = 0$ such that $W(t_n)(\omega) = 0$ for all n .

SOLUTION. (i) Let $X = (X_t)_{t \geq 0}$ an $(E, \mathcal{E})$ -valued stochastic process defined on the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $B \in \mathcal{E}$. Given that τ_0 is the *first hitting time* of the set $\{0\}$, by definition we have

$$\tau_0 := \inf \{t > 0 \mid X_t \in \{0\}\}.$$

To show τ_0 is measurable, we have to show that for all $t \geq 0$, the set $\{\omega \in \Omega \mid \tau_0(\omega) \leq t\}$ is in $\mathcal{F}$. Notice that $\{\tau_0 \leq t\}$ can be expressed as the countable union of sets:

$$\{\tau_0 \leq t\} = \bigcup_{q \in [0, t] \cap \mathbb{Q}} \{X_q \in \{0\}\},$$

where for each $q \in [0, t] \cap \mathbb{Q}$, the event $\{X_q \in \{0\}\}$ is $\mathcal{F}$ -measurable because X_q is a $(E, \mathcal{E})$ -valued random variable and $\{0\} \in \mathcal{E}$. Since rational numbers q are dense in $\mathbb{R}$, and the countable union of measurable sets is measurable, thereby showing that $\{\tau_0 \leq t\}$ is $\mathcal{F}$ -measurable and hence τ_0 is measurable.

(ii) By definition, the expression $\{L_x < t\}$ denotes the last time $W_t = x$ occurs before time t , then there must not be any exit from x after time t . Therefore, for every $n > t$, the process W_t must not equal x for all $s \in [t, n]$, then we can derive the expression

$$\{L_x < t\} = \bigcap_{n > t} \{|W_s - x| > 0, t \leq s \leq n\}. \quad (1.5)$$

To show L_x is measurable for all x , we have to show that for any t , the set $\{L_x < t\}$ is in $\mathcal{F}$. Given that W_s is a random variable for $s \in [t, n]$, the set $\{\omega : |W_s(\omega) - x| > 0\}$ is the pre-image under W_s of the Borel set $\mathbb{R} \setminus \{x\}$ (open set), and thus is in $\mathcal{F}$. Therefore, the condition $|W_s - x| > 0$ defines an $\mathcal{F}$ -measurable set. Furthermore, since the intersection of $\mathcal{F}$ -measurable sets is also $\mathcal{F}$ -measurable, the set defined on the right-hand side of (1.5) is $\mathcal{F}$ -measurable. This implies that $\{L_x < t\}$ is measurable for all t and hence L_x is measurable for all x .

(iii) By [Corollary 1.27](#), given that W is a Brownian motion whose paths are all continuous, the set

$$A = \left\{ \sup_{t \geq 0} W_t = \infty, \inf_{t \geq 0} W_t = -\infty \right\}$$

has probability $P(A) = 1$, which implies that, almost surely, the Brownian motion paths are unbounded in both directions, crossing every real number x an infinite number of times over the interval $[0, \infty)$ by the property of continuity. Then given any real number x and any finite time T , there exist times $T_1, T_2 > T$ such that $W_{T_1} > x$ and $W_{T_2} < x$. By the continuity of the paths, it follows from the *Intermediate Value Theorem* that there must exist a t such that $W_t = x$. Then for any finite candidate t' for the last exit time from x , one can always find a $t'' > t'$ such that $W_{t''} = x$. Thus, the last exit time L_x must be infinite almost surely.

(iv) Given that W is a Brownian motion, the time-inversion property states that the process $X = (X_t)_{t \geq 0}$ with $X_0 = 0$ and $X_t = tW_{1/t}$ for $t > 0$ is also a Brownian motion. By (iii), $L_x = \infty$ almost surely for $W(t)$, indicating that $W(t)$ crosses any x , including 0, infinitely often as t increases without bound, therefore the time-inverted process X_t also crossing 0 infinitely many times. Then there must exist a time $s_n > n$ such that $X_{s_n} = 0$ with $\lim_n s_n \rightarrow \infty$, where s_n forms a strictly increasing sequence. Then we define

$$t_n := \frac{1}{s_n} \in \left[\frac{1}{n+1}, \frac{1}{n} \right], \quad \forall n > 0$$

such that $W_{t_n} = 0$ with $\lim_n t_n = 0$ and hence t_n forms a strictly decreasing sequence converging to 0. Since $X(s_n) = 0$ implies $s_n W(1/s_n) = \frac{1}{t_n} W(t_n) = 0 \Rightarrow W(t_n) = 0$, then for all $\omega \in \Omega$, there must exist such strictly decreasing sequence $\{t_n(\omega)\}_n$ with $\lim_n t_n(\omega) = 0$ such that $W(t_n)(\omega) = 0$ for all n .

□

EXERCISE 1.5. Let W be a BM (with continuous paths) with respect to its natural filtration $(\mathcal{F}_t^W)_t$. Define for $a > 0$

$$S_a = \inf \{t \geq 0 : W_t > a\}.$$

(i) Is S_a an optional time? Justify your answer.

(ii) Let now $\sigma_a = \inf \{t \geq 0 : W_t = a\}$ be the first entrance time of a and let

$$M_a = \sup \{t \geq 0 : W_t = at\},$$

be the last time that W_t equals at . Is M_a a stopping time? Justify your answer. Show that $M_a < \infty$ with probability 1 (you could use time-inversion for BM).

(iii) Show that M_a has the same distribution as $\frac{1}{\sigma_a}$.

EXERCISE 1.6 (Further Properties of the Zero Set of Brownian Motion). Consider Brownian motion $W = (W_t)_{t \geq 0}$ with continuous paths. Assume that W is defined on the underlying filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P})$. Let $Z : \Omega \rightarrow \mathcal{B}([0, \infty))$ be given by $Z(\omega) = \{t \geq 0 : W_t(\omega) = 0\}$.

(i) Show that for every $r \geq 0$, the random variable

$$\tau_r(\omega) = \inf \{t \geq r : W_t(\omega) = 0\}$$

is an *a.s.* finite stopping time.

(ii) Using the strong Markov property, show that the same assertion holds for the first zero of Brownian motion following r .

(iii) Show that for almost all $\omega \in \Omega$, every $t \in Z(\omega)$ is an accumulation point of other points in $Z(\omega)$.

Martingales

SECTION 2.1

| Conditional Expectation

THEOREM 2.1 (Conditional Expectation).

Suppose we have a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a real-valued, integrable random variable, i.e. $\mathbb{E}[|X|] < \infty$. Let $\mathcal{A}$ be a sub- σ -algebra of $\mathcal{F}$. Then there exists a a.s. unique random variable Y such that

- (i) Y is $\mathcal{A}/\mathcal{B}(\mathbb{R})$ -measurable;
- (ii) Y is integrable, i.e. $\mathbb{E}[|Y|] < \infty$;
- (iii) for each $A \in \mathcal{A}$ we have

$$\int_A Y d\mathbb{P} = \int_A X d\mathbb{P}.$$

We can also write

$$\mathbb{E}[\mathbf{1}_A Y] = \int \mathbf{1}_A Y d\mathbb{P} = \int \mathbf{1}_A X d\mathbb{P}.$$

Remark. Suppose we have constructed an $\mathcal{A}$ -measurable random variable Z , with $\mathbb{E}|Z| < \infty$, such that (iii) holds for all $A \in \mathcal{A}$, i.e., (iii) holds on a π -system $\mathcal{J}$ generating $\mathcal{A}$, containing the whole space ($\Omega \in \mathcal{J}$, $\sigma(\mathcal{J}) = \mathcal{A}$). Then (iii) holds for all $A \in \mathcal{A}$, and so Z is a version of the conditional expectation $\mathbb{E}[X | \mathcal{A}]$.

LEMMA 2.2.

Suppose X, Y are random variables defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then $X \geq Y$, a.s. if and only if

$$\int_F X d\mathbb{P} = \mathbb{E}[\mathbf{1}_F X] \geq \mathbb{E}[\mathbf{1}_F Y] = \int_F Y d\mathbb{P}$$

for all $F \in \mathcal{F}$.

Proof. “ $\Leftarrow$ ” Let $F = \{X \leq Y\} \in \mathcal{F}$ and suppose $\mathbb{P}(F) > 0$. Then

$$F = \bigcup_n \left\{ X \leq Y - \frac{1}{n} \right\}.$$

□

EXAMPLE 2.3. Let $\mathcal{A} = \{\emptyset, A, A^c, \Omega\}$, choose random variable $Y : (\Omega, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with $Y = \{c_1 \mathbb{1}_A + c_2 \mathbb{1}_{A^c} \mid (c_1 \neq c_2)\}$. Then

$$\mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X \mid \sigma(Y)] = \mathbb{E}[X \mid Y] = \alpha_1 \mathbb{1}_A + \alpha_2 \mathbb{1}_{A^c}.$$

Compute α_1, α_2 :

$$\mathbb{E}[\mathbb{1}_A]X = \mathbb{E}[\mathbb{1}_A]\mathbb{E}[X \mid Y] = \mathbb{E}[\mathbb{1}_A]\alpha_1 = \alpha_1 \mathbb{P}(A).$$

Then

$$\alpha_1 = \frac{\mathbb{E}[\mathbb{1}_A]X}{\mathbb{P}(A)}, \quad \alpha_2 = \frac{\mathbb{E}[\mathbb{1}_{A^c}]X}{\mathbb{P}(A^c)}.$$

Generically, $\mathbb{E}[X \mid Y] = h(Y)$ with $h(c_1) = \alpha_1$ and $h(c_2) = \alpha_2$.

THEOREM 2.4.

Let $(\Omega_1, \mathcal{F}_1)$ be a measure space with the property that $\mathcal{F}_1$ is the σ -algebra generated by a map g on Ω_1 with values in a measure space $(\Omega_2, \mathcal{F}_2)$, i.e. $\mathcal{F}_1 = \sigma(g)$. Then a real-valued function $f : \Omega_1 \rightarrow \mathbb{R}$ is $\mathcal{F}_1/\mathcal{B}(\mathbb{R})$ -measurable if and only if there exists a real-valued measurable function h on $(\Omega_2, \mathcal{F}_2)$, such that $f = h(g)$.

COROLLARY 2.5 (Doob-Dynkin Lemma).

Let $Y = (Y_1, \dots, Y_k)$, $Y_i : \Omega \rightarrow E$, be a random vector, and let X be an integrable real-valued random variable, both defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then there exists a real-valued $\mathcal{E}^k/\mathcal{B}(\mathbb{R})$ -measurable function $h : E^k \rightarrow \mathbb{R}$, such that $\mathbb{E}[X \mid Y_1, \dots, Y_k] = h(Y_1, \dots, Y_k)$.

Let $Y = (Y_t)_{t \in T}$ be an $(E, \mathcal{E})$ -valued random variable defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then there exists a measurable function $h : E^S \rightarrow \mathbb{R}$, $S = \{t_1, \dots\}$, such that $\mathbb{E}[X \mid Y] = h(Y_{t_1}, \dots)$.

LEMMA 2.6.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{F}_1, \mathcal{F}_2 \subseteq \mathcal{F}$ be independent σ -algebras. Let $(E_1, \mathcal{E}_1)$, $(E_2, \mathcal{E}_2)$ be measure spaces. Suppose that $X : \Omega \rightarrow E_1$ is $\mathcal{F}_1$ -measurable and that $Y : \Omega \rightarrow E_2$ is $\mathcal{F}_2$ -measurable. Let further $f : E_1 \times E_2 \rightarrow \mathbb{R}$ be $\mathcal{E}_1 \times \mathcal{E}_2$ -measurable with $\mathbb{E}|f(X, Y)| < \infty$. Then there exists an $\mathcal{E}_2$ -measurable function $g : E_2 \rightarrow \mathbb{R}$, such that $\mathbb{E}[f(X, Y) \mid \mathcal{F}_2] = g(Y)$, where

$$g(y) = \int_{\Omega} f(X(\omega), y) d\mathbb{P}(\omega) = \int_x f(x, y) d\mathbb{P}_X(x).$$

EXAMPLE 2.7.

LEMMA 2.8.

Let Y be a $\sigma(\eta_1, \dots, \eta_n)$ measurable function. Then there is an admissible gambling strategy $\alpha_1, \dots, \alpha_t$ such that

$$Y - \mathbb{E}[Y] = \sum_{j=1}^n \alpha_j \eta_j.$$

Proof. (i) To show $E(\alpha_j \eta_j \mid \mathcal{F}_{j-1}) = 0$, we first recall that in the proof of Lemma 2.8, we define α_j by

$$\alpha_j \eta_j = E(Y \mid \mathcal{F}_j) - E(Y \mid \mathcal{F}_{j-1}),$$

where $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Then, we can rewrite $E(\alpha_j \eta_j \mid \mathcal{F}_{j-1})$ as:

$$\begin{aligned} E(\alpha_j \eta_j \mid \mathcal{F}_{j-1}) &= E(E(Y \mid \mathcal{F}_j) - E(Y \mid \mathcal{F}_{j-1}) \mid \mathcal{F}_{j-1}) \\ &= E(E(Y \mid \mathcal{F}_j) \mid \mathcal{F}_{j-1}) - E(E(Y \mid \mathcal{F}_{j-1}) \mid \mathcal{F}_{j-1}) \\ &\quad \text{(linearity of conditional expectation)} \end{aligned}$$

$$= E(Y | \mathcal{F}_{j-1}) - E(Y | \mathcal{F}_{j-1}) = 0. \quad (\text{tower property of conditional expectation})$$

Therefore, we have shown that $E(\alpha_j \eta_j | \mathcal{F}_{j-1}) = 0$.

- (ii) *Corollary 2.5* (Doob–Dynkin) states that for any integrable real-valued random variable X and a random vector $\mathbf{Y}$, there exists a measurable function h such that $E(X | \mathbf{Y}) = h(\mathbf{Y})$. Then in this case when $X = \alpha_j \eta_j$ and $\mathbf{Y} = (\eta_1, \dots, \eta_{j-1})$, there exists a function f such that

$$E(\alpha_j \eta_j | \eta_1, \dots, \eta_{j-1}) = f(\eta_1, \dots, \eta_{j-1}, \eta_j).$$

From (i) we know that $E(\alpha_j \eta_j | \mathcal{F}_{j-1}) = 0$. Since $\mathcal{F}_{j-1}$ is generated by $\eta_1, \dots, \eta_{j-1}$, we have

$$f(\eta_1, \dots, \eta_{j-1}, \eta_j) = E(\alpha_j \eta_j | \eta_1, \dots, \eta_{j-1}) = E(\alpha_j \eta_j | \mathcal{F}_{j-1}) = 0.$$

This means that the function f must be equal to zero for any value of η_j including 1 and -1 . Therefore we conclude

$$f(\eta_1, \dots, \eta_{j-1}, 1) = f(\eta_1, \dots, \eta_{j-1}, -1) = 0$$

as required.

- (iii) The preceding steps show that α_j can be expressed as a function of $\eta_1, \dots, \eta_{j-1}$ because f does not depend on the value of η_j . Since $\mathcal{F}_{j-1} = \sigma(\eta_1, \dots, \eta_{j-1})$, α_j is $\mathcal{F}_{j-1}$ -measurable. Thus $\alpha_1, \dots, \alpha_n$ is an admissible gambling strategy and

$$Y - \mathbb{E}[Y] = \sum_{j=1}^n \alpha_j \eta_j.$$

□

SECTION 2.2

| Definition and Examples

DEFINITION 2.9 (Martingales).

An $\mathcal{F}_t$ -adapted, real-valued process M is called a *martingale* (with respect to the filtration $\mathcal{F}_t$) if:

- (i) $\mathbb{E}|M_t| < \infty$ for all $t \in T$;
- (ii) $\mathbb{E}[M_t | \mathcal{F}_s] \stackrel{a.s.}{=} M_s$ for $s \leq t$.

If property (ii) holds with ' $\geq$ ' (resp. ' $\leq$ ') instead of ' $=$ ', then M is called a *sub-martingale* (resp. *super-martingale*).

EXAMPLE 2.10 (Doob Martingale). Suppose that X is an integrable random variable and $(\mathcal{F}_t)_{t \in T}$ a filtration. For $t \in T$, define $M_t = \mathbb{E}[X | \mathcal{F}_t]$, or, more precisely, let M_t be a version of $\mathbb{E}[X | \mathcal{F}_t]$. Then $M = (M_t)_{t \in T}$ is an $\mathcal{F}_t$ -martingale and M is uniformly integrable.

EXAMPLE 2.11. Suppose that M is a martingale and that ϕ is a convex function such that $\mathbb{E}|\phi(M_t)| < \infty$ for $t \in T$. Then the process $\phi(M)$ is a *submartingale*. The same is true if M is a submartingale and ϕ is an increasing, convex function.

Proof.

□

SECTION 2.3

| Discrete-Time Martingales

In this section we restrict ourselves to martingales (and filtrations) that are indexed by (a subinterval of) $\mathbb{Z}^+$. We will assume the underlying filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$ to be fixed.

Martingale Transforms

THEOREM 2.12.

Let M be a $(\mathcal{F}_n)_n$ sub- or supermartingale and τ an $(\mathcal{F}_n)_n$ -stopping time. Then the stopped process M^τ is an $(\mathcal{F}_{\tau \wedge n})_n$ sub- or supermartingale as well.

THEOREM 2.13.

Let M be a (sub)martingale and let σ, τ be two stopping times such that $\sigma \leq \tau \leq K$, for some constant $K > 0$. Then

$$\mathbb{E}[M_\tau | \mathcal{F}_\sigma](\geq) = M_\sigma, \quad a.s.$$

An adapted integrable process M is a martingale if and only if $\mathbb{E}M_\tau = \mathbb{E}M_\sigma$ for any pairs of bounded stopping times $\sigma \leq \tau$.

Doob Decomposition

THEOREM 2.14.

Let X be a sub- or supermartingale. For all $\lambda > 0$ and $n \in \mathbb{Z}^+$:

$$\lambda \mathbb{P} \left\{ \max_{k \leq n} |X_k| \geq 3\lambda \right\} \leq 4\mathbb{E}|X_0| + 3\mathbb{E}|X_n|.$$

Convergence Theorems

THEOREM 2.15 (Doob's Martingale Convergence Theorem).

If M is a supermartingale that is bounded in $\mathcal{L}^1$, then M_n converges a.s. to a finite $\mathcal{F}_\infty$ -measurable limit M_∞ as $n \rightarrow \infty$, with $\mathbb{E}|M_\infty| < \infty$.

DEFINITION 2.16 (Uniformly Integrable).

Let C be an arbitrary collection of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We call the collection *uniformly integrable* if to every $\epsilon > 0$ there exists a constant $K \geq 0$ such that

$$\int_{|X| > K} |X| d\mathbb{P} \leq \epsilon, \quad \forall X \in C.$$

LEMMA 2.17.

Let C be a collection of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- (i) If the class C is bounded in $\mathcal{L}^p(\mathbb{P})$ for some $p > 1$, then C is uniformly integrable.
- (ii) If C is uniformly integrable, then C is bounded in $\mathcal{L}^1(\mathbb{P})$.

THEOREM 2.18.

Let $(X_n)_{n \in \mathbb{Z}^+}$ and X be integrable random variables. Then $X_n \xrightarrow{\mathcal{L}^1} X$ if and only if $X_n \xrightarrow{\mathbb{P}} X$ and the sequence $(X_n)_n$ is uniformly integrable.

LEMMA 2.19.

Let X be an integrable random variable. Then the class

$$C = \{\mathbb{E}(X \mid \mathcal{A}) \mid \mathcal{A} \text{ is sub-}\sigma\text{-algebra of } \mathcal{F}\}$$

is uniformly integrable.

COROLLARY 2.20.

Doob martingale $(M_n = \mathbb{E}[X \mid \mathcal{F}_n])_n$ is uniformly integrable.

THEOREM 2.21.

Let M be a supermartingale that is bounded in $\mathcal{L}^1$. Then $M_n \xrightarrow{\mathcal{L}^1} M_\infty$, $n \rightarrow \infty$, if and only if $\{M_n \mid n \in \mathbb{Z}^+\}$ is uniformly integrable, where M_∞ is integrable and $\mathcal{F}_\infty$ -measurable. In that case

$$\mathbb{E}[M_\infty \mid \mathcal{F}_n] \leq M_n, \quad \text{a.s.} \quad (2.1)$$

If in addition M is a martingale, then there is equality in (2.1), in other words, M is a Doob martingale.

THEOREM 2.22 (Lévy's Upward Theorem).

Let X be an integrable random variable, defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $(\mathcal{F}_n)_n$ be a filtration, $\mathcal{F}_n \subset \mathcal{F}$, for all n . Then as $n \rightarrow \infty$

$$\mathbb{E}[X \mid \mathcal{F}_n] \rightarrow \mathbb{E}[X \mid \mathcal{F}_\infty],$$

a.s. and in $\mathcal{L}^1$.

Proof.

□

THEOREM 2.23 (Lévy-Doob Downward Theorem).

Let $(\mathcal{F}_{-n} \mid n \in \mathbb{Z}^+)$ be a collection of σ -algebras, such that $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$ for every n , and let $M = (\dots, M_{-2}, M_{-1})$ be a supermartingale, i.e.,

$$\mathbb{E}[M_{-m} \mid \mathcal{F}_{-n}] \leq M_{-n} \quad \text{a.s.}, \quad \forall -n \leq -m \leq -1.$$

If $\sup \mathbb{E}M_{-n} < \infty$, then the process M is uniformly integrable and the limit $M_{-\infty} = \lim_{n \rightarrow \infty} M_{-n}$ exists a.s. and in $\mathcal{L}^1$. Moreover,

$$\mathbb{E}[M_{-n} \mid \mathcal{F}_{-\infty}] \leq M_{-\infty} \quad \text{a.s.} \quad (2.2)$$

If M is a martingale, we have equality in (2.2) and in particular $M_{-\infty} = \mathbb{E}[M_{-1} \mid \mathcal{F}_{-\infty}]$.

Proof.

□

Optional Sampling Theorems

THEOREM 2.24 (Optional Sampling Theorem).

Let M be a uniformly integrable (super-)martingale. Then the family of random variables

$$\{M_\tau \mid \tau \text{ is a finite stopping time}\}$$

is uniformly integrable and for all stopping times $\sigma \leq \tau$ we have

$$\mathbb{E}[M_\tau \mid \mathcal{F}_\sigma] = (\leq) M_\sigma \quad a.s.$$

THEOREM 2.25.

SECTION 2.4

| Continuous-Time Martingales

We consider general martingales indexed by a subset T of $\mathbb{R}^+$. If the martingale $M = (M_t)_{t \geq 0}$ has ‘nice’ sample paths, for instance they are right-continuous, then M can be ‘approximated’ accurately by a discrete-time martingale.

Upcrossings in Continuous Time

For a continuous-time process X we define the number of upcrossings of the interval $[a, b]$ in the bounded set of time points $T \subset \mathbb{R}^+$ as follows. For a finite set $F = \{t_1, \dots, t_n\} \subseteq T$ we define $U_F[a, b]$ as the number of upcrossings of $[a, b]$ of the discrete-time process $(X_{t_i})_{i=1, \dots, n}$. We put

$$U_T[a, b] = \sup \{U_F[a, b] \mid F \subseteq T, F \text{ finite}\}.$$

Doob’s upcrossing lemma has the following extension.

LEMMA 2.26.

Let M be a supermartingale and let $T \subseteq \mathbb{R}_+$ be a countable, bounded set. Then for all $a < b$, the number of upcrossings $U_T[a, b]$ of the interval $[a, b]$ by M satisfies

$$(b - a)\mathbb{E}U_T[a, b] \leq \sup_{t \in T} \mathbb{E}(M_t - a)^-.$$

COROLLARY 2.27.

$U_T[a, b]$ is a.s. finite, and $\mathcal{F}_t$ -measurable if $t = \sup \{s \mid s \in T\}$.

Regularisation

THEOREM 2.28.

Let M be a supermartingale. Then there exists a set $\Omega_s^* \in \mathcal{F}_s$ of probability 1, such that for all $\omega \in \Omega_s^*$ the limits

$$\lim_{q \uparrow t} M_q(\omega) \quad \text{and} \quad \lim_{q \downarrow t} M_q(\omega)$$

exist and are finite for every $t \in (0, s]$ and $t \in [0, s)$ respectively, for any $s \leq \infty$.

Proof. □

COROLLARY 2.29.

There exists an $\mathcal{F}_\infty$ -measurable set Ω^* , $\mathbb{P}\{\Omega^*\} = 1$, such that every sample path of a right-continuous supermartingale is cadlag on Ω^* .

Our aim is now to construct a modification of a supermartingale that is a supermartingale with a.s. cadlag sample paths itself, under suitable conditions. Given a supermartingale M , define for every $t \geq 0$,

$$M_{t+}(\omega) = \begin{cases} \lim_{q \downarrow t, q \in \mathbb{Q}} M_q(\omega), & \text{if this limit exists and is finite} \\ 0, & \text{otherwise} \end{cases}.$$

LEMMA 2.30.

Let M be a supermartingale.

(i) Then $\mathbb{E}|M_{t+}| < \infty$ for every t and

$$\mathbb{E}[M_{t+} \mid \mathcal{F}_t] \leq M_t, \quad a.s.$$

If in addition $t \rightarrow \mathbb{E}M_t$ is right-continuous, then this inequality is an equality.

(ii) The process $(M_{t+})_{t \geq 0}$ is a supermartingale with respect to the filtration $(\mathcal{F}_{t+})_{t \geq 0}$ and it is a martingale if M is a martingale.

EXAMPLE 2.31. Let $\Omega = \{-1, 1\}$, $\mathcal{F}_t = \{\Omega, \emptyset\}$ for $t \leq 1$ and $\mathcal{F}_t = \{\Omega, \emptyset, \{1\}, \{-1\}\}$ for $t > 1$. Let $\mathbb{P}(\{1\}) = \mathbb{P}(\{-1\}) = \frac{1}{2}$.

THEOREM 2.32 (Doob's Regularity Theorem).

Let M be a supermartingale with respect to the filtration $(\mathcal{F}_t)_t$ satisfying the usual conditions. Then M has a cadlag modification $\tilde{M}$ (such that $\{M_t - \tilde{M} \neq 0\}$ is a null set contained in $\mathcal{F}_0$) if and only if $t \rightarrow \mathbb{E}M_t$ is right-continuous. In that case $\tilde{M}$ is a supermartingale with respect to $(\mathcal{F}_t)_{t \geq 0}$ as well. If M is a martingale then $\tilde{M}$ is a martingale.

Proof. " $\Leftarrow$ " By Theorem 2.28 and the fact that $(\mathcal{F}_t)$ satisfies the usual conditions, there exists an event $\Omega^* \in \mathcal{F}_0$ of probability 1, on which the limits

$$M_{t-} = \lim_{q \uparrow t} M_q, \quad M_{t+} = \lim_{q \downarrow t} M_q$$

exist for every t . Define the process $\tilde{M}$ by

$$\tilde{M}_t(\omega) = M_{t+}(\omega) \mathbb{1}_{\{\Omega^*\}}(\omega)$$

for $t \geq 0$. Then $\tilde{M}_t = M_{t+}$ a.s. and they differ at most on a null-set contained in $\mathcal{F}_0$. Since M_{t+} is $\mathcal{F}_{t+}$ -measurable, we have that $\tilde{M}_t$ is $\mathcal{F}_{t+}$ -measurable. It follows that

$$\tilde{M}_t = \mathbb{E}[M_{t+} \mid \mathcal{F}_{t+}] \quad a.s.$$

By right-continuity of the filtration, right-continuity of the map $t \rightarrow \mathbb{E}M_t$ and the preceding lemma, we get that

$$\tilde{M}_t = \mathbb{E}[M_{t+} \mid \mathcal{F}_{t+}] = \mathbb{E}[M_t \mid \mathcal{F}_t] = M_t \quad a.s.$$

In other words, $\tilde{M}$ is a modification of M . Right-continuity of the filtration implies further that $\tilde{M}$ is adapted to $(\mathcal{F}_t)$. The process $\tilde{M}$ is cadlag as well as a supermartingale.

" $\Rightarrow$ " We have to show that if a supermartingale M has a cadlag modification $\tilde{M}$, then $t \rightarrow \mathbb{E}M_t$ must be right-continuous. Given that M is a supermartingale with respect to the filtration $(\mathcal{F}_t)_t$ satisfying the usual conditions, by Theorem 2.28, there exists an event $\Omega^* \in \mathcal{F}_0$ of probability 1, on which the limit $M_{t+} = \lim_{q \downarrow t} M_q$ exists for every t . Since $\tilde{M}$ is a cadlag modification of M , it follows that a.s. for every $\omega \in \Omega^*$, we have $M_t(\omega) = \tilde{M}_t(\omega)$ for all $t \geq 0$. Then

$$M_t = \tilde{M}_t = \lim_{q \downarrow t} \tilde{M}_q = \lim_{q \downarrow t} M_q \quad a.s.$$

Taking expectations on both sides (M as a supermartingale is integrable) and by the dominated convergence theorem, we get

$$\mathbb{E}[M_t] = \mathbb{E}\left[\lim_{q \downarrow t} M_q\right] = \lim_{q \downarrow t} \mathbb{E}[M_q].$$

Thus, the function $t \rightarrow \mathbb{E}M_t$ is right-continuous. Again, right-continuity of the filtration implies that $\tilde{M}$ is adapted to $(\mathcal{F}_t)$, indicating the process $\tilde{M}$ is a supermartingale. $\square$

Convergence Theorems

THEOREM 2.33.

Let M be a right-continuous supermartingale that is bounded in $\mathcal{L}^1$. Then M_t converges a.s. to a finite $\mathcal{F}_\infty$ -measurable limit M_∞ , as $t \rightarrow \infty$, with $\mathbb{E}|M_\infty| < \infty$.

Proof.

□

THEOREM 2.34.

Let M be a right-continuous supermartingale that is bounded in $\mathcal{L}^1$.

(i) If M is uniformly integrable, then $M_t \rightarrow M_\infty$ a.s. and in $\mathcal{L}^1$, and

$$\mathbb{E}[M_\infty | \mathcal{F}_t] \leq M_t \quad \text{a.s.}$$

with equality if M is a martingale.

(ii) If M is a martingale and $M_t \rightarrow M_\infty$ in $\mathcal{L}^1$ as $t \rightarrow \infty$, then M is uniformly integrable.

Proof.

□

Optional Sampling

THEOREM 2.35 (Optional Sampling Theorem).

Let M be a right-continuous, uniformly integrable super-martingale. Then for all stopping times $\sigma \leq \tau$ we have that M_τ and M_σ are integrable and

$$\mathbb{E}[M_\tau | \mathcal{F}_\sigma] \leq M_\sigma \quad \text{a.s.}$$

with equality if M is martingale.

Proof.

□

COROLLARY 2.36.

If M is a right-continuous (super)martingale and τ is an $(\mathcal{F}_t)_t$ stopping time, then the stopped process is a (super)martingale as well.

Proof.

□

SECTION 2.5

| Applications to Brownian Motion

Distribution of Hitting Times

Let W be a standard Brownian motion and, for $a > 0$, let

$$\tau_a := \inf \{t > 0 \mid W_t = a\}$$

be the (a.s. finite) hitting time/adapted stopping time of level a .

THEOREM 2.37.

For $a > 0$ the Laplace transform of the hitting time τ_a is given by

$$\mathbb{E}e^{-\lambda\tau_a} = e^{-a\sqrt{2\lambda}}, \quad \lambda \geq 0.$$

Proof.

□

COROLLARY 2.38.

For every $a > 0$ it holds that $\mathbb{E}\tau_a = \infty$.

Proof.

□

SECTION 2.6

| Poisson Process and the PASTA Property

Let N be a right continuous Poisson process with parameter μ on $(\Omega, \mathcal{F}, \mathbb{P})$. Here the space Ω are right-continuous, non-decreasing integer valued paths, such that for each path ω one has $\omega_0 = 0$ as well as $\omega_t \leq \lim_{s \uparrow t} \omega_s + 1$, for all $t > 0$. The path properties imply all paths in Ω to have at most finitely many discontinuities in each bounded time interval. The σ -algebra $\mathcal{F}$ is the associated σ -algebra that makes the projections on the t -coordinate measurable.

LEMMA 2.39.

$$\frac{N_t}{t} \xrightarrow{a.s.} \mu, \quad t \rightarrow \infty.$$

The structure of Poisson paths implies that each path can be viewed as to represent the ‘distribution function’ of a counting measure, that gives measure 1 to each point where $N(\omega)$ has a discontinuity. In fact, the measure ν_ω , with $\nu_\omega([0, t]) = N_t(\omega)$ is a Lebesgue-Stieltjes measure generated by the trajectory $N(\omega)$, and $\nu_\omega(A)$ ‘counts’ the number of jumps of $N(\omega)$ occurring in $A \in \mathcal{B}[0, \infty)$.

Denote these successive jumps by $S_n(\omega)$, $n = 1, \dots$. Let $f : [0, t] \rightarrow \mathbb{R}$ be any bounded or non-negative measurable function. Then for each $\omega \in \Omega$, $t \geq 0$, we define

$$\int_0^t f(s) dN_s(\omega) = \sum_{n=0}^{N_t(\omega)} f(S_n(\omega)) = \sum_{n=0}^{\infty} f(S_n(\omega)).$$

This can be derived using the *standard machinery*.

The PASTA Property

Let us now consider some $(E, \mathcal{E})$ -valued stochastic process X on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P})$, where $(E, \mathcal{E})$ is some measure space. Let $B \in \mathcal{E}$.

The aim is to compare the fraction of time that X -process spends in set B , with the fraction of time points generated by the Poisson process that the X -process is in B . We need to introduce some notation: $U_t = \mathbb{1}_{\{X_t \in B\}}$, and, as usual, λ stands for the Lebesgue measure on $(\mathbb{R}^+, \mathcal{B}(\mathbb{R}^+))$.

LEMMA 2.40.

U has *ladcag* paths, in other words, all paths of U are left continuous and have right limits.

We further define

$$\begin{aligned} \bar{U}_t &= \frac{1}{t} \int_0^t U_s d\lambda(s), \\ A_t &= \int_0^t U_s dN_s, \\ \bar{A}_t &= \frac{A_t}{N_t} \mathbb{1}_{\{N_t > 0\}}, \\ \mathcal{F}_t &= \sigma(U_s, s \leq t, N_s, s \leq t). \end{aligned}$$

Here $\bar{U}_t$ stands for the fraction of time during $(0, t]$ that X spends in set B ; A_t is the amount of Poisson time points before t at which X is in set B , and $\bar{A}_t$ is the fraction of Poisson time points up to time t at which X is in B .

LEMMA 2.41.

Lack of anticipation property $\sigma(N_{t+u} - N_t, u \geq 0)$ and $\mathcal{F}_t$ are independent for all $t \geq 0$.

THEOREM 2.42.

Under Lemma 2.40 and Lemma 2.41, there exists a finite random variable $\bar{U}_\infty$ such that $\bar{U}_t \xrightarrow{a.s.} \bar{U}_\infty$, iff there exists a finite random variable $\bar{A}_\infty$ such that $A_t \xrightarrow{a.s.} \bar{A}_\infty$ and then $\bar{A}_\infty \stackrel{a.s.}{=} \bar{U}_\infty$.

Proof.

□

Exercises

EXERCISE 2.1. Consider the unit interval $I = [0, 1]$ equipped with the Borel- σ -algebra $\mathcal{B}([0, 1])$ and the Lebesgue measure. Let f be an integrable function on I . Let for $n = 1, 2, \dots$

$$f_n(x) = 2^n \int_{(k-1)2^{-n}}^{k2^{-n}} f(y) dy, \quad (k-1)2^{-n} \leq x < k2^{-n}, \quad k = 1, \dots, 2^n,$$

and define $f_n(1) = 1$ (the value $f_n(1)$ is not important). Finally, we define $\mathcal{F}_n$ as the σ -algebra on $[0, 1]$ generated by intervals of the form $[(k-1)2^{-n}, k2^{-n})$, $1 \leq k \leq 2^n$.

- (i) Argue that $\mathcal{F}_n$ is an increasing sequence of σ -algebras.
- (ii) Show that $(f_n)_n$ is a martingale.
- (iii) Use Lévy's Upward Theorem to prove that $f_n \rightarrow f$, a.s. and in $\mathcal{L}_1$, as $n \rightarrow \infty$.

SOLUTION. (i) For $\mathcal{F}_n$ to be an increasing sequence of σ -algebras, we have to show that $\mathcal{F}_n \subset \mathcal{F}_{n+1}$ for all $n \in \mathbb{N}$. By definition, $\mathcal{F}_n$ is generated by intervals partition the unit interval $[0, 1]$ into 2^n equal-length sub-intervals. When moving to $\mathcal{F}_{n+1}$, the unit interval $[0, 1]$ is partitioned into 2^{n+1} equal-length sub-intervals of the form $[(j-1)2^{-(n+1)}, j2^{-(n+1)})$ where $j \in [1, 2^{n+1}]$, each of length $2^{-(n+1)}$. And we have

$$[(k-1)2^{-n}, k2^{-n}) = [(2k-2)2^{-(n+1)}, (2k-1)2^{-(n+1)}) \cup [(2k-1)2^{-(n+1)}, 2k2^{-(n+1)})],$$

which shows that any interval generating $\mathcal{F}_n$ can be represented by the union of two generating intervals of $\mathcal{F}_{n+1}$. Since σ -algebras are closed under finite union, every element of $\mathcal{F}_n$ can be constructed using the intervals that generate $\mathcal{F}_{n+1}$, proving that $\mathcal{F}_n$ is an increasing sequence of σ -algebras.

- (ii) Since f is given as an integrable function on $I = [0, 1]$ with respect to the Lebesgue measure, it follows that $E|f| < \infty$. For each n , $f_n(x)$ is defined as:

$$f_n(x) = 2^n \int_{(k-1)2^{-n}}^{k2^{-n}} f(y) dy,$$

where $(k-1)2^{-n} \leq x < k2^{-n}$ for $k = 1, \dots, 2^n$. The function f_n represents the average value of f over each sub-interval $[(k-1)2^{-n}, k2^{-n})$ scaled by the number of sub-intervals 2^n . The integral of $|f_n|$ over each interval is finite because f is integrable, and thus $E|f_n|$ is also finite for all n .

For any $x \in [(k-1)2^{-n}, k2^{-n}) \subset \mathcal{F}_n$, by (i), f_{n+1} is the average of f over the two sub-intervals that constitute it in $\mathcal{F}_{n+1}$:

$$\begin{aligned} E(f_{n+1} | \mathcal{F}_n)(x) &= 2^{n+1} \left(\int_{(2k-2)2^{-(n+1)}}^{(2k-1)2^{-(n+1)}} f(y) dy + \int_{(2k-1)2^{-(n+1)}}^{2k2^{-(n+1)}} f(y) dy \right) \cdot \frac{1}{2} \\ &= 2^n \int_{(k-1)2^{-n}}^{k2^{-n}} f(y) dy = f_n(x). \end{aligned}$$

Thus, f_n satisfies the martingale property with respect to the filtration $\mathcal{F}_n$, showing that $(f_n)_n$ is a martingale.

- (iii) The function f is an integrable random variable on the probability space $(I = [0, 1], \mathcal{B}([0, 1]), P)$, where P is Lebesgue measure. By (ii), $f_n = E(f | \mathcal{F}_n)$ for every n , and $\mathcal{F}_\infty = \sigma(\bigcup_n \mathcal{F}_n)$. Therefore, Lévy's upward theorem gives

$$f_n = E(f | \mathcal{F}_n) \xrightarrow{a.s.} E(f | \mathcal{F}_\infty) = f$$

in $\mathcal{L}_1$ as $n \rightarrow \infty$, since f is $\mathcal{F}_\infty$ -measurable. Thus, $f_n \rightarrow f$ almost surely and in $\mathcal{L}_1$. □

EXERCISE 2.2 (Martingale Formulation of Bellman's Optimality Principle). Suppose your winning per unit stake on game n are ϵ_n , where the ϵ_n are *i.i.d.* random variables with

$$\mathbb{P}\{\epsilon_n = 1\} = p = 1 - \mathbb{P}\{\epsilon_n = -1\},$$

with $p > \frac{1}{2}$. Your bet α_n on game n must lie between 0 and Z_{n-1} , your capital at time $n-1$. Your object is to maximise your *interest rate* $\mathbb{E} \log \left(\frac{Z_N}{Z_0} \right)$, where $N = \text{length of the game}$ is finite and Z_0 is a given constant. Let $\mathcal{F}_n = \sigma(\epsilon_1, \dots, \epsilon_n)$ be your *history* up to time n . Let $\{\alpha_n\}_n$ be an admissible strategy, in other words a predictable sequence. Show that $\log(Z_n) - n\alpha$ is a supermartingale with α the entropy given by

$$\alpha = p \log p + (1-p) \log(1-p) + \log 2.$$

Hence $\mathbb{E} \log \left(\frac{Z_N}{Z_0} \right) \leq n\alpha$. Show also that for some strategy $\log(Z_n) - n\alpha$ is a martingale. What is the best strategy?

EXERCISE 2.3 (Galton-Watson Process). This is a simple model for population growth, growth of the number of cells, etc.

A population of cells evolves as follows. In every time step, every cell splits into 2 cells with probability p or it dies with probability $1-p$, independently of the other cells and of the population history. Let N_t denote the number of cells at time t , $t = 0, 1, 2, \dots$. Initially, there is only 1 cell, i.e., $N_0 = 1$.

We can describe this model formally by defining Z_t^n , $n = 1, \dots, N_t$, $t = 0, 1, \dots$, to be *i.i.d.* random variables with

$$\mathbb{P}(Z_t^n = 2) = p = 1 - \mathbb{P}(Z_t^n = 0),$$

and then $N_{t+1} = \sum_{n=1}^{N_t} Z_t^n$. Let $\{\mathcal{F}_t\}_{t=0,1,\dots}$ be the natural filtration generated by $\{N_t\}_{t=0,1,\dots}$.

Argue or prove that

$$\begin{aligned} \mathbb{P}(\mathbf{1}_{N_{t+1}=2y} | \mathcal{F}_t) &= \mathbb{E}[\mathbf{1}_{N_{t+1}=2y} | \mathcal{F}_t] \\ &= \mathbb{E}[\mathbf{1}_{N_{t+1}=2y} | N_t] \end{aligned}$$

$$= \mathbb{P}(N_{t+1} = 2y \mid N_t) = \binom{N_t}{y} p^y (1-p)^{N_t-y}.$$

Hence, conditional on $\mathcal{F}_t$, $\frac{N_{t+1}}{2}$ has a binomial distribution with parameters N_t and p .

SOLUTION. □

EXERCISE 2.4 (Empirical Distributions). Let $X_1, \dots, X_n$ be *i.i.d.* random variables, each with the uniform distribution on $[0, 1]$. For $0 \leq t < 1$ define

$$G_n(t) = \frac{1}{n} \#\{k \leq n \mid X_k \leq t\} = \frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k \leq t\}}.$$

In words, $G_n(t)$ is the fraction of X_k that have value at most t . Denote

$$\mathcal{F}_n(t) = \sigma(\mathbb{1}_{\{X_1 \leq s\}}, \dots, \mathbb{1}_{\{X_n \leq s\}}, s \leq t)$$

and $\mathcal{G}_n(t) = \sigma(G_n(s), s \leq t)$.

(i) Prove that for $0 \leq t \leq u < 1$ we have

$$E\{G_n(u) \mid \mathcal{F}_n(t)\} = G_n(t) + (1 - G_n(t)) \frac{u - t}{1 - t}.$$

Show that this implies that

$$E\{G_n(u) \mid \mathcal{G}_n(t)\} = G_n(t) + (1 - G_n(t)) \frac{u - t}{1 - t}.$$

(ii) Show that the stochastic process $M_n = (M_n(t))_{t \in [0, 1]}$ defined by

$$M_n(t) = \frac{G_n(t) - t}{1 - t}$$

is a continuous-time martingale with respect to $\mathcal{G}_n(t)$, $t \in [0, 1]$.

(iii) Is M_n uniform integrable?

Hint. compute $\lim_{t \uparrow 1} M_n(t)$.

(iv) We extend the process M_n to $[0, \infty)$, by putting $M_n(t) = 1$ and $\mathcal{G}_n(t) = \sigma(X_1, \dots, X_n)$, for $t \geq 1$. Show that $(M_n(t))_{t \in [0, \infty)}$ is an $\mathcal{L}^1$ -bounded submartingale relative to $(\mathcal{G}_n(t))_{t \in [0, \infty)}$, that converges in $\mathcal{L}^1$, but is not uniform integrable.

SOLUTION. (i) Given that

$$G_n(t) = \frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k \leq t\}},$$

we can write

$$G_n(u) = \frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k \leq u\}} = \frac{1}{n} \left(\sum_{k=1}^n \mathbb{1}_{\{X_k \leq t\}} \right) + \frac{1}{n} \left(\sum_{k=1}^n \mathbb{1}_{\{X_k \in (t, u]\}} \right),$$

Since $X_1, \dots, X_n$ are *i.i.d.* random variables, each with the uniform distribution on $[0, 1]$, we know $P(t < X_k \leq u \mid X_k > t) = \frac{u-t}{1-t}$. Therefore, we have

$$\begin{aligned} E\{G_n(u) \mid \mathcal{F}_n(t)\} &= E\left\{ \frac{1}{n} \left(\sum_{k=1}^n \mathbb{1}_{\{X_k \leq t\}} \right) + \frac{1}{n} \left(\sum_{k=1}^n \mathbb{1}_{\{X_k \in (t, u]\}} \right) \mid \mathcal{F}_n(t) \right\} \\ &= G_n(t) + E\left\{ \frac{1}{n} \left(\sum_{k=1}^n \mathbb{1}_{\{X_k \in (t, u]\}} \right) \mid \mathcal{F}_n(t) \right\} \end{aligned} \quad (\text{linearity})$$

$$= G_n(t) + \frac{1}{n} \sum_{k=1}^n E\{\mathbb{1}_{\{X_k \in (t, u]\}} \mid \mathcal{F}_n(t)\} \quad (i.i.d.)$$

$$= G_n(t) + \frac{1}{n} \sum_{k=1}^n E\{\mathbb{1}_{\{X_k \in (t, u]\}} \mid \sigma(\mathbb{1}_{\{X_k \leq s\}}, s \leq t)\} \quad (*)$$

$$= G_n(t) + \frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k > t\}} \frac{u-t}{1-t} = G_n(t) + (1 - G_n(t)) \frac{u-t}{1-t}.$$

(*) Follows from $\mathbb{1}_{\{X_k > t\}} \frac{u-t}{1-t}$ being $\sigma(\mathbb{1}_{\{X_k \leq s\}}, s \leq t)$ -measurable and having the same integral as $\mathbb{1}_{\{X_k \in (t, u]\}}$ over sets $\{X_k \leq s\}, \{X_k > s\}, s \in [0, t]$ which generate that sigma algebra.

Since $\mathcal{G}_n(t) = \sigma(G_n(s), s \leq t)$ and $G_n(s) = \frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k \leq s\}}$, we notice that by definition, each value of $G_n(s)$ can be computed directly from the indicator functions in $\mathcal{F}_n(t)$, hence $\mathcal{G}_n(t) \subseteq \mathcal{F}_n(t)$. Therefore, using the tower property and $G_n(t)$ being $\mathcal{G}_n(t)$ -measurable, we can conclude that the equation

$$E\{G_n(u) \mid \mathcal{F}_n(t)\} = G_n(t) + (1 - G_n(t)) \frac{u-t}{1-t}$$

implies

$$E\{G_n(u) \mid \mathcal{G}_n(t)\} = G_n(t) + (1 - G_n(t)) \frac{u-t}{1-t}.$$

(ii) By (i), we have

$$E\{G_n(u) \mid \mathcal{G}_n(t)\} = G_n(t) + (1 - G_n(t)) \frac{u-t}{1-t}.$$

Given that

$$M_n(t) = \frac{G_n(t) - t}{1-t},$$

we can write

$$\begin{aligned} E\{M_n(u) \mid \mathcal{G}_n(t)\} &= E\left\{\frac{G_n(u) - u}{1-u} \mid \mathcal{G}_n(t)\right\} \\ &= \frac{1}{1-u} (E\{G_n(u) \mid \mathcal{G}_n(t)\} - u) \\ &= \frac{1}{1-u} \left(G_n(t) + (1 - G_n(t)) \frac{u-t}{1-t} - u\right) \\ &= \frac{1}{1-t} \cdot \frac{(1-t)G_n(t) + (u-t)(1 - G_n(t)) - u(1-t)}{1-u} \\ &= \frac{1}{1-t} \cdot \frac{(1-u)G_n(t) - (1-u)t}{1-u} \\ &= \frac{G_n(t) - t}{1-t} = M_n(t). \end{aligned}$$

Therefore the stochastic process $M_n = (M_n(t))_{t \in [0,1]}$ satisfies the martingale property with respect to the filtration $\mathcal{G}_n(t)$. And by definition we have

$$E(M_n(t)) = E\left(\frac{G_n(t) - t}{1-t}\right) = \frac{E(G_n(t)) - t}{1-t} = \frac{t-t}{1-t} = 0,$$

which implies $M_n(t)$ has finite expectation, hence $M_n = (M_n(t))_{t \in [0,1]}$ is indeed a continuous-time martingale.

(iii) By the definition of $M_n(t)$, we calculate

$$\lim_{t \uparrow 1} M_n(t) = \lim_{t \uparrow 1} \frac{G_n(t) - t}{1-t}.$$

As $t \uparrow 1$, all X_k 's will be less than or equal to t , and $G_n(t) \rightarrow 1$. Thus the limit

$$\lim_{t \uparrow 1} M_n(t) = \frac{1-t}{1-t} = 1,$$

which is a bounded limit. Since a bounded martingale is UI, but there exists a sudden change as $t \uparrow 1$ (we will prove this rigorously in (iv) later), we conclude that M_n is not UI.

(iv) For $u \geq 1, u \geq t, M_n(u) = 1$, we have

$$E\{M_n(u) \mid \mathcal{G}_n(t)\} = 1 \geq M_n(t),$$

and we have shown that the equality holds if $1 > u \geq t$ in ii). $(M_n(t))_{t \in [0, \infty)}$ is therefore a submartingale. To show L^1 -boundedness, we calculate the expected value

$$E(M_n(t)) = \begin{cases} \frac{E(G_n(t)-1)}{1-t}, & t < 1 \\ 1, & t \geq 1 \end{cases}.$$

When $t < 1$, we notice that $E(G_n(t)) = \frac{1}{n} \cdot nt = t$, then

$$\frac{E(G_n(t) - 1)}{1 - t} = \frac{t - 1}{1 - t} = 0.$$

Therefore,

$$E(M_n(t)) = \begin{cases} 0, & t < 1 \\ 1, & t \geq 1 \end{cases},$$

which implies $M_n(t)$ is an L^1 -bounded submartingale relative to $\mathcal{G}_n(t)$, that converges to 1 in L^1 . But $M_n(t)$ is not UI because $E(M_n(t)) = 0$ when $t < 1$ and there exists a sudden change at $t = 1$. We will show from the definition that $M_n(t)$ is not UI. We have

$$\mathbb{P}\left(G_n(t) \leq \frac{n-1}{n}\right) = 1 - \mathbb{P}(G_n(t) = 1) = 1 - t^n.$$

Whenever this holds we have

$$|M_n(t)| \geq \frac{t - \frac{n-1}{n}}{1-t} =: c(t).$$

Let $K > 0$ and $\epsilon = \frac{1}{2n}$, let us take t big enough so that $c(t) > K$ and $t > \frac{2n-1}{2n}$. In that case we have:

$$\begin{aligned} \int_{\{|M_n(t)| > K\}} |M_n(t)| &\geq \int_{\{|M_n(t)| \geq c(t)\}} |M_n(t)| \\ &\geq c(t)(1 - t^n) \\ &\geq t - \frac{n-1}{n} > \epsilon. \end{aligned}$$

□

EXERCISE 2.5 (Brownian Motion with Drift). Let $(W_t)_t$ be a standard BM, and define

$$X_t = W_t + ct,$$

for some constant c . The process X_t is called *BM with drift*. Fix some $\lambda > 0$.

(i) Show that

$$M_t := e^{\theta X_t - \lambda t}$$

is a martingale (with respect to the natural filtration) if and only if $\theta = \sqrt{c^2 + 2\lambda} - c$ or $\theta = -\sqrt{c^2 + 2\lambda} - c$.

(ii) Next, let

$$H_x = \inf \{t > 0 \mid X_t = x\}.$$

Argue for $x \neq 0$ that H_x is a stopping time.

(iii) Show that

$$\mathbb{E}[e^{-\lambda H_x}] = \begin{cases} e^{-x(\sqrt{c^2+2\lambda}-c)}, & x > 0 \\ e^{-x(-\sqrt{c^2+2\lambda}-c)}, & x < 0 \end{cases}.$$

(iv) Use the result from (iii) to prove that for $x > 0$,

$$\mathbb{P}\{H_x < \infty\} = \begin{cases} 1, & c \geq 0 \\ e^{-2|c|x}, & c < 0 \end{cases}.$$

(v) Explain why this result is reasonable.

SOLUTION. (i) Given that $X_t = W_t + ct$ for some constant c , we have $X_t = W_t + ct$. Observe that X_s is $\mathcal{F}_s$ -measurable, and by the definition of BM, $W_t - W_s$ is independent of $\mathcal{F}_s$. Then we can write $X_t - X_s = W_t - W_s + c(t - s)$. We verify that

$$\begin{aligned} \mathbb{E}[M_t | \mathcal{F}_s] &= \mathbb{E}[\exp(\theta X_t - \lambda t) | \mathcal{F}_s] \\ &= \mathbb{E}[\exp(\theta X_s + \theta(W_t - W_s) + c\theta(t - s) - \lambda t) | \mathcal{F}_s] \\ &= \exp(\theta X_s + c\theta(t - s) - \lambda t) \mathbb{E}[\exp(\theta(W_t - W_s))] \\ &= \exp\left(\theta X_s + c\theta(t - s) - \lambda t + \frac{1}{2}\theta^2(t - s)\right), \end{aligned}$$

where the moment generating function for $W_t - W_s \sim \mathcal{N}(0, t - s)$ is

$$\mathbb{E}[\exp(\theta(W_t - W_s))] = \exp\left(\frac{1}{2}\theta^2(t - s)\right).$$

For $(M_t)_t$ to be a martingale, we require that $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $s \leq t$. It is equivalent for the equality

$$\exp\left(\theta X_s + c\theta(t - s) - \lambda t + \frac{1}{2}\theta^2(t - s)\right) = \exp(\theta X_s - \lambda s)$$

to hold a.s. for $s \leq t$. Rearranging, we have

$$\begin{aligned} \theta X_s + c\theta(t - s) - \lambda t + \frac{1}{2}\theta^2(t - s) &= \theta X_s - \lambda s \\ \implies c\theta(t - s) - \lambda(t - s) + \frac{1}{2}\theta^2(t - s) &= 0 \\ \implies \frac{1}{2}\theta^2 + c\theta - \lambda &= 0 \iff \theta = -c \pm \sqrt{c^2 + 2\lambda}, \end{aligned}$$

Notice that now

$$\mathbb{E}[M_t] = \mathbb{E}[\exp(\theta(W_t + ct) - \lambda t)] = \exp\left(c\theta t - \lambda t + \frac{1}{2}\theta^2 t\right) = 1 < \infty.$$

Therefore, the process $(M_t)_t$ is a martingale if and only if $\theta = \sqrt{c^2 + 2\lambda} - c$ or $\theta = -\sqrt{c^2 + 2\lambda} - c$.

(ii) **Lemma 1.38.** Let (E, d) be a metric space and let $\mathcal{B}(E)$ be the Borel- σ -algebra of open sets compatible with the metric d . Suppose that $X = (X_t)_{t \geq 0}$ is a continuous, $(E, \mathcal{B}(E))$ -valued stochastic process and that B is closed in E . Then the first hitting time $\sigma_B = \inf\{t \geq 0 \mid X_t \in B\}$ is an $\mathcal{F}_t^X$ -stopping time.

Given that X_t is continuous and $\{x\}$ is a closed set, we claim that H_x is a stopping time.

(iii) **Theorem 2.24 (Optional Sampling Theorem).** Let M be a uniformly integrable (super-)martingale. Then the family of random variables $\{M_\tau \mid \tau \text{ is a finite stopping time}\}$ is uniformly integrable and for all stopping times $\sigma \leq \tau$ we have $\mathbb{E}[M_\tau | \mathcal{F}_\sigma] = (\leq) M_\sigma$ a.s.

Without loss of generality, let $x > 0$. In this case, we choose $\theta = \sqrt{c^2 + 2\lambda} - c$. Given that by (ii) H_x is a stopping time, and by (i) M_t is a uniformly integrable martingale, we apply [Theorem 2.24](#) and obtain

$$\mathbb{E}[M_{H_x}] = M_0 = 1,$$

which is equivalent to

$$\mathbb{E}[\exp(\theta X_{H_x} - \lambda H_x)] = \mathbb{E}[\exp(\theta x - \lambda H_x)] = \mathbb{E}[e^{-\lambda H_x}] e^{\theta x} = 1.$$

Therefore, we have

$$\mathbb{E}[e^{-\lambda H_x}] = e^{-\theta x} = e^{-x(\sqrt{c^2+2\lambda}-c)}, \quad x > 0.$$

For $x < 0$ we proceed similarly by choosing $\theta = -\sqrt{c^2+2\lambda}-c$. Hence, we have shown that

$$\mathbb{E}[e^{-\lambda H_x}] = \begin{cases} e^{-x(\sqrt{c^2+2\lambda}-c)}, & x > 0 \\ e^{-x(-\sqrt{c^2+2\lambda}-c)}, & x < 0 \end{cases}.$$

(iv) Observe that

$$\lim_{\lambda \rightarrow 0} e^{-\lambda H_x} = \begin{cases} 1, & H_x < \infty \\ 0, & H_x = \infty \end{cases}.$$

Then by the *monotone convergence theorem*,

$$\lim_{\lambda \rightarrow 0} \mathbb{E}[e^{-\lambda H_x}] = \mathbb{E}\left[\lim_{\lambda \rightarrow 0} e^{-\lambda H_x}\right] = \mathbb{E}[\mathbf{1}_{\{H_x < \infty\}}] = \mathbb{P}(H_x < \infty).$$

For $x > 0$, by (iii), $\mathbb{E}[e^{-\lambda H_x}] = e^{-x(\sqrt{c^2+2\lambda}-c)}$.

- When $c \geq 0$:

$$\lim_{\lambda \rightarrow 0} \mathbb{E}[e^{-\lambda H_x}] = e^{-x(c-c)} = 1.$$

- When $c \leq 0$:

$$\lim_{\lambda \rightarrow 0} \mathbb{E}[e^{-\lambda H_x}] = e^{-x(-c-c)} = e^{-2|c|x}.$$

Therefore for $x > 0$,

$$\mathbb{P}\{H_x < \infty\} = \begin{cases} 1, & c \geq 0 \\ e^{-2|c|x}, & c < 0 \end{cases}.$$

(v) For $c \geq 0$, when the drift is non-negative, the process $X_t = W_t + ct$ moves upward. Then for any $x > 0$, X_t will eventually hit x with probability one. When the drift is negative, X_t tends to move downward. There exists a non-zero probability $e^{-2|c|x}$ preventing X_t from reaching x , where the larger the negative drift $|c|$ the smaller the probability.

□

EXERCISE 2.6. Let $X_1, X_2, \dots$ be a sequence of *i.i.d.* $\mathbb{R}(\mathcal{B}(\mathbb{R}))$ -valued and integrable random variables, defined on the probability space $(\Omega, \mathcal{F}, P)$. Assume that $\mathbb{E}(X_i) = 0$ and $\mathbb{E}[X_i^2] = \alpha < \infty$, $i = 1, \dots$. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, and let further $M_0 = 0$, $M_n = \sum_{i=1}^n X_i$, $i \geq 1$. Then, as easily checked $(M_n)_{n \geq 0}$ and $(M_n^2 - n\alpha)_{n \geq 0}$ are $(\mathcal{F}_n)_n$ -adapted martingales. Finally, let τ be an $(\mathcal{F}_n)_n$ -adapted stopping time on $(\Omega, \mathcal{F}, P)$ with $\mathbb{E}\tau < \infty$. Prove that $\mathbb{E}M_\tau = 0$ and that $\mathbb{E}M_\tau^2 = \mathbb{E}\tau \cdot \alpha$.

SOLUTION. Given that $M_n = \sum_{i=1}^n X_i$ is a martingale adapted to the filtration $\mathcal{F}_n$, where X_i are *i.i.d.* random variables with $\mathbb{E}(X_i) = 0$, it follows that

$$\mathbb{E}(M_n) = \mathbb{E}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathbb{E}(X_i) = 0.$$

Since τ is a finite stopping time with $\mathbb{E}\tau < \infty$, by the *optional sampling theorem*, for any stopping times $\sigma \leq \tau$ we have $\mathbb{E}M_\tau \stackrel{a.s.}{=} \mathbb{E}M_\sigma$. Setting $\sigma = 0$, we obtain $\mathbb{E}M_\tau = \mathbb{E}M_0 = 0$ because $M_0 = 0$.

Since $(M_n^2 - n\alpha)_{n \geq 0}$ is a martingale and $\mathbb{E}(M_n^2) = n\alpha$, the optional sampling theorem gives

$$\mathbb{E}(M_\tau^2 - \tau\alpha) = \mathbb{E}(M_0^2) = 0.$$

Therefore, $\mathbb{E}M_\tau^2 = \alpha \mathbb{E}\tau$.

In conclusion, we have proved that $EM_\tau = 0$ and that $EM_\tau^2 = E\tau \cdot \alpha$. □

Markov Processes

DEFINITION 3.1 (Transition Kernel).

Let $(E, \mathcal{E})$ be a measurable space. A *transition kernel* on E is a map $P : E \times \mathcal{E} \rightarrow [0, 1]$ such that:

- (i) for every $x \in E$, the map $B \mapsto P(x, B)$ is a probability measure on $(E, \mathcal{E})$;
- (ii) for every $B \in \mathcal{E}$, the map $x \mapsto P(x, B)$ is $\mathcal{E}/\mathcal{B}[0, 1]$ -measurable.

DEFINITION 3.2 (Markov Chain).

Let X be an $(E, \mathcal{E})$ valued stochastic process defined on some underlying probability space $(\Omega, \mathcal{F}, \mathbb{P}_\nu)$. Then X is a *Markov chain with initial distribution ν* if:

- (i) $\mathbb{P}_\nu(X_0 \in B) = \nu(B)$ for all $B \in \mathcal{E}$;
- (ii) if there exists a transition kernel P such that the *Markov property* holds:

$$\mathbb{P}_\nu(X_{n+1} \in B \mid \mathcal{F}_n^X) = \mathbb{P}_\nu(X_{n+1} \in B \mid \sigma(X_n)) = P(X_n, B)$$

for all $B \in \mathcal{E}$ and $n = 0, 1, 2, \dots$

Remark. We would like to infer that

$$\mathbb{P}_\nu(X_{n+m} \in B \mid \mathcal{F}_n^X) = P^m(X_n, B)$$

or all $B \in \mathcal{E}$ and $n = 0, 1, 2, \dots$, where P^m is defined inductively by

$$P^m(X_n, B) = \int P(y, B) P^{m-1}(X_n, dy), \quad m = 2, 3, \dots$$

NOTATION 3.3.

Integrals of the form $\int f d\nu$ are often written in operator notation as νf . A similar notation for transition kernels is as follows:

$$P\mathbb{1}_B(x) = \int \mathbb{1}_B(y) P(x, dy) = \int_B P(x, dy) = P(x, B).$$

The Markov property can now be reformulated as

$$\mathbb{E}_\nu[\mathbb{1}_B(X_{n+1}) \mid \mathcal{F}_n^X] = \mathbb{E}_\nu[\mathbb{1}_B(X_{n+1}) \mid \sigma(X_n)] = P(X_n, B) = P\mathbb{1}_B(X_n).$$

Let $b\mathcal{E}$ for the space of bounded, measurable functions $f : E \rightarrow \mathbb{R}$ and $m\mathcal{E}$ for the $\mathcal{E}/\mathcal{B}(\mathbb{R})$ -measurable functions $f : E \rightarrow \mathbb{R}$. Since P is a transition kernel, $P\mathbb{1}_B \in b\mathcal{E}$. Applying the *standard machinery* yields that $Pf \in b\mathcal{E}$ for all $f \in b\mathcal{E}$, in particular Pf is measurable. Applying the *standard machinery* once more, yields

$$\mathbb{E}_\nu[\mathbb{1}_B(X_{n+1}) \mid \mathcal{F}_n^X] = Pf(X_n).$$

This has two consequences. The first is the immediate implication

$$\begin{aligned} P(X_{n+2} \in B \mid \mathcal{F}_n^X) &= \mathbb{E}_\nu[\mathbb{1}_B(X_{n+2}) \mid \mathcal{F}_n^X] \\ &= \mathbb{E}_\nu[\mathbb{E}_\nu[\mathbb{1}_B(X_{n+2}) \mid \mathcal{F}_{n+1}^X] \mid \mathcal{F}_n^X] && \text{(Tower Property)} \\ &= \mathbb{E}_\nu[P\mathbb{1}_B(X_{n+1}) \mid \mathcal{F}_n^X] = P(P\mathbb{1}_B)(X_n). \end{aligned}$$

If $X_n = x$, the latter equals

$$\int_E P\mathbb{1}_B(y)P(x, dy) = \int_E P(y, B)P(x, dy) = P^2(x, B).$$

Let us now go to the continuous time case for $T = [0, \infty)$.

DEFINITION 3.4.

Let $(E, \mathcal{E})$ be a measurable space. A collection of transition kernels $(P_t)_{t \geq 0}$ is called a *(homogeneous) transition function* if for all $s, t \geq 0$, $x \in E$, and $B \in \mathcal{E}$

$$P_{t+s}(x, B) = \int P_t(y, B)P_s(x, dy)$$

This relation is known as the *Chapman-Kolmogorov relation*.

Translated to operator notation, for a transition function $(P_t)_{t \geq 0}$ it holds that for every non-negative (or bounded) measurable function f and $s, t \geq 0$ we have

$$P_{t+s}f = P_t(P_sf) = P_s(P_tf).$$

In other words, the linear operators $(P_t)_{t \geq 0}$ form a *semi-group* of operators on the space of non-negative (or bounded) functions on E .

Let f, g, h be bounded (non-negative) measurable functions on E . As argued before, P_tf is bounded and measurable. Hence multiplying by g gives gP_tf , which is bounded and measurable. Here

$$gP_tf(x) = g(x)P_tf(x) = \int_y f(y)P_t(x, dy).$$

Then we can apply P_s to this function, yielding the bounded, measurable function P_sgP_tf , with

$$P_s(gP_tf)(x) = \int_y g(y)P_tf(y)P_s(x, dy) = \int_y g(y) \int_z f(z)P_t(y, dz)P_s(x, dy).$$

And νP_sgP_tf is again bounded, measurable and we can integrate over the probability distribution ν on $(E, \mathcal{E})$:

$$\nu h P_sgP_tf = \int_x h(x)P_sgP_tf(x)\nu(dx).$$

DEFINITION 3.5.

Let $(E, \mathcal{E})$ be a measurable space and let X be an $(E, \mathcal{E})$ -valued stochastic process that is adapted to some underlying filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, P_\nu)$. X is a *Markov process with initial distribution ν* , if

- (i) $P_\nu(X_0 \in B) = \nu(B)$ for every $B \in \mathcal{E}$;
- (ii) *Markov property*: there exists a transition function $(P_t)_t$, such that for all $s, t \geq 0$ and $f \in b\mathcal{E}$:

$$\mathbb{E}_\nu[f(X_{t+s}) \mid \mathcal{F}_s] = P_tf(X_s) = \mathbb{E}_{X_s}f(X_t), \quad P_\nu\text{-a.s.}$$

Remark.

$$\mathbb{E}_{X_s} f(X_t) = \int_y f(y) P_t(X_s, dy) = \int_y f(y) P_t(X, dy)$$

on $X_s = X$.

DEFINITION 3.6.

Let $(E, \mathcal{E})$ be a measurable space and let $X : (\Omega, \mathcal{F}) \rightarrow (E^{\mathbb{R}_+}, \mathcal{E}^{\mathbb{R}_+})$ be a map that is adapted to the filtration $(\mathcal{F}_t)_t$, with $\mathcal{F}_t \subset \mathcal{F}, t \geq 0$. X is a *Markov process*, if there exists a transition function $(P_t)_t$, such that for each distribution ν on $(E, \mathcal{E})$ there exists a probability distribution P_ν on $(\Omega, \mathcal{F})$ with

- (i) $P_\nu\{X_0 \in B\} = \nu(B)$ for every $B \in \mathcal{E}$;
- (ii) *Markov property*: for all $s, t \geq 0$ and $f \in b\mathcal{E}$:

$$\mathbb{E}_\nu[f(X_{t+s}) \mid \mathcal{F}_s] = P_t f(X_s) = \mathbb{E}_{X_s} f(X_t), \quad P_\nu\text{-a.s.}$$

LEMMA 3.7 (Characterisation Lemma).

Let X be an $(E, \mathcal{E})$ -valued stochastic process with transition function $(P_t)_{t \geq 0}$, adapted to the filtration $(\mathcal{F}_t)_t$. Let ν be a distribution on $(E, \mathcal{E})$. If X is Markov with initial distribution ν , then for all $0 = t_0 < t_1 < \dots < t_n$, and all functions $f_0, \dots, f_n \in b\mathcal{E}, n \in \mathbb{Z}^+$,

$$\mathbb{E}_\nu \prod_{i=0}^n f_i(X_{t_i}) = \nu f_0 P_{t_1-t_0} \dots P_{t_n-t_{n-1}} f_n. \quad (3.1)$$

Vice versa, suppose that (3.1) holds for all $0 = t_0 < t_1 < \dots < t_n$, and all functions $f_0, \dots, f_n \in b\mathcal{E}, n \in \mathbb{Z}^+$, then X is Markov with respect to the natural filtration $(\mathcal{F}_t^X)_t$.

Remark. The proof of Lemma 3.7 shows that it is sufficient to check (3.1) for indicator functions:

$$\mathbb{E}_\nu \prod_{i=0}^n \mathbb{1}_{A_i}(X_{t_i}) = \nu \mathbb{1}_{A_0} P_{t_1-t_0} \mathbb{1}_{A_1} \dots P_{t_n-t_{n-1}} \mathbb{1}_{A_n}$$

for all $0 < t_1 < \dots < t_n < s$, and $A_0, \dots, A_n \in \mathcal{E}$. Then

$$\prod_{i=0}^n \mathbb{1}_{A_i}(X_{t_i}) = \{X_{t_0} \in A_0, \dots, X_{t_n} \in A_n\}$$

for all $0 < t_1 < \dots < t_n < s, A_i \in \mathcal{E}, i = 0, \dots, n$, and $n = 0, \dots$ is a π -system generating $\mathcal{F}_t^X$.

Proof.

□

Lemma 3.7 means that

$$P_\nu(X_{t+s} \in B \mid \sigma(X_s)) = P_t(X_s, B).$$

And this is not necessary a Markov process.

EXAMPLE 3.8 (Not a Markov Process). Consider the following space

$$S = \{(1, 1, 1), (2, 2, 2), (3, 3, 3), (1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$$

with σ -algebra $\mathcal{S} = 2^S$. Putting $P(x) = \frac{1}{9}$ defines a probability measure on this space. Define a sequence of i.i.d. random vectors $Z_k = (X_{3k}, X_{3k+1}, X_{3k+2}, \dots), k = 0, \dots$ on $(S, \mathcal{S}, \mathbb{P})$. Then

the sequence $\{X_n\}_n$ is an $(E = \{1, 2, 3\}, \mathcal{E} = 2^E)$ -valued stochastic process on $(S, \mathcal{S}, \mathbb{P})$. Then

$$P(X_{n+1} = j \mid \sigma(X_n)) = \frac{1}{3}, \quad \forall j = \{1, 2, 3\},$$

meaning that the motion is determined by the 3×3 stochastic matrix with all elements equal to $\frac{1}{3}$. We have to check that

$$\mathbb{E} \mathbb{1}_j(X_{n+1}) \mathbb{1}_i(X_n) = \frac{1}{3} \mathbb{E} \mathbb{1}_i(X_n), \quad \forall i.$$

Then $\{X_n\}_n$ is a transition kernel since $P(X_2 = 1 \mid \sigma(X_0, X_1)) = f(X_0, X_1)$ with

$$f(X_0, X_1) = \begin{cases} 1, & (X_0, X_1) \in \{(1, 1), (2, 3), (3, 2)\} \\ 0, & \text{otherwise} \end{cases}.$$

Hence $f(1, 1) \neq f(2, 1)$, thus showing that the Markov property lacks.

EXAMPLE 3.9 (A BM Process). Let W be a standard BM on an underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X_0 be a measurable random variable with distribution $\nu = \delta_x$, for some $x \in \mathbb{R}$, independent of W . Define $X_t = X_0 + W_t, t \geq 0$. Then $X = (X_t)_t$ is a Markov process with initial distribution ν with respect to its natural filtration. Note that $X_t - X_s = W_t - W_s$ is independent of $\mathcal{F}_s^X$.

Proof.

□

EXAMPLE 3.10 (Ornstein-Uhlenbeck Process). Let W be a standard BM. Let $\alpha, \sigma^2 > 0$ and let X_0 be a $\mathbb{R}$ -valued random variable with distribution ν that is independent of $\sigma(W_t, t \geq 0)$. Define the scaled BM by

$$X_t = e^{-\alpha t} \left(X_0 + W_{\sigma^2 \left(\frac{e^{2\alpha t} - 1}{2\alpha} \right)} \right).$$

If $\nu = \delta_x$, $X = (X_t)_t$ a Markov process with the P_ν distribution of X_t a normal distribution with mean $e^{-\alpha t} x$ and variance $\frac{\sigma^2(1 - e^{-2\alpha t})}{2\alpha}$. Note that $X_t \xrightarrow{D} \mathcal{N}\left(0, \frac{\sigma^2}{2\alpha}\right)$. If $X_0 \stackrel{d}{=} \mathcal{N}\left(0, \frac{\sigma^2}{2\alpha}\right)$, then X_t is a Gaussian, Markov process with mean $m(t) = 0$ and covariance function $r(s, t) = \frac{\sigma^2 e^{-\alpha|t-s|}}{2\alpha}$.

EXAMPLE 3.11 (Geometric BM). Let W be a standard BM on an underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X_0 be a measurable positive random variable with distribution ν , independent of W . Let $\mu \in \mathbb{R}$ and $\sigma^2 \in (0, \infty)$ and define

$$X_t = X_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t}, \quad t \geq 0.$$

Then X is a Markov process. Depending on the value μ , it is a (super/sub) martingale. Geometric BM is used in financial mathematics to model stock prices.

LEMMA 3.12.

Let X be a Markov process with state space $(E, \mathcal{E})$, initial distribution ν and transition function $(P_t)_t$, defined on an underlying filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P_\nu)$. Suppose that $(E', \mathcal{E}')$ is a measurable space and let $\phi : E \rightarrow E'$ be measurable and onto. If $(Q_t)_t$ is a collection of transition kernels such that

$$P_t(f \circ \phi) = (Q_t f) \circ \phi$$

for all bounded, measurable functions f on E' , then $Y = \phi(X)$ is a Markov process with respect to $(\mathcal{F}_t)_{t \geq 0}$, with state space $(E', \mathcal{E}')$, initial measure ν' , with $\nu'(B') = \nu(\phi^{-1}(B'))$, $B' \in \mathcal{E}'$, and transition function (Q_t) .

Hint. Let f be a bounded, measurable function on E' . By assumption and the semi-group property of (P_t) ,

$$(Q_t Q_s f) \circ \phi = P_t((Q_s f) \circ \phi) = P_t P_s(f \circ \phi) = P_{t+s}(f \circ \phi) = (Q_{t+s} f) \circ \phi.$$

Since ϕ is onto, this implies that $(Q_t)_t$ is a semi-group. It is easily verified that Y has the Markov property.

Proof. Let f be a bounded, measurable function on E' . By assumption and the semi-group property of (P_t) ,

$$(Q_t Q_s f) \circ \phi = P_t((Q_s f) \circ \phi) = P_t P_s(f \circ \phi) = P_{t+s}(f \circ \phi) = (Q_{t+s} f) \circ \phi.$$

Since ϕ is onto, this implies that $(Q_t)_t$ is a semi-group.

For $Y = \phi(X)$ to be a Markov process with initial measure ν' and transition function (Q_t) , we need to show that for all $s, t \geq 0$,

$$\mathbb{E}_{\nu'}(f(Y_{t+s}) \mid \mathcal{F}_s) = Q_t f(Y_s), \quad P_{\nu}\text{-a.s.}$$

Given the initial distribution ν of X , the initial measure ν' of Y is defined by

$$\nu'(B') = \nu(\phi^{-1}(B')), \quad B' \in \mathcal{E}'.$$

For any $B' \in \mathcal{E}'$, we have

$$P_{\nu}(Y_0 \in B') = P_{\nu}(X_0 \in \phi^{-1}(B')) = \nu(\phi^{-1}(B')) = \nu'(B') = P_{\nu'}(Y_0 \in B').$$

Since the distributions ν and ν' are induced by the same probability distribution on the underlying probability space Ω , i.e. $P_{\nu} = P_{\nu'}$. Together with the definition and X is a Markov process, we can write:

$$\begin{aligned} \mathbb{E}_{\nu'}(f(Y_{t+s}) \mid \mathcal{F}_s) &= \mathbb{E}_{\nu}(f(Y_{t+s}) \mid \mathcal{F}_s) && (P_{\nu} = P_{\nu'}) \\ &= \mathbb{E}_{\nu}(f(\phi(X_{t+s})) \mid \mathcal{F}_s) && (Y = \phi(X)) \\ &= \mathbb{E}_{\nu}((f \circ \phi)(X_{t+s}) \mid \mathcal{F}_s) \\ &= P_t(f \circ \phi)(X_s) && (\text{Markov property of } X) \\ &= ((Q_t f) \circ \phi)(X_s) && (P_t(f \circ \phi) = (Q_t f) \circ \phi) \\ &= Q_t f(\phi(X_s)) = Q_t f(Y_s), \quad P_{\nu}\text{-a.s.} \end{aligned}$$

Hence, Y satisfies the Markov property with transition kernels $(Q_t)_t$, and therefore is a Markov process on the state space $(E', \mathcal{E}')$ with initial measure ν' . $\square$

EXAMPLE 3.13 (W_t^2 is a Markov Process).

SECTION 3.1

| Existence of a Canonical Version

For a given transition function $(P_t)_t$ and probability measure ν on a measurable space $(E, \mathcal{E})$, we can construct a so-called canonical Markov process X which has initial distribution ν and transition function $(P_t)_t$.

An E -valued process can be viewed as a random variable of the space $E^{\mathbb{R}^+}$ of E -valued functions f on $\mathbb{R}^+$, or of a subspace $\Gamma \subset E^{\mathbb{R}^+}$ if X is known to have more structure. The σ -algebra $\Gamma \cap \mathcal{E}^{\mathbb{R}^+}$ is the smallest σ -algebra that makes all projections $f \rightarrow f(t)$ measurable.

Let $\Omega = \Gamma$ and $\mathcal{F} = \Gamma \cap \mathcal{E}^{\mathbb{R}^+}$. Consider the process $X = (X_t)_{t \geq 0}$ defined as the identity map $X(\omega) = \omega$, so that $X_t(\omega) = \omega_t$ is projection on the t -th coordinate.

The *Kolmogorov consistency theorem* validates the existence of a process on $(\Omega, \mathcal{F})$ with given *fdd's*. Hence, we have to specify appropriate *fdd's* based on the given transition function $(P_t)_t$ and initial distribution ν .

COROLLARY 3.14.

Let $(P_t)_t$ be a transition function and let ν be a probability measure on $(E, \mathcal{E})$. Then there exists a unique probability measure P_ν on $(\Omega, \mathcal{F})$ such that under P_ν , the canonical process X is a Markov process with initial distribution ν and transition function $(P_t)_t$ with respect to its natural filtration $(\mathcal{F}_t^X)_t$.

Proof.

□

LEMMA 3.15.

Let Z be an $\mathcal{F}_\infty^X$ -measurable random variable, that is either non-negative or bounded. Then the map $x \rightarrow \mathbb{E}_x Z$ is $\mathcal{E}/\mathcal{B}$ -measurable and for every initial distribution ν ,

$$\mathbb{E}_\nu Z = \int_x \mathbb{E}_x Z \nu(dx).$$

This lemma allows to formulate a more general version of the Markov property. For any $t \geq 0$ we define the *translation or shift operator* $\theta_t : E^{\mathbb{R}^+} \rightarrow E^{\mathbb{R}^+}$ by

$$(\theta_t \omega)_s = \omega_{t+s}, \quad s \geq 0, \quad \omega \in E^{\mathbb{R}^+}.$$

So θ_t just cuts off the part of ω before time t and shifts the remainder to the origin. Clearly $\theta_t \circ \theta_s = \theta_{t+s}$.

THEOREM 3.16 (Generalised Markov Property for Canonical Process).

Assume that X is a canonical Markov process with respect to a filtration $(\mathcal{F}_t)_t$. Let Z be an $\mathcal{F}_\infty^X$ -measurable random variable, non-negative or bounded. Then for every $t > 0$ and any initial distribution ν :

$$\mathbb{E}_\nu[Z \circ \theta_t \mid \mathcal{F}_t] = \mathbb{E}_{X_t} Z, \quad P_\nu\text{-a.s.}$$

SECTION 3.2

| Strong Markov Property

DEFINITION 3.17 (Strong Markov Property).

X is said to have the *strong Markov property* if for every function $f \in b\mathcal{E}$, any adapted stopping time σ and any initial distribution ν and any $t \geq 0$:

$$\mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_\nu[f(X_{\sigma+t}) \mid \mathcal{F}_\sigma] = \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_{X_\sigma} f(X_t), \quad \mathbb{P}_\nu\text{-a.s.} \quad (3.2)$$

LEMMA 3.18.

Let X be an $(E, \mathcal{E})$ -valued Markov process. Then (3.2) holds for any function $f \in b\mathcal{E}$, any initial distribution ν and any stopping time σ , for which there exists a countable subset $S \subset [0, \infty)$ such that $\sigma \in S \cup \{\infty\}$.

Proof. □

COROLLARY 3.19.

Any discrete time Markov chain has the strong Markov property.

THEOREM 3.20.

Let X be an $(E, \mathcal{E})$ -valued Markov process adapted to the filtration $(\mathcal{F}_t)_t$ with E a Polish space and $\mathcal{E}$ the Borel- σ -algebra, and with right-continuous paths. Suppose that $x \mapsto \mathbb{E}_x f(X_s) = P_s f(x)$ is continuous, or, more generally, that $t \mapsto \mathbb{E}_{X_t} f(X_s)$ is right-continuous (everywhere) for each bounded continuous function f . Then the strong Markov property holds.

Proof. □

Sometimes the strong Markov property is defined through optional times. In other words, the strong Markov property is defined to hold if for every function $f \in b\mathcal{E}$, any adapted optional time σ and any initial distribution ν and any $t \geq 0$:

$$\mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_\nu[f(X_{\sigma+t}) \mid \mathcal{F}_{\sigma^+}] = \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_{X_\sigma} f(X_t), \quad \mathbb{P}_\nu\text{-a.s.} \quad (3.3)$$

COROLLARY 3.21.

Assume the conditions of Theorem 3.20. Then

- (i) X is Markov with respect to $(\mathcal{F}_{t^+})_t$;
- (ii) Let τ be an $(\mathcal{F}_t)_t$ -optional time. Then (3.3) holds for any function $f \in b\mathcal{E}$;
- (iii) Let τ be a finite adapted optional time. Let Y be a bounded $\mathcal{F}_{\tau^+}$ -measurable random variable. Then

$$\mathbb{E}_\nu Y f(X_{\tau+t}) = \mathbb{E}_\nu [Y \mathbb{E}_{X_\tau} f(X_t)].$$

Proof. (i) We want to show that X is Markov with respect to $(\mathcal{F}_{t^+})_t$, i.e.,

$$\mathbb{E}_\nu[f(X_{t+s}) \mid \mathcal{F}_{s^+}] = \mathbb{E}_{X_s} f(X_t).$$

By the strong Markov property of X , for stopping time $s < \infty$, we have

$$\mathbb{E}_\nu[f(X_{s+t}) \mid \mathcal{F}_s] = \mathbb{E}_{X_s} f(X_t), \quad \mathbb{P}_\nu\text{-a.s.}$$

Note that the filtration $\mathcal{F}_{t^+}$ is defined by

$$\mathcal{F}_{t^+} = \bigcap_{n=1}^{\infty} \mathcal{F}_{t+\frac{1}{n}},$$

so we have $\mathcal{F}_t \subseteq \mathcal{F}_{t^+}$. Then by the tower property, it also holds

$$\begin{aligned} \mathbb{E}_v[f(X_{s+t}) \mid \mathcal{F}_s] &= \mathbb{E}_v[\mathbb{E}_v[f(X_{s+t}) \mid \mathcal{F}_{s^+}] \mid \mathcal{F}_s] \\ &= \mathbb{E}_v[f(X_{s+t}) \mid \mathcal{F}_{s^+}]. \end{aligned}$$

Hence X is Markov adapted with respect to the filtration $(\mathcal{F}_{t^+})_t$.

- (ii) Since τ is an $(\mathcal{F}_t)_t$ -optional time, which means for any $t \geq 0$, the set $\{\tau \leq t\}$ is $\mathcal{F}_t$ -measurable. From $\mathcal{F}_t \subseteq \mathcal{F}_{t^+}$, it's also $\mathcal{F}_{t^+}$ -measurable, which implies τ is a $(\mathcal{F}_{t^+})_t$ -stopping time. This fact, together with part (a), allows us to apply [Theorem 3.20](#) on X with filtration $(\mathcal{F}_{t^+})_t$ to get the desired result.
- (iii) Since $\tau < \infty$ by assumption, the previous two parts give us:

$$\mathbb{E}_v[f(X_{\tau+t}) \mid \mathcal{F}_{\tau^+}] = \mathbb{E}_{X_\tau}[f(X_t)], \quad P_v\text{-a.s.}$$

Then similarly, we have

$$\begin{aligned} \mathbb{E}_v[Yf(X_{\tau+t})] &= \mathbb{E}_v[Y\mathbb{E}_v[f(X_{\tau+t}) \mid \mathcal{F}_{\tau^+}]] \\ &= \mathbb{E}_v[Y\mathbb{E}_{X_\tau}[f(X_t)]] \end{aligned}$$

as required. □

LEMMA 3.22.

Assume the conditions of [Theorem 3.20](#).

- (i) **Blumenthal's 0-1 Law.** If $A \in \mathcal{F}_{0+}^X$ then $P_x(A) = 0$ or 1 for all $x \in E$.
- (ii) If τ is an $(\mathcal{F}_t^X)_t$ -optional time, then $P_x\{\tau = 0\} = 0$ or 1 , for all $x \in E$.

Proof. (i) Notice the identity

$$\mathbb{E}[\mathbb{1}_A g(B_{t_1}, B_{t_2}, \dots, B_{t_n})] = P(A) \mathbb{E}[g(B_{t_1}, B_{t_2}, \dots, B_{t_n})],$$

where $A \in \mathcal{F}_{0+}^X$ and g is a continuous bounded function. Then for any open set $U \subseteq \mathbb{R}^n$, there exists a sequence of continuous bounded functions $(g_k)_{k \in \mathbb{N}}$ such that $g_k \uparrow \mathbb{1}_U$. By the *monotone convergence theorem*, we have

$$\begin{aligned} \mathbb{E}[\mathbb{1}_A \mathbb{1}_U g(B_{t_1}, B_{t_2}, \dots, B_{t_n})] &= \sup_{k \geq 1} \mathbb{E}[\mathbb{1}_A g_k(B_{t_1}, B_{t_2}, \dots, B_{t_n})] \\ &= P(A) \sup_{k \geq 1} \mathbb{E}[g_k(B_{t_1}, B_{t_2}, \dots, B_{t_n})] \\ &= P(A) \mathbb{E}[\mathbb{1}_U(B_{t_1}, B_{t_2}, \dots, B_{t_n})]. \end{aligned}$$

Open sets form a generator of the Borel σ -algebra $\mathcal{B}(\mathbb{R}^n)$, by the *monotone class theorem*, the result extends to all Borel sets $F \in \mathcal{B}(\mathbb{R}^n)$ and we get

$$\mathbb{E}[\mathbb{1}_A \mathbb{1}_F g(B_{t_1}, B_{t_2}, \dots, B_{t_n})] = P(A) \mathbb{E}[\mathbb{1}_F(B_{t_1}, B_{t_2}, \dots, B_{t_n})].$$

Thus, for any $A \in \mathcal{F}_{0+}^X$ and $F \in \mathcal{B}(\mathbb{R}^n)$, we have shown that:

$$P_x(A \cap \{(B_{t_1}, B_{t_2}, \dots, B_{t_n}) \in F\}) = P_x(A)P_x((B_{t_1}, B_{t_2}, \dots, B_{t_n}) \in F).$$

Let $F = \mathbb{R}^n$, then

$$P_x(A) = P_x(A)P_x((B_{t_1}, B_{t_2}, \dots, B_{t_n}) \in F),$$

where $P_x((B_{t_1}, B_{t_2}, \dots, B_{t_n}) \in F) = 1$. Therefore $P_x(A)$ must be 0 or 1, and this completes the proof of *Blumenthal's 0-1 Law*.

(ii) Since

$$\{\tau = 0\} = \bigcap_{s>0} \{\tau < s\} \in \mathcal{F}_t, \quad \forall t > 0,$$

we know that $\{\tau = 0\} \in \mathcal{F}_{0+}$. By *Blumenthal's 0-1 Law*, for any event $A \in \mathcal{F}_{0+}$, the probability $P_x(A)$ is either 0 or 1 for all $x \in E$. Therefore, $P_x\{\tau = 0\} = 0$ or 1, for all $x \in E$. $\square$

LEMMA 3.23.

Let X be a canonical, right-continuous Markov process with values in a Polish state space E , equipped with the Borel σ -algebra $\mathcal{E}$. Define

$$S = \inf \{t > 0 \mid X_t \neq X_0\}.$$

This is the 'sojourn time' in state x , when $X_0 = x$.

(i) Then for each $x \in E$, there exists $\alpha_x \in [0, \infty]$, such that

$$\mathbb{P}_x(S > t) = \mathbb{P}_x(S \geq t) = e^{-\alpha_x t}, \quad t \geq 0.$$

If $\alpha_x = 0$, then x is called an absorption point or a trap. If $\alpha_x \in (0, \infty)$, then x is called a holding point, and if $\alpha_x = \infty$, then x is a regular point.

(ii) Suppose that E only contains holding and absorption points. Then, there exists a transition kernel q on $E \times \mathcal{E}$, such that

$$\mathbb{P}_x\{X_S \in B, t < S < \infty\} = e^{-\alpha_x t} q(x, B),$$

where x is a holding point and $B \in \mathcal{E}$.

Proof. $\square$

LEMMA 3.24.

Let X be a canonical Markov process. Then

$$\mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_v [Z \circ \theta_\sigma \mid \mathcal{F}_\sigma^X] = \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_{X_\sigma} Z, \quad \mathbb{P}_v\text{-a.s.} \quad (3.4)$$

for any bounded or non-negative $\mathcal{F}_\infty^X$ -measurable random variable Z , any initial distribution v , and any stopping time σ , for which there exists a countable subset $S \subset [0, \infty)$ such that $\sigma \in S \cup \{\infty\}$.

THEOREM 3.25.

Let X be an $(E, \mathcal{E})$ -valued canonical Markov process with E a Polish space and $\mathcal{E}$ the Borel- σ -algebra, and with right-continuous paths. Suppose that $x \mapsto \mathbb{E}_x f(X_s) = P_s f(x)$ is continuous, or, more generally, $t \mapsto \mathbb{E}_{X_t} f(X_s)$ is right-continuous everywhere for each bounded continuous function f . Then (3.4) holds for any bounded or non-negative $\mathcal{F}_\infty^X$ -measurable random variable Z , any initial distribution v and any stopping time σ .

Proof. Monotone Class Theorem (Halmos). Let $\mathcal{H}$ be a class of bounded functions from a set $\mathcal{S}$ to $\mathbb{R}$, satisfying the following conditions:

- (i) $\mathcal{H}$ is a vector space over $\mathbb{R}$.
- (ii) If $f_n, n = 1, \dots$, is a sequence of non-negative functions in $\mathcal{H}$, such that $f_n \uparrow f$ for a bounded function f , then $f \in \mathcal{H}$.
- (iii) The constant function belongs to $\mathcal{H}$.

If $\mathcal{H}$ contains the indicator function of every set in a π -system $\mathcal{I}$, then $\mathcal{H}$ contains every bounded $\sigma(\mathcal{I})$ -measurable function.

The shift operator θ_σ is given by

$$(\theta_\sigma \omega)_t = \omega_{t+\sigma}, \quad t \geq 0.$$

We write $Z(\omega) := f(X_{t_1}(\omega), X_{t_2}(\omega), \dots, X_{t_n}(\omega))$, then

$$\begin{aligned} \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_v[Z \circ \theta_\sigma \mid \mathcal{F}_\sigma] &= \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_v[f(X_{t_1+\sigma}, X_{t_2+\sigma}, \dots, X_{t_n+\sigma}) \mid \mathcal{F}_\sigma] \\ &= \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_{X_\sigma}[f(X_{t_1}(\omega), X_{t_2}(\omega), \dots, X_{t_n}(\omega))] \end{aligned} \quad (\text{strong Markov property})$$

for all such functions Z . Let

$$\mathcal{H} := \{Z : \Omega \rightarrow \mathbb{R} \mid Z \text{ is bounded or non-negative } \mathcal{F}_\infty^X\text{-measurable, and (3.4) holds true}\},$$

which is a π -system that generates the σ -algebra $\mathcal{F}_\infty^X$. We verify:

- (i) For $Z_1, Z_2 \in \mathcal{H}$, and any scalar c , we have

$$\mathbb{E}_v[(Z_1 + \alpha Z_2) \circ \theta_\sigma \mid \mathcal{F}_\sigma] = \mathbb{E}_{X_\sigma}[Z_1 + \alpha Z_2],$$

then $\mathcal{H}$ is a vector space.

- (ii) Suppose $(Z_n)_n$ is an increasing sequence of functions in $\mathcal{H}$ and $Z = \lim_{n \rightarrow \infty} Z_n$. Then by the *monotone convergence theorem*,

$$\mathbb{E}_v[Z \circ \theta_\sigma \mid \mathcal{F}_\sigma] = \mathbb{E}_v \left[\lim_{n \rightarrow \infty} Z_n \circ \theta_\sigma \mid \mathcal{F}_\sigma \right] = \lim_{n \rightarrow \infty} \mathbb{E}_v[Z_n \circ \theta_\sigma \mid \mathcal{F}_\sigma],$$

$$\mathbb{E}_{X_\sigma} Z = \mathbb{E}_{X_\sigma} \left[\lim_{n \rightarrow \infty} Z_n \right] = \lim_{n \rightarrow \infty} \mathbb{E}_{X_\sigma} [Z_n].$$

Therefore we can pass the limit from

$$\mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_v[Z_n \circ \theta_\sigma \mid \mathcal{F}_\sigma] = \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_{X_\sigma} Z_n, \quad \mathbb{P}_v\text{-a.s.},$$

and show (3.4). Hence $\mathcal{H}$ is a monotone class.

- (iii) For any constant function $Z = c$, we have

$$\mathbb{E}_v[c \mid \mathcal{F}_\sigma] = c = \mathbb{E}_{X_\sigma}[c],$$

then the constant function belongs to $\mathcal{H}$.

By the *monotone class theorem*, every bounded or non-negative $\mathcal{F}_\infty^X$ -measurable functions belongs to $\mathcal{H}$, which completes the proof. $\square$

LEMMA 3.26.

Let E be a separable Banach space. Let X be a right-continuous, canonical process with values in $(E, \mathcal{B}(E))$, defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P})$. Suppose that X has stationary, independent increments.

- (i) X is a Markov process with initial distribution $\mathbb{P}\{X_0 \in \cdot\}$.
- (ii) Let τ be a finite $(\mathcal{F}_t)$ -stopping time. Then the process $X(\tau) = (X_{\tau+t} - X_\tau)_{t \geq 0}$ is independent of $\mathcal{F}_\tau$. It is a Markov process, adapted to the filtration $(\mathcal{F}_{\tau+t})_t$, and to $(\mathcal{F}_t^{X(\tau)})_t$. The distribution $\mathbb{P}_\tau$ of $X(\tau)$ is the same as the distribution of $X - X_0$ under $\mathbb{P}_0$.

Proof.

$\square$

SECTION 3.3

| Applications to Brownian Motion

THEOREM 3.27 (Reflection Principle).

Let W be a BM with continuous paths. Let $x \in \mathbb{R}$ be given. Define the process W' by

$$W'_t = \begin{cases} W_t, & t \leq \tau_x \leq \infty \\ 2x - W_t, & t > \tau_x \end{cases}.$$

Then W' is a standard BM with continuous paths.

Proof. □

COROLLARY 3.28.

Let W be a standard BM and S its running maximum. Then

$$P\{W_t \leq x, S_t \geq y\} = P\{W_t \leq x - 2y\}, \quad x \leq y.$$

The pair (W_t, S_t) has joint density

$$(x, y) \mapsto \frac{(2y - x)e^{-\frac{(2y-x)^2}{2t}}}{\sqrt{\frac{\pi t^3}{2}}} \mathbb{1}_{\{x \leq y\}},$$

with respect to the Lebesgue measure.

| Exercises

EXERCISE 3.1. Consider the Ornstein Uhlenbeck process in [Example 3.10](#).

- (i) Show that the defined process is a Markov process which converges in distribution to an $\mathcal{N}\left(0, \frac{\sigma^2}{2\alpha}\right)$ distributed random variable.
- (ii) If $X_0 \stackrel{d}{=} \mathcal{N}\left(0, \frac{\sigma^2}{2\alpha}\right)$, show that $X_t \stackrel{d}{=} \mathcal{N}\left(0, \frac{\sigma^2}{2\alpha}\right)$ (in other words: the $\mathcal{N}\left(0, \frac{\sigma^2}{2\alpha}\right)$ distribution is an *invariant distribution* for the Markov process).
- (iii) Show that X_t is a Gaussian process with the given mean and covariance functions.

SOLUTION. (i) By definition, we have

$$X_t = e^{-\alpha t} \left(X_0 + W_{\sigma^2 \left(\frac{e^{2\alpha t} - 1}{2\alpha} \right)} \right) = e^{-\alpha t} X_0 + e^{-\alpha t} W_{\sigma^2 \left(\frac{e^{2\alpha t} - 1}{2\alpha} \right)}. \quad (3.5)$$

Let

$$f(t) = \sigma^2 \left(\frac{e^{2\alpha t} - 1}{2\alpha} \right),$$

we calculate the derivative

$$f'(t) = \sigma^2 e^{2\alpha t}.$$

By Itô's lemma, the differential of $W_{f(t)}$ can be represented as

$$dW_{f(t)} = \sqrt{f'(t)} dW_t = \sigma e^{\alpha t} dW_t.$$

Then we have

$$W_{\sigma^2 \left(\frac{e^{2\alpha t} - 1}{2\alpha} \right)} = W_{f(t)} = \sigma \int_0^t e^{\alpha s} dW_s. \quad (3.6)$$

Plugging (3.6) into (3.5), we get

$$X_t = e^{-\alpha t} \left(X_0 + W_{\sigma^2 \left(\frac{e^{2\alpha t} - 1}{2\alpha} \right)} \right) = e^{-\alpha t} X_0 + \sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW_s.$$

To show the Markov property, we write

$$\begin{aligned} X_{t+s} &= e^{-\alpha(t+s)} X_0 + \sigma e^{-\alpha(t+s)} \int_0^{t+s} e^{\alpha u} dW_u \\ &= e^{-\alpha(t+s)} X_0 + \sigma e^{-\alpha(t+s)} \left(\int_0^t e^{\alpha u} dW_u + \int_t^{t+s} e^{\alpha u} dW_u \right) \quad (\text{substitute } u = t + s) \\ &= e^{-\alpha s} X_t + \sigma e^{-\alpha(t+s)} \int_t^{t+s} e^{\alpha u} dW_u. \end{aligned}$$

Since the increment $\int_t^{t+s} e^{\alpha u} dW_u$ is independent of the filtration $\mathcal{F}_t$, we have shown that X_{t+s} depends only on the current state X_t , which proves that it is a Markov process. Then we calculate the mean and variance of X_t :

$$\begin{aligned} \mathbb{E}[X_t] &= e^{-\alpha t} \mathbb{E}[X_0] = e^{-\alpha t} x \xrightarrow{t \rightarrow \infty} 0, \\ \text{Var}(X_t) &= \mathbb{E}[(X_t - \mathbb{E}X_t)^2] \\ &= \sigma^2 e^{-2\alpha t} \int_0^t e^{2\alpha s} ds \\ &= \frac{\sigma^2 (1 - e^{-2\alpha t})}{2\alpha} \xrightarrow{t \rightarrow \infty} \frac{\sigma^2}{2\alpha}. \end{aligned}$$

Therefore, we can conclude that $X_t \xrightarrow{D} N\left(0, \frac{\sigma^2}{2\alpha}\right)$.

(ii) Given that $X_0 \stackrel{d}{=} N\left(0, \frac{\sigma^2}{2\alpha}\right)$, and

$$X_t = e^{-\alpha t} X_0 + \sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW_s.$$

Again, we calculate the mean and variance:

$$\begin{aligned} \mathbb{E}[X_t] &= e^{-\alpha t} \cdot 0 = 0, \\ \text{Var}(X_t) &= \text{Var}(e^{-\alpha t} X_0) + \text{Var}\left(\sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW_s\right) \\ &= e^{-2\alpha t} \frac{\sigma^2}{2\alpha} + \frac{\sigma^2 (1 - e^{-2\alpha t})}{2\alpha} \\ &= \frac{\sigma^2}{2\alpha}. \end{aligned}$$

Thus, $X_t \stackrel{d}{=} N\left(0, \frac{\sigma^2}{2\alpha}\right)$, confirming that $N\left(0, \frac{\sigma^2}{2\alpha}\right)$ is an invariant distribution for the Markov process X_t .

(iii) Given that

$$X_t = e^{-\alpha t} X_0 + \sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW_s.$$

We calculate its mean function

$$m(t) = \mathbb{E}[X_t] = 0,$$

and the covariance function

$$\begin{aligned}
 r(s, t) &= \mathbb{E}[X_s X_t] \\
 &= \mathbb{E}\left[\left(e^{-\alpha s} X_0 + \sigma e^{-\alpha s} \int_0^s e^{\alpha u} dW_u\right)\left(e^{-\alpha t} X_0 + \sigma e^{-\alpha t} \int_0^t e^{\alpha u} dW_u\right)\right] \\
 &= e^{-\alpha(s+t)} \mathbb{E}[X_0^2] + \sigma^2 e^{-\alpha(s+t)} \int_0^{s \wedge t} e^{2\alpha u} du \\
 &= e^{-\alpha(s+t)} \frac{\sigma^2}{2\alpha} + \frac{\sigma^2 e^{-\alpha(s+t)}}{2\alpha} (e^{2\alpha(s \wedge t)} - 1) \\
 &= e^{-\alpha(s+t)} \frac{\sigma^2}{2\alpha} + \frac{\sigma^2}{2\alpha} (e^{-\alpha|s-t|} - e^{-\alpha(s+t)}) = \frac{\sigma^2 e^{-\alpha|t-s|}}{2\alpha}.
 \end{aligned}$$

Therefore, X_t is a Gaussian process with the given mean and covariance functions. $\square$

EXERCISE 3.2 (Geometric Brownian Motion: Markov and Martingale Properties). Consider geometric BM in [Example 3.11](#). Show that it is a Markov process. Characterise the values of μ for which it is a super-, a sub-, or a mere martingale. Compute the transition function.

SOLUTION.

(i) Given that

$$X_t = X_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right),$$

we have

$$\begin{aligned}
 X_{t+s} &= X_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)(t+s) + \sigma W_{t+s}\right) \\
 &= X_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)s + \sigma W_s\right) \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma(W_{t+s} - W_s)\right) \\
 &= X_s \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma(W_{t+s} - W_s)\right),
 \end{aligned}$$

where $W_{t+s} - W_s \sim \mathcal{N}(0, t)$ and independent of $\mathcal{F}_s$. Let

$$Y := \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma(W_{t+s} - W_s)\right).$$

– By [Definition 3.5](#), we verify the Markov property:

$$\mathbb{E}[f(X_{t+s}) \mid \mathcal{F}_s] = \mathbb{E}[f(X_s Y) \mid \mathcal{F}_s] = g(X_s),$$

where

$$\begin{aligned}
 g(x) &= \mathbb{E}[f(xY)] \\
 &= \mathbb{E}\left[f\left(x \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma(W_{t+s} - W_s)\right)\right)\right] \\
 &= \mathbb{E}\left[f\left(x \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma\sqrt{t}Z\right)\right)\right], \quad Z \sim \mathcal{N}(0, 1).
 \end{aligned}$$

Therefore, X is a Markov process.

– We calculate

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = \mathbb{E}\left[X_s \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)(t-s) + \sigma(W_t - W_s)\right) \mid \mathcal{F}_s\right]$$

$$\begin{aligned}
&= X_s \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) (t - s) \right) \mathbb{E}[\exp(\sigma(W_t - W_s))] \\
&= X_s \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) (t - s) \right) \exp \left(\frac{1}{2} \sigma^2 (t - s) \right) = X_s e^{\mu(t-s)}.
\end{aligned}$$

Then if $\mu > 0$, $e^{\mu(t-s)} > 1$ and $\mathbb{E}[X_t | \mathcal{F}_s] > X_s$, X is a submartingale; if $\mu = 0$, $e^{\mu(t-s)} = 1$ and X is a martingale; if $\mu < 0$, $e^{\mu(t-s)} < 1$ and $\mathbb{E}[X_t | \mathcal{F}_s] < X_s$, X is a supermartingale.

- By

$$X_{t+s} = X_s \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma(W_{t+s} - W_s) \right),$$

condition on $X_s = x$, we have

$$\ln X_{t+s} = \ln x + \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma(W_{t+s} - W_s).$$

Then

$$\ln \frac{X_{t+s}}{x} \sim \mathcal{N} \left(\left(\mu - \frac{\sigma^2}{2} \right) t, \sigma^2 t \right).$$

Let $X_{t+s} = y$, we have

$$\ln \frac{y}{x} = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma z,$$

solving for z gives

$$z = \frac{1}{\sigma} \left(\ln \frac{y}{x} - \left(\mu - \frac{\sigma^2}{2} \right) t \right).$$

Then the density of the random variable

$$Z = \frac{1}{\sigma} \left(\ln \frac{y}{x} - \left(\mu - \frac{\sigma^2}{2} \right) t \right) \sim \mathcal{N}(0, t)$$

is

$$\varphi(z) = \frac{1}{\sqrt{2\pi t}} \exp \left(-\frac{z^2}{2t} \right).$$

Observe

$$y = x \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma z \right).$$

Differentiate with respect to z :

$$\frac{dy}{dz} = \sigma x \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma z \right) = \sigma y.$$

Then the transition density

$$\begin{aligned}
p(t, x, y) &= \varphi(z) \left| \frac{dz}{dy} \right| \\
&= \varphi(z) \frac{1}{\sigma y} \\
&= \frac{1}{\sqrt{2\pi t}} \exp \left(-\frac{1}{2t} \left(\frac{\ln \frac{y}{x} - \left(\mu - \frac{\sigma^2}{2} \right) t}{\sigma} \right)^2 \right) \frac{1}{\sigma y} \\
&= \frac{1}{y \sigma \sqrt{2\pi t}} \exp \left(-\frac{1}{2\sigma^2 t} \left(\ln \frac{y}{x} - \left(\mu - \frac{\sigma^2}{2} \right) t \right)^2 \right).
\end{aligned}$$

The transition function is defined for $x > 0$ and any Borel set $B \subset (0, \infty)$ by

$$P_t(x, B) = \mathbb{P}(X_{t+s} \in B \mid X_s = x) = \int_B p(t, x, y) dy.$$

□

EXERCISE 3.3. Let X be a Markov process with state space E and transition function $(P_t)_{t \geq 0}$. Show that for every bounded, measurable function f on E and for all $t \geq 0$, the process $(P_{t-s}f(X_s))_{s \in [0, t]}$ is a martingale.

SOLUTION. First, we verify that $(P_{t-s}f(X_s))_{s \in [0, t]}$ is a stochastic process, i.e., for any fixed $s \in [0, t]$, $\omega \rightarrow P_{t-s}f(X_s(\omega))$ is a random variable. Note that for any $A \in \mathcal{E}$ and s , $\omega \rightarrow P_{t-s}(X_s(\omega), A)$ is a composition of 2 random variables, and therefore a random variable itself.

Let $f = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$ be a simple function. Then,

$$P_{t-s}f(X_s(\omega)) = \int_E f(x) P_{t-s}(X_s(\omega), dx) = \sum_{i=1}^n \alpha_i P_{t-s}(X_s(\omega), A_i).$$

This is also measurable as a linear combination of measurable functions. Any non-negative measurable function f can be represented as an increasing limit of non-negative simple functions $f = \lim_{n \rightarrow \infty} f_n$. For such f we have:

$$P_{t-s}f(X_s(\omega)) = \int_E f(x) P_{t-s}(X_s(\omega), dx) = \lim_{n \rightarrow \infty} \int_E f_n(x) P_{t-s}(X_s(\omega), dx) \quad (\text{Monotone convergence theorem})$$

which is again measurable, as it is a limit of measurable functions.

Finally, if f is any bounded measurable function, it can be decomposed into the difference of two non-negative bounded measurable functions $f = f_+ - f_-$, and the measurability of $P_{t-s}f(X_s)$ follows from the linearity of integration and the previous discussion. We conclude that $(P_{t-s}f(X_s))_{s \in [0, t]}$ is indeed a stochastic process for any bounded measurable function f .

To show that $(P_{t-s}f(X_s))_{s \in [0, t]}$ is a martingale, we have to verify that for all $0 \leq u \leq s < t$, we have

$$\mathbb{E}(P_{t-s}f(X_s) \mid \mathcal{F}_u) = P_{t-u}f(X_u).$$

Since the transition function

$$P_{t-s}f(X_s) = \mathbb{E}(f(X_{s+(t-s)}) \mid X_s) = \mathbb{E}(f(X_t) \mid X_s),$$

we can write:

$$\mathbb{E}(P_{t-s}f(X_s) \mid \mathcal{F}_u) = \mathbb{E}(\mathbb{E}(f(X_t) \mid X_s) \mid \mathcal{F}_u) = \mathbb{E}(f(X_t) \mid \mathcal{F}_u) = P_{t-u}f(X_u). \quad (\text{tower and Markov properties})$$

And $(P_{t-s}f(X_s))$ has finite expectation because it is a bounded random variable:

$$|P_{t-s}f(X_s)| \leq \int_E |f(x)| P_{t-s}(X_s, dx) \leq \int_E \|f\|_\infty P_{t-s}(X_s, dx) = \|f\|_\infty.$$

Therefore, the process $(P_{t-s}f(X_s))_{s \in [0, t]}$ is indeed a martingale. □

EXERCISE 3.4. (i) Derive the expression for the joint density of BM and its running maximum given in [Corollary 3.28](#).

(ii) Let W be a standard BM and S_t its running maximum. Show that for all $t \geq 0$ and $x > 0$:

$$P\{S_t \geq x\} = P\{\tau_x \leq t\} = 2P\{W_t \geq x\} = P\{|W_t| \geq x\}.$$

SOLUTION. (i) Let W_t be a standard BM, and $S_t = \sup_{s \leq t} W_s$ be the running maximum of W_t . Consider the event $\{W_t \leq x, S_t \geq y\}$, by the reflection principle (details in the first part of the proof of [Corollary 3.28](#) of LN), we get

$$P\{W_t \leq x, S_t \geq y\} = P\{W_t \leq x - 2y\}.$$

Then for $x \leq y$, the joint density function of W_t and S_t is given by:

$$\begin{aligned} \frac{\partial^2}{\partial x \partial y} P\{W_t \leq x, S_t \leq y\} &= \frac{\partial^2}{\partial x \partial y} (P\{W_t \leq x\} - P\{W_t \leq x - 2y\}) \\ &= \frac{\partial^2}{\partial x \partial y} \left(\int_{-\infty}^x \frac{1}{\sqrt{2\pi t}} e^{-\frac{w^2}{2t}} dw - \int_{-\infty}^{x-2y} \frac{1}{\sqrt{2\pi t}} e^{-\frac{w^2}{2t}} dw \right) \\ &= \frac{\partial}{\partial y} \left(\frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} - \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-2y)^2}{2t}} \right) \\ &= -\frac{1}{\sqrt{2\pi t}} \cdot \frac{\partial}{\partial y} \left(e^{-\frac{(x-2y)^2}{2t}} \right) \\ &= -\frac{1}{\sqrt{2\pi t}} \cdot \frac{2(x-2y)}{t} e^{-\frac{(x-2y)^2}{2t}} = \frac{(2y-x)e^{-\frac{(2y-x)^2}{2t}}}{\sqrt{\frac{\pi t^3}{2}}} \mathbb{1}_{\{x \leq y\}}. \end{aligned}$$

(ii) We show the expression

$$P\{S_t \geq x\} \stackrel{(a)}{=} P\{\tau_x \leq t\} \stackrel{(b)}{=} 2P\{W_t \geq x\} \stackrel{(c)}{=} P\{|W_t| \geq x\}$$

in the following parts. The first hitting time of x by W_t is defined as

$$\tau_x = \inf \{s \geq 0 \mid W_s = x\},$$

and $S_t = \sup_{s \leq t} W_s$ is the running maximum of W_t .

- If $S_t \geq x$, then $\sup_{s \leq t} W_s \geq x$, this implies that $W_s = x$ for some $s \leq t$, then the first hitting time τ_x must be less than or equal to t , and hence $\{S_t \geq x\} \subseteq \{\tau_x \leq t\}$. If $\tau_x \leq t$, there exists some $s \leq t$ such that $W_s = x$ satisfying $S_t \geq x$, and hence $\{\tau_x \leq t\} \subseteq \{S_t \geq x\}$. Therefore, $P\{S_t \geq x\} = P\{\tau_x \leq t\}$ is proved.

- By the symmetry of BM, we get $P(W_t \geq x) = P(W_t \leq -x)$. By the reflection principle, we have

$$P(\tau_x \leq t) = P(S_t \geq x) = P\left(\sup_{s \leq t} W_s \geq x\right) = P(W_t \geq x) + P(W_t \leq -x) = 2P(W_t \geq x).$$

where the event that W_t reaches x and event that W_t reaches $-x$ are equally likely. Therefore, $P\{\tau_x \leq t\} = 2P\{W_t \geq x\}$ is proved.

- By the symmetry of BM, we get $P(W_t \geq x) = P(W_t \leq -x)$. Then

$$P(|W_t| \geq x) = P(W_t \geq x) + P(W_t \leq -x) = 2P(W_t \geq x).$$

where the event that W_t reaches x and event that W_t reaches $-x$ are equally likely. Therefore, $2P\{W_t \geq x\} = P\{|W_t| \geq x\}$ is proved.

□

EXERCISE 3.5. Assume the conditions of [Theorem 3.20](#). Let Z be a bounded $\mathcal{F}_\infty^X$ -measurable random variable. Prove for any initial distribution ν and $t \geq 0$, that

$$\mathbb{E}[Z | \mathcal{F}_{t^+}^X] = \mathbb{E}[Z | \mathcal{F}_t^X], \quad P_\nu\text{-a.s.}$$

SOLUTION. Since the paths of X are right-continuous and the filtration $(\mathcal{F}_t^X)_{t \geq 0}$ is right-continuous, we have

$$\mathcal{F}_t^X = \bigcap_{s>t} \mathcal{F}_s^X.$$

And the σ -algebra $\mathcal{F}_{t^+}^X$ is the right-limit σ -algebra just after time t , defined as

$$\mathcal{F}_{t^+}^X = \bigcap_{n=1}^{\infty} \mathcal{F}_{t+\frac{1}{n}}^X,$$

which implies that $\mathcal{F}_t^X \subseteq \mathcal{F}_{t^+}^X$, and hence $\mathbb{E}[Z | \mathcal{F}_{t^+}^X]$ is $\mathcal{F}_t^X$ -measurable. By the tower property of conditional expectations, we have

$$\mathbb{E}[Z | \mathcal{F}_t^X] = \mathbb{E}[\mathbb{E}[Z | \mathcal{F}_{t^+}^X] | \mathcal{F}_t^X] = \mathbb{E}[Z | \mathcal{F}_{t^+}^X],$$

which completes the proof. $\square$

EXERCISE 3.6. Assume the conditions of [Lemma 3.23](#), and define $\sigma_x = \inf \{t > 0 \mid X_t \neq x\}$.

(i) Show that σ_x is an optional time.

(ii) Assume that x is a ‘holding point’ (see [Lemma 3.23](#). In the proof it is erroneously mentioned that σ_x is a stopping time, but this has no impact on the validity of the result). Show that

$$\{X_{\sigma_x} = x, \sigma_x < \infty\} \subseteq \{\sigma_x \circ \theta_{\sigma_x} = 0, \sigma_x < \infty\}.$$

(iii) Using the strong Markov property, show that

$$P_x\{X_{S_x} = x, \sigma_x < \infty\} = P_x\{X_{S_x} = x, S_x < \infty\}P_x\{S_x = 0\}.$$

(iv) Conclude that $P_x\{X_{S_x} = x, S_x < \infty\} = 0$, i.e., a canonical Markov process with right-continuous paths, satisfying the strong Markov property can only leave a holding point by a jump.

SOLUTION. (i) By the definition $\sigma_x = \inf \{t > 0 \mid X_t \neq x\}$, in the context of right-continuous filtration, we have to show for any $t \geq 0$, $\{\sigma_x \leq t\} \in \mathcal{F}_t$. For any $s \leq t$, we have

$$\{\sigma_x \leq t\} = \bigcup_{s \leq t} \{X_s \neq x\}.$$

Since X_s is $\mathcal{F}_s$ -measurable, the pre-image $\{X_s \neq x\}$ of the Borel set $E \setminus \{x\}$ under X_s is also in $\mathcal{F}_s$. So $\{X_s \neq x\}$ is also in $\mathcal{F}_t$ since $\mathcal{F}_s \subseteq \mathcal{F}_t$, it follows that $\{\sigma_x \leq t\} \in \mathcal{F}_t$ (σ -algebra is closed under countable unions), and hence σ_x is an optional time.

(ii) Given that $\sigma_x = \inf \{t > 0 \mid X_t \neq x\} < \infty$ a stopping time, and at time σ_x , $X_{\sigma_x} = x$, i.e., the process is back in state x immediately after time σ_x . By the definition of the shift operator θ_{σ_x} , we have $(\theta_{\sigma_x} X)_s = X_{\sigma_x+s}$. Then we have

$$\sigma_x \circ \theta_{\sigma_x} = \inf \{t > 0 \mid X_{\sigma_x+s} \neq x\} = 0.$$

Therefore, we have proved

$$\{X_{\sigma_x} = x, \sigma_x < \infty\} \subseteq \{\sigma_x \circ \theta_{\sigma_x} = 0, \sigma_x < \infty\}.$$

(iii) Given that

$$S_x = \inf \{t > 0 \mid X_t \neq X_0\} = \inf \{t > 0 \mid X_t \neq x\}$$

is the sojourn time in the initial state $X_0 = x$, and $\sigma_x = \inf \{t > 0 \mid X_t \neq x\} < \infty$ a stopping time, by the strong Markov property at σ_x , we get

$$\mathbb{E}_x[f(X_{\sigma_x+t}) \mid \mathcal{F}_{\sigma_x}] = \mathbb{E}_{X_{\sigma_x}}f(X_t)$$

for any bounded Borel function f . Let $f(y) = \mathbb{1}_{\{y=x\}}$ be the indicator function that $X_t = x$, then

$$\mathbb{E}_x[\mathbb{1}_{\{X_{\sigma_x+t}=x\}} \mid \mathcal{F}_{\sigma_x}] = \mathbb{E}_{X_{\sigma_x}}\mathbb{1}_{\{X_t=x\}} = 0$$

since $X_t \neq x$ a.s. This implies $P_x(X_{\sigma_x+t} = x \mid \mathcal{F}_{\sigma_x}) = 0$ for all $t > 0$. Or we can use the [Theorem 3.16](#) from LN by setting $Z = \mathbb{1}_{\{X_{S_x}=x\}}$ to have

$$\mathbb{E}_x[\mathbb{1}_{\{X_{S_x}=x\}} \circ \theta_{\sigma_x}] = \mathbb{E}_{X_{\sigma_x}}\mathbb{1}_{\{X_{S_x}=x\}}, \quad P_x\text{-a.s.}$$

Therefore the only scenario where $X_{\sigma_x} = x$ is $\sigma_x = 0$. The probability of the event can be rewritten as:

$$\begin{aligned} P_x\{X_{S_x} = x, \sigma_x < \infty\} &= P_x\{X_{\sigma_x} = x, \sigma_x < \infty, \sigma_x = 0\} \\ &= P_x\{X_{\sigma_x} = x, \sigma_x < \infty\}P_x\{\sigma_x = 0\} \\ &= P_x\{X_{S_x} = x, S_x < \infty\}P_x\{S_x = 0\}, \end{aligned} \quad (\sigma_x = S_x)$$

which completes the proof.

(iv) The event $\{S_x = 0\}$ implies that the process leaves the state x immediately at $t = 0$. Given that the paths of the process are right-continuous, the probability of leaving the state x at exactly $t = 0$ is zero, i.e., $P_x\{S_x = 0\} = 0$. Therefore, by part (iii), we conclude

$$P_x\{X_{S_x} = x, \sigma_x < \infty\} = P_x\{X_{S_x} = x, S_x < \infty\}P_x\{S_x = 0\} = P_x\{X_{S_x} = x, S_x < \infty\} \cdot 0 = 0.$$

□

Feller-Dynkin Processes

SECTION 4.1

| Semi-Groups

The starting point is a transition function $\{P_t\}_{t \geq 0}$ on the measurable space $(E, \mathcal{E})$, where again E is Polish and $\mathcal{E}$ the Borel- σ -algebra on E . Let $\mathcal{S}$ be a Banach space of real-valued measurable functions on E , and let $\|\cdot\|$ denote the corresponding norm. By virtue of the Chapman-Kolmogorov equations, $\{P_t\}_t$ is a so-called semigroup.

DEFINITION 4.1.

The semigroup $\{P_t\}_t$ is a *strongly continuous semigroup* on $\mathcal{S}$ (shorthand notation: $\text{SCSG}(\mathcal{S})$), if

(i) $P_t : \mathcal{S} \rightarrow \mathcal{S}$ is a *bounded linear operator* for each $t \geq 0$. In other words,

$$\|P_t\| := \sup_{f \in \mathcal{S}} \frac{\|P_t f\|}{\|f\|} < \infty$$

for $t \geq 0$;

(ii) $\lim_{t \downarrow 0} \|P_t f - f\| = 0$ for each $f \in \mathcal{S}$.

A main notion that we will use is *closedness*.

EXAMPLE 4.2.

EXAMPLE 4.3.

LEMMA 4.4.

Let $\{P_t\}_{t \geq 0}$ be a $\text{SCSG}(\mathcal{S})$. There are constants $M \geq 1, \alpha \geq 0$, such that $\|P_t\| \leq M e^{\alpha t}$.

COROLLARY 4.5.

$t \mapsto P_t f$ is continuous (in other words $\lim_{s \rightarrow t} \|P_t f - P_s f\| = 0$) for all $f \in \mathcal{S}$, and $t \geq 0$.

Proof.

□

Let next

$$\mathcal{D} = \left\{ f \in \mathcal{S} \mid \exists g \in \mathcal{S} \text{ such that } \lim_{t \downarrow 0} \left\| \frac{P_t f - f}{t} - g \right\| = 0 \right\}.$$

A priori it is not clear whether $\mathcal{D}$ is even non-empty. For each $f \in \mathcal{D}$ we write

$$Af = g = \lim_{t \downarrow 0} \frac{P_t f - f}{t}.$$

$A : \mathcal{D} \rightarrow \mathcal{S}$ is a (generally unbounded) linear operator, with domain $\mathcal{D}(A) = \mathcal{D}$, A is called the *generator*.

EXAMPLE 4.6. We claim that for the BM process we have

$$\mathcal{D}(\mathcal{A}) = \{f \in C^2(\mathbb{R}) \mid f', f'' \in C_0(\mathbb{R})\}.$$

One can show that $f, f'' \in C_0(\mathbb{R})$ implies that $f' \in C_0(\mathbb{R})$. Furthermore, for $f \in \mathcal{D}(A)$ we have $\mathcal{A}f = \frac{f''}{2}$. Notice that this implies that $\mathbf{1}_{\{B\}}, B \in \mathcal{B}$, are not even contained in the domain, and so these functions do not satisfy the Kolmogorov forward and backward equations.

Proof. □

THEOREM 4.7.

Let $\{P_t\}_t$ be an SCSG(S).

(i) Let $f \in S, t \geq 0$. Then $\int_0^t P_s f ds \in \mathcal{D}(A)$ and

$$P_t f - f = A \int_0^t P_s f ds.$$

(ii) Let $f \in \mathcal{D}(A)$ and $t \geq 0$. Then $P_t f \in \mathcal{D}(A)$. The function $t \mapsto P_t f$ is differentiable in S and the Kolmogorov backward and forward equations hold:

$$\frac{d}{dt} P_t f = A P_t f = P_t A f.$$

More precisely,

$$\lim_{h \downarrow 0} \left\| \frac{P_{t+h} f - P_t f}{h} - P_t A f \right\| = \lim_{h \downarrow 0} \left\| \frac{P_{t+h} f - P_t f}{h} - A P_t f \right\| = 0.$$

(iii) Let $f \in \mathcal{D}(A), t \geq 0$. Then

$$P_t f - f = \int_0^t P_s A f ds = \int_0^t A P_s f ds.$$

Proof. □

COROLLARY 4.8.

Let $\{P_t\}_t$ be an SCSG(S). Then $\overline{\mathcal{D}(A)} = S$ and A is a closed operator.

Proof. □

SECTION 4.2

| The Generator Determines the Semi-Group

By virtue of [Corollary 4.5](#), the map $t \mapsto P_t f$ is continuous for each $f \in \mathcal{S}$. Recall that $\|P_t\| \leq M e^{\alpha t}$ for some constants $M \geq 1$ and $\alpha \geq 0$. By the *Integration Lemma*, for all $\lambda > \alpha$ we may define

$$R_\lambda f(x) = \int_0^\infty e^{-\lambda t} P_t f(x) dt.$$

R_λ is simply the Laplace transform of the semigroup calculated at the ‘frequency’ λ . The next lemma collects preliminary properties of the operators R_λ . In particular, it states that for all $\lambda > 0$, R_λ is in fact an operator that maps $\mathcal{S}$ into itself. It is called the *resolvent of order λ* .

LEMMA 4.9.

Let $\{P_t\}_t$ be a SCSG($\mathcal{S}$).

(i) $\|R_\lambda\| \leq \frac{M}{\lambda - \alpha}.$

(ii) The resolvent equation

$$R_\mu - R_\lambda + (\mu - \lambda)R_\mu R_\lambda = 0$$

holds for all $\lambda, \mu > \alpha$.

Proof. □

THEOREM 4.10.

Let $\{P_t\}_t$ be a SCSG($\mathcal{S}$) with $\|P_t\| \leq M \cdot e^{\alpha t}$. For all $\lambda > \alpha$ the following hold.

(i) $R_\lambda \mathcal{S} = \mathcal{D}(A).$

(ii) $\lambda \mathbf{I} - A : \mathcal{D}(A) \rightarrow \mathcal{S}$ is a 1-1 linear operator with $(\lambda \mathbf{I} - A)\mathcal{D}(A) = \mathcal{S}.$

(iii) $(\lambda \mathbf{I} - A)^{-1} : \mathcal{S} \rightarrow \mathcal{D}(A)$ exists as a bounded linear operator. In particular $(\lambda \mathbf{I} - A)^{-1} = R_\lambda.$

(iv) $R_\lambda(\lambda \mathbf{I} - A)f = f$ for all $f \in \mathcal{D}(A).$

(v) $(\lambda \mathbf{I} - A)R_\lambda g = g$ for all $g \in \mathcal{S}.$

Proof. □

THEOREM 4.11 (Hille-Yosida Theorem).

Suppose that there exists a linear operator $A : \mathcal{D}(A) \rightarrow \mathcal{S}$, where $\mathcal{D}(A)$ is a linear subspace of $\mathcal{S}$. Then A is the generator of a SCSG($\mathcal{S}$) semigroup if and only if the following three properties hold:

(i) $\overline{\mathcal{D}(A)} = \mathcal{S};$

(ii) A is $(0, 1)$ -dissipative. In other words:

$$\|\lambda f - Af\| \geq \lambda \|f\|, \quad \forall f \in \mathcal{D}(A), \lambda > 0;$$

(iii) $(\lambda \mathbf{I} - A)\mathcal{D}(A) = \mathcal{S}$ for some $\lambda > 0.$

Proof. □

LEMMA 4.12.

Let $\{P_t\}_t$ be a SCSG($\mathcal{S}$) with $\|P_t\| \leq M e^{\alpha t}$. Suppose that the linear operator $\mathcal{B}$ is an extension of

the generator $\mathcal{A}$. In other words, $\mathcal{B} : \mathcal{D} \rightarrow \mathcal{S}$ is a linear operator with $\mathcal{D} \supset \mathcal{D}(\mathcal{A})$, and $\mathcal{B}f = \mathcal{A}f$ for $f \in \mathcal{D}(\mathcal{A})$. If $\lambda\mathbf{I} - \mathcal{B}$ is 1-1 for some value $\lambda > \alpha$, then $\mathcal{B}$ equals $\mathcal{A}$, that is, $\mathcal{D} = \mathcal{D}(\mathcal{A})$. A sufficient condition for $\lambda\mathbf{I} - \mathcal{B}$ to be 1-1 is that there is no non-trivial $f \in \mathcal{D}$ with $\mathcal{B}f = \lambda f$.

Proof.

□

SECTION 4.3

| Feller-Dynkin Transition Functions

DEFINITION 4.13.

The transition function $(P_t)_{t \geq 0}$ is called a *Feller-Dynkin transition function* if

- (i) $P_t C_0(E) \subseteq C_0(E)$, for all $t \geq 0$.
- (ii) $P_t f(x) \rightarrow f(x)$, $t \downarrow 0$, for every $f \in C_0(E)$ and $x \in E$.

A Markov process with Feller-Dynkin transition function is called a *Feller-Dynkin process*.

THEOREM 4.14.

Suppose that $\{P_t\}_t$ is a Feller-Dynkin transition function. Then $\overline{R_\lambda C_0(E)} = C_0(E)$, and $\{P_t\}_t$ is a SCCSG($C_0(E)$).

Proof.

□

EXAMPLE 4.15.

LEMMA 4.16.

Let X be a Feller-Dynkin process with state space $(E, \mathcal{E})$, initial distribution ν and transition function $(P_t)_t$ defined on an underlying filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P}_\nu)$. Suppose that $(E', \mathcal{E}')$ is a measurable space. Let $\phi : E \rightarrow E'$ be continuous and onto, and such that $\|\phi(x_n)\| \rightarrow \infty$ if and only if $\|x_n\| \rightarrow \infty$. Suppose that $(Q_t)_t$ is a collection of transition kernels, such that

$$P_t(f \circ \phi) = (Q_t f) \circ \phi$$

for all $f \in b\mathcal{E}'$. Then $Y = \phi(X)$ is a Feller-Dynkin process with state space $(E', \mathcal{E}')$, initial measure ν' , with $\nu'(B') = \nu(\phi^{-1}(B'))$, $B' \in \mathcal{E}'$, and transition function $(Q_t)_t$. The generator $\mathcal{B}$ of Y satisfies

$$\mathcal{D}(\mathcal{B}) = \{f \in C_0(E') \mid f \circ \phi \in \mathcal{D}(\mathcal{A})\}$$

and $\mathcal{A}(f \circ \phi) = (\mathcal{B}f) \circ \phi$ for $f \in \mathcal{D}(\mathcal{B})$.

EXAMPLE 4.17. In [Example 3.13](#) we have seen that W_t^2 is a Markov process. W_t^2 is also a Feller-Dynkin process with generator $\mathcal{B}f(x) = 2xf''(x) + f'(x)$, $f \in \mathcal{D}(\mathcal{B}) = C_0^2(\mathbb{R}_+)$.

Generators provide an important link between Feller-Dynkin processes and martingales.

THEOREM 4.18.

Let X be a right-continuous Feller-Dynkin process, defined on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_t)$. For every $f \in \mathcal{D}(A)$ and initial probability measure ν , the process

$$M_t^f = f(X_t) - f(X_0) - \int_0^t A f(X_s) ds,$$

is a $\mathbb{P}_\nu$ -martingale.

COROLLARY 4.19 (Dynkin's Formula).

Let X be a right-continuous Feller-Dynkin process. For every $f \in \mathcal{D}(A)$ and every $\{\mathcal{F}_t^X\}$ -stopping

time τ with $\mathbb{E}_x \tau < \infty$, we have

$$\mathbb{E}_x f(X_\tau) = f(x) + \mathbb{E}_x \int_0^\tau Af(X_s) ds, \quad x \in E.$$

Proof. □

EXAMPLE 4.20.

We will now give a probabilistic interpretation of the *characteristic generator*. Call a point $x \in E$ *absorbing* if for all $t \geq 0$ it holds that $P_t(x, \{x\}) = 1$. This means that if the process starts at an absorbing point x , it never leaves x .

Let X be a right-continuous, Feller-Dynkin process. For $r > 0$, define the $\{\mathcal{F}_t\}_t$ -stopping time

$$\eta_r = \inf \{t \geq 0 \mid \|X_t - X_0\| \geq r\}.$$

LEMMA 4.21.

If $x \in E$ is not absorbing, then $\mathbb{E}_x \eta_r < \infty$ for all $r > 0$ sufficiently small.

Proof. Let $B_x(\epsilon) = \{y \mid \|y - x\| \leq \epsilon\}$ be the closed ball of radius ϵ around the point x . If x is not absorbing, then $P_t(x, B_x(\epsilon)) < p < 1$ for some $t, \epsilon > 0$. By the Feller-Dynkin property of the semi-group P_t we have that $P_t(y, \cdot) \xrightarrow{w} P_t(x, \cdot)$ as $y \rightarrow x$. Hence, the *Portmanteau theorem*, and the fact that $B_x(\epsilon)$ is closed imply that

$$\limsup_{y \rightarrow x} P_t(y, B_x(\epsilon)) \leq P_t(x, B_x(\epsilon)).$$

Let $\hat{p} \in (p, 1)$. It follows that for all y sufficiently close to x , say $y \in B_x(r)$ for some $r \in (0, \epsilon)$, we have $P_t(y, B_x(r)) \leq \hat{p}$. Using the Markov property, we prove $\mathbb{P}_x\{\eta_r > nt\} \leq \hat{p}^n$ by induction. The assertion is immediate for $n = 0$. If it holds for n , then

$$\mathbb{P}_x\{\eta_r > (n+1)t\} = \mathbb{E}_x[\mathbb{1}_{\{\eta_r > nt\}} \mathbb{P}_{X_{nt}}\{\eta_r > t\}] \leq \hat{p} \mathbb{P}_x\{\eta_r > nt\} \leq \hat{p}^{n+1},$$

because $X_{nt} \in B_x(r)$ on $\{\eta_r > nt\}$. Hence,

$$\mathbb{E}_x \eta_r = \int_0^\infty \mathbb{P}_x\{\eta_r \geq s\} ds \leq t \sum_{n=0}^\infty \mathbb{P}_x\{\eta_r \geq nt\} \leq \frac{t}{1 - \hat{p}} < \infty.$$

This completes the proof. □

We can now prove the following alternative description of the generator.

THEOREM 4.22.

Let X be a right-continuous Feller-Dynkin process. For $f \in \mathcal{D}(A)$ we have $Af(x) = 0$ if x is absorbing, and otherwise

$$Af(x) = \lim_{r \downarrow 0} \frac{\mathbb{E}_x f(X_{\eta_r}) - f(x)}{\mathbb{E}_x \eta_r} \quad (4.1)$$

pointwise.

Proof. □

The operator defined by the right-hand side of (4.1) is called the *characteristic operator* of the Markov process X .

Exercises

EXERCISE 4.1 (Generator of Geometric Brownian Motion). Consider canonical geometric BM from [Example 3.11](#) with continuous paths. Define

$$\mathcal{D} = \{f \in C_0(\mathbb{R}^+) \mid x \mapsto f'(x), f''(x), xf'(x), x^2f''(x), x \in \mathbb{R}^+ \in C_0(E)\}.$$

Geometric BM has generator

$$\mathcal{A}f(x) = \mu xf'(x) + \frac{1}{2}\sigma^2 x^2 f''(x), \quad x \in \mathbb{R}^+, \quad (5.7.2)$$

on $f \in \mathcal{D} \subset \mathcal{D}(\mathcal{A})$.

- (i) Show that this is true for $\mu = \frac{\sigma^2}{2}$ and characterise all of $\mathcal{D}(\mathcal{A})$ in this case. You may use the results for BM (use the corresponding Brownian-motion result).
- (ii) Give the main steps in the proof of (5.7.2) in the general case, for functions in $\mathcal{D}$.

SOLUTION. (i) Given that

$$X_t = X_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right),$$

and the density

$$p(t, x, y) = \frac{1}{y\sigma\sqrt{2\pi t}} \exp\left(-\frac{1}{2\sigma^2 t} \left(\ln \frac{y}{x} - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2\right),$$

we have

$$\begin{aligned} P_t f(x) &= \int_0^\infty f(y) p(t, x, y) dy \\ &= \int_{-\infty}^\infty f(xe^z) p(t, x, xe^z) x e^z dz \quad (z = \ln \frac{y}{x} \implies y = x e^z) \\ &= \int_{-\infty}^\infty f(xe^z) \frac{1}{x e^z \sigma \sqrt{2\pi t}} \exp\left(-\frac{1}{2\sigma^2 t} \left(z - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2\right) x e^z dz \\ &= \int_{-\infty}^\infty f(xe^z) \frac{1}{\sigma \sqrt{2\pi t}} \exp\left(-\frac{1}{2\sigma^2 t} (z - m)^2\right) dz, \end{aligned}$$

which is normally distributed with mean $m = \left(\mu - \frac{\sigma^2}{2}\right)t$ and variance $\sigma^2 t$.

- (ii) Since $f \in C^2$, by *Taylor's theorem* at $z = 0$ we have

$$f(xe^z) = f(x) + \frac{d}{dz} f(xe^z) \Big|_{z=0} z + \frac{1}{2} \cdot \frac{d^2}{dz^2} f(xe^z) \Big|_{z=0} z^2 + O(z^3),$$

where the derivatives:

$$\frac{d}{dz} f(xe^z) \Big|_{z=0} = f'(x)x, \quad \frac{d^2}{dz^2} f(xe^z) \Big|_{z=0} = f''(x)x^2 + f'(x)x.$$

Then we can write

$$f(xe^z) = f(x) + f'(x)xz + \frac{1}{2}(f''(x)x^2 + f'(x)x)z^2 + O(z^3).$$

Substitute into the semigroup:

$$P_t f(x) = \int_{-\infty}^\infty \left(f(x) + f'(x)xz + \frac{1}{2}(f''(x)x^2 + f'(x)x)z^2 + O(z^3) \right) \varphi_t(z) dz$$

$$= f(x) \underbrace{\int_{-\infty}^{\infty} \varphi_t(z) dz}_{=1} + f'(x)x \underbrace{\int_{-\infty}^{\infty} z \varphi_t(z) dz}_{=m} + \frac{1}{2} \left(f''(x)x^2 + f'(x)x \right) \underbrace{\int_{-\infty}^{\infty} z^2 \varphi_t(z) dz}_{=m^2 + \sigma^2 t} + \xi,$$

where

$$\xi = \int_{-\infty}^{\infty} O(z^3) \varphi_t(z) dz = O(t^{\frac{3}{2}}).$$

By (a), $m = \left(\mu - \frac{\sigma^2}{2}\right)t$, then

$$\begin{aligned} P_t f(x) &= f(x) + x f'(x) \left(\mu - \frac{\sigma^2}{2}\right)t + \frac{1}{2} \left(f''(x)x^2 + f'(x)x \right) \left(\left(\mu - \frac{\sigma^2}{2}\right)^2 t^2 + \sigma^2 t \right) + O(t^{\frac{3}{2}}) \\ &= f(x) + t \left(\mu x f'(x) + \frac{1}{2} \sigma^2 x^2 f''(x) \right) + O(t^{3/2}), \end{aligned}$$

where we ignore the terms of order t^2 . By the definition of the generator,

$$\begin{aligned} \mathcal{A}f(x) &= \lim_{t \downarrow 0} \frac{P_t f(x) - f(x)}{t} \\ &= x f'(x) \left(\mu - \frac{\sigma^2}{2}\right) + \frac{1}{2} \sigma^2 \left(f''(x)x^2 + f'(x)x \right) \\ &= x f'(x) \left(\left(\mu - \frac{\sigma^2}{2}\right) + \frac{1}{2} \sigma^2 \right) + \frac{1}{2} \sigma^2 x^2 f''(x) \\ &= \mu x f'(x) + \frac{1}{2} \sigma^2 x^2 f''(x), \quad x > 0, \end{aligned}$$

which completes the proof. $\square$

EXERCISE 4.2. We want to construct a standard BM in $\mathbb{R}^d$ ($d < \infty$): this is an $\mathbb{R}^d$ -valued process $W = (W^1, \dots, W^d)$, where $W^1, \dots, W^d$ are independent standard BM in $\mathbb{R}$.

- (i) Sketch how to construct d -dimensional BM.
- (ii) Show that W has stationary, independent increments.
- (iii) Show that W is a Feller-Dynkin process with respect to the natural filtration, with transition function

$$P_t f(x) = \frac{1}{(2\pi t)^{\frac{d}{2}}} \int_{\mathbb{R}^d} f(y) e^{-\frac{\|y-x\|^2}{2t}} dy,$$

where $y = (y_1, \dots, y_d)$, $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ and

$$\|y - x\| = \sqrt{\sum_{i=1}^d (y_i - x_i)^2}$$

is the $L^2(\mathbb{R}^d)$ -norm.

EXERCISE 4.3. Let X be an $\mathbb{R}^d$ -valued canonical continuous Feller-Dynkin process, such that $X_t = X_0 + W_t$, where X_0 is an $\mathbb{R}^d$ -valued random variable and $(W_t)_t$ a standard d -dimensional BM that is independent of X_0 . Notice that X is strong Markov. We would like to show that the generator is defined by

$$Af(x) = \frac{1}{2} \Delta f(x), \quad (4.2)$$

where $\Delta f(x) = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} f(x)$ is the Laplacian of f , for

$$f \in \mathcal{D} = \left\{ f : \mathbb{R}^d \rightarrow \mathbb{R} \mid f, \frac{\partial}{\partial x_i} f, \frac{\partial^2}{\partial x_i \partial x_j} f \in C_0(\mathbb{R}^d), i, j = 1, \dots, d \right\}.$$

We again want to use the characteristic operator. To this end, define for $r > 0$

$$\tau_r = \inf \{t \geq 0 \mid \|X_t - X_0\| \geq r\}.$$

- (i) Argue that τ_r is a finite $(\mathcal{F}_t)_t$ -stopping time. Show that $\mathbb{E}_x \tau_r = \frac{r^2}{d}$ (by using optional stopping). Argue that X_{τ_r} has the uniform distribution on $\{y \mid \|y - x\| = r\}$.
- (ii) Show the validity of (4.2) for $f \in \mathcal{D}$ (use the characteristic operator). Argue that this implies $\mathcal{D}(A) \supset \mathcal{D}$.
- (iii) For $0 < a < \|x\| < b$, show that

$$P_x \{T_a < T_b\} = \begin{cases} \frac{\log b - \log \|x\|}{\log b - \log a}, & d = 2 \\ \frac{\|x\|^{2-d} - b^{2-d}}{a^{2-d} - b^{2-d}}, & d \geq 3 \end{cases},$$

where $T_a = \inf \{t \geq 0 \mid \|X_t\| \leq a\}$ and $T_b = \inf \{t \geq 0 \mid \|X_t\| \geq b\}$.

- (iv) Compute $P_x \{T_a < \infty\}$ for x with $a < \|x\|$.

EXERCISE 4.4 (Strong Markov Property of Feller–Dynkin Processes). In view of [Theorem 3.20](#), answer the following questions.

- (i) Explain why it is not immediately clear from the theorem that every Feller–Dynkin process has the strong Markov property.
- (ii) Prove that every Feller–Dynkin process has the strong Markov property.

SOLUTION. (i) In a Feller–Dynkin process, the semigroup maps $C_0(E)$ into itself and is strongly continuous on deterministic times. But it is not obvious whether the mapping

$$t \mapsto \mathbb{E}_{X_t}[f(X_s)] = P_s f(X_t)$$

is continuous at a random stopping time σ . One must be able to pass to the limit

$$\mathbb{E}_{X_{\sigma_n}}[f(X_t)] \rightarrow \mathbb{E}_{X_\sigma}[f(X_t)],$$

which requires additional continuity assumptions.

- (ii) Let σ be a $\mathcal{F}_t$ -stopping time. By [Lemma 1.43](#), there exists a sequence of stopping times σ_n taking values in a countable subset $S \subset [0, \infty)$ such that

$$\sigma_n \downarrow \sigma, \quad a.s.$$

Let $f : E \rightarrow \mathbb{R}$ be a bounded continuous function. For any stopping time σ_n , by [Lemma 3.18](#), the strong Markov property holds:

$$\mathbb{1}_{\{\sigma_n < \infty\}} \mathbb{E}_V[f(X_{\sigma_n+t}) \mid \mathcal{F}_{\sigma_n}] = \mathbb{1}_{\{\sigma_n < \infty\}} \mathbb{E}_{X_{\sigma_n}} f(X_t), \quad a.s.$$

Since X_t is right-continuous, we have

$$X_{\sigma_n} \rightarrow X_\sigma, \quad a.s.$$

Given that $P_t f \in C_0(E)$, $x \mapsto P_t f(x)$ is continuous and hence

$$P_t f(X_{\sigma_n}) \rightarrow P_t f(X_\sigma), \quad a.s.$$

Since f is bounded, by the *dominated convergence theorem*, as $n \rightarrow \infty$,

$$\mathbb{E}_v[f(X_{\sigma_n+t}) \mid \mathcal{F}_{\sigma_n}] \rightarrow \mathbb{E}_v[f(X_{\sigma+t}) \mid \mathcal{F}_\sigma].$$

Then by [Theorem 3.20](#), the strong Markov property holds:

$$\mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_v[f(X_{\sigma+t}) \mid \mathcal{F}_\sigma] = \mathbb{1}_{\{\sigma < \infty\}} P_t f(X_\sigma) = \mathbb{1}_{\{\sigma < \infty\}} \mathbb{E}_{X_\sigma}[f(X_t)], \quad a.s.$$

□

EXERCISE 4.5 (Feller–Dynkin Property of Geometric Brownian Motion). Prove that geometric BM is also a Feller–Dynkin process.

SOLUTION. Let $f \in C_0((0, \infty))$ and define

$$P_t f(x) = \int_0^\infty f(y) p(t, x, y) dy,$$

where

$$p(t, x, y) = \frac{1}{y\sigma\sqrt{2\pi t}} \exp\left(-\frac{1}{2\sigma^2 t} \left(\ln \frac{y}{x} - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2\right).$$

- Since the density is jointly continuous in (x, y) for fixed $t > 0$ and f is bounded, it follows that

$$f(y)p(t, x_n, y) \rightarrow f(y)p(t, x, y).$$

Notice that

$$\int_{-\infty}^\infty p(t, x, y) dy = 1,$$

then by the *dominated convergence theorem*,

$$\lim_{n \rightarrow \infty} P_t f(x_n) = \int_0^\infty f(y) p(t, x, y) dy = P_t f(x),$$

which implies $P_t f$ is continuous.

Let $\varepsilon > 0$. Since $f \in C_0((0, \infty))$ is continuous and vanishes at infinity, there exists a compact interval $K = [a, b] \subset (0, \infty)$ such that $|f(y)| < \varepsilon$ for all $y \notin K$. Notice that

$$\lim_{x \rightarrow 0, x \rightarrow \infty} \int_K p(t, x, y) dy = 0.$$

Then

$$|P_t f(x)| \leq \int_K |f(y)| p(t, x, y) dy + \int_{(0, \infty) \setminus K} p(t, x, y) dy \leq \varepsilon + \varepsilon = 2\varepsilon,$$

which shows that $P_t f(x) \rightarrow 0$ as $x \rightarrow 0$ or $x \rightarrow \infty$.

- Let $z = \ln \frac{y}{x}$, then $dy = x e^z dz$. The density is

$$p(t, x, y) dy = \frac{1}{\sigma\sqrt{2\pi t}} \exp\left(-\frac{1}{2\sigma^2 t} \left(z - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2\right) \sim \mathcal{N}\left(\left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right).$$

As $t \downarrow 0$, $\left(\mu - \frac{\sigma^2}{2}\right)t \rightarrow 0$ and $\sigma^2 t \rightarrow 0$, which implies $p(t, x, y)dy \rightarrow \delta_0(dz)$, i.e., $p(t, x, y)dy \rightarrow \delta_x(dy)$. Then, by the continuity of f ,

$$\lim_{t \downarrow 0} P_t f(x) = \int_0^\infty f(y) \delta_x(dy) = f(x),$$

which proves the strong continuity of the semigroup.

In conclusion, the geometric BM is a Feller-Dynkin process. □

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